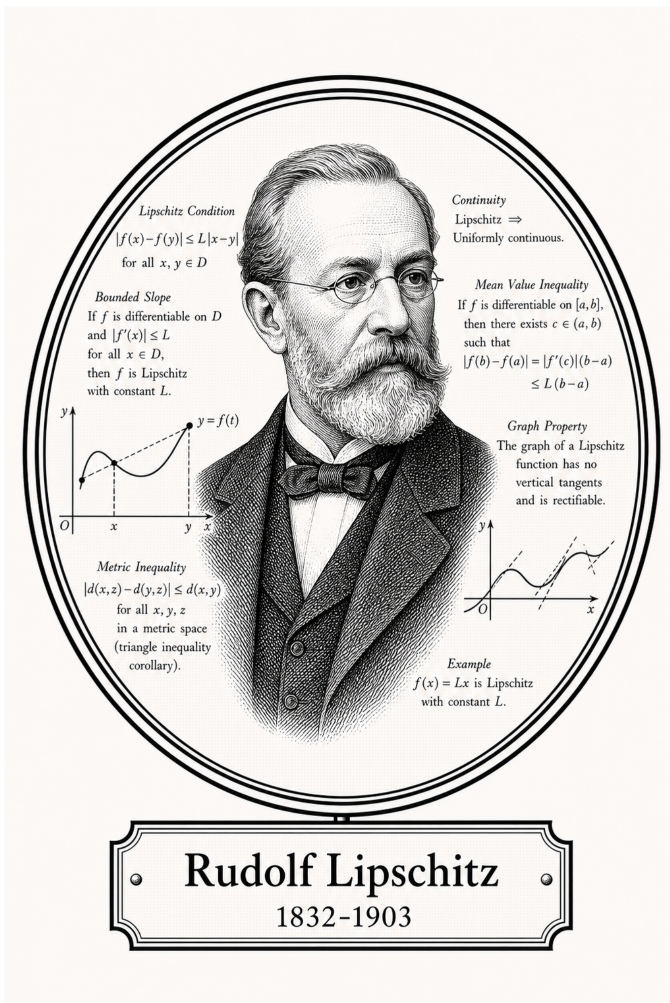
From Cantor to Ito

# Classical Analysis

## Functions, Continuity, and Differentiation

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1832-1903

*Self-Study Notes*

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*For every reader learning to make mathematics their own.*

# Contents

# Chapter 1

# Elementary Functions

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## 1.1 Hyperbolic Functions

**Hyperbolic Functions — Quick Reference**

| Concept | Meaning |
|---|---|
| Hyperbolic functions | sinh and cosh from even/odd exponential parts |
| Hyperbolic Pythagorean identity | $\cosh^2 x - \sinh^2 x = 1$ |
| Addition formulas | Hyperbolic angle-addition identities |
| Tanh limits | $\tanh x \to \pm 1$ as $x \to \pm\infty$ |
| Inverse hyperbolic functions | Logarithmic inverse formulas |

### Hyperbolic Functions

**Definition (Hyperbolic Functions)**

**Definition 1.1.1** (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \qquad \sinh x := \frac{e^x - e^{-x}}{2}, \qquad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \ \sinh x = \frac{e^x - e^{-x}}{2}, \ e^x = \cosh x + \sinh x.$$

*Remark* (Definition predicate reading).

$$\mathrm{HyperbolicCosh}(x,y) \;:\equiv\; y = \frac{e^x + e^{-x}}{2}. \qquad \mathrm{HyperbolicSinh}(x,y) \;:\equiv\; y = \frac{e^x - e^{-x}}{2}.$$

$$\mathrm{HyperbolicTanh}(x,y) \;:\equiv\; \cosh x \neq 0 \;\wedge\; y = \frac{\sinh x}{\cosh x}.$$

*Remark* (Definition predicate reading).

$\mathrm{HyperbolicPythagorean}(x) \;:\equiv\; \mathrm{HyperbolicCosh}(x,c) \wedge \mathrm{HyperbolicSinh}(x,s) \wedge c^2 - s^2 = 1.$

*Remark* (Negated quantified statement).

$$\neg\, \mathrm{HyperbolicTanh}(x,y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note $\cosh x > 0$ for all $x$, so HyperbolicTanh is defined on all of $\mathbb{R}$.

*Remark* (Contrast with trigonometric functions).

| **Trigonometric** | **Hyperbolic** |
|---|---|
| $\cos^2 x + \sin^2 x = 1$ | $\cosh^2 x - \sinh^2 x = 1$ |
| $(\cos x)' = -\sin x$ | $(\cosh x)' = \sinh x$ |
| $(\sin x)' = \cos x$ | $(\sinh x)' = \cosh x$ |
| $e^{ix} = \cos x + i \sin x$ | $e^x = \cosh x + \sinh x$ |
| Parametrises unit circle | Parametrises unit hyperbola |

The minus sign in $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$ reflects the substitution $x \mapsto ix$ in Euler's formula.

**Proposition (Hyperbolic Pythagorean Identity)**

**Proposition 1.1.2** (Hyperbolic Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \cosh^2 x - \sinh^2 x = 1.$$

*Remark* (Negated quantified statement).

$$\neg\, (\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

*Remark* (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

*Remark* (Dependencies).

- Exponential Function
- $\mathbb{R}$ Is a Field

**Proposition (Addition Formulas for Hyperbolic Functions)**

**Proposition 1.1.3** (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

*Remark* (Definition predicate reading).

$$\mathrm{HyperbolicAddition}(x, y) \ :\equiv \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x$$

*Remark* (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R} : \ \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y.$$

The negation has no witness.

*Remark* (Proof strategy). Substitute the definitions and expand, or note the formulas follow from $e^{x+y} = e^x e^y$ by taking even and odd parts.

*Remark* (Comparison with trigonometric addition formulas). The sinh formula matches $\sin(x+y)$ exactly. The cosh formula differs from $\cos(x+y)$ only in the sign of the last term: $+\sinh x \sinh y$ versus $-\sin x \sin y$. The sign difference is a direct consequence of $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$.

*Remark* (Dependencies).

- Hyperbolic Functions
- Exponential Function
- $\mathbb{R}$ Is a Field

**Proposition (Limits of $\tanh$)**

**Proposition 1.1.4** (Limits of tanh)**.**

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ \wedge\ \mathrm{LimitAtInfinity}(\tanh x,\ -\infty,\ -1).$$

*Remark* (Definition predicate reading).

$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(\tanh x, 1, \varepsilon).$ $\blacksquare$

*Remark* (Negated quantified statement).

$$\neg\left(\lim_{x\to+\infty} \tanh x = 1\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ |\tanh x - 1| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Write $\tanh x = (1 - e^{-2x})/(1 + e^{-2x})$. As $x \to +\infty$, $e^{-2x} \to 0$, giving $\tanh x \to 1$. The limit at $-\infty$ follows by oddness: $\tanh(-x) = -\tanh(x)$.

*Remark* (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

**Definition**

**Definition 1.1.5** (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R} \to \mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse arccosh is defined on $[1, \infty)$. Since tanh is strictly increasing with range $(-1, 1)$, arctanh is defined on $(-1, 1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R}.$$

$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1.$$

$$\operatorname{arctanh} x = \tfrac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right), \quad |x| < 1.$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R}:\ \sinh(\operatorname{arcsinh} x) = x\ \wedge\ \operatorname{arcsinh}(\sinh x) = x.$$

$$\forall x \geq 1:\ \cosh(\operatorname{arccosh} x) = x. \qquad \forall |x| < 1:\ \tanh(\operatorname{arctanh} x) = x.$$

*Remark* (Definition predicate reading).

$$\mathrm{Arcsinh}(y, x)\ :\equiv\ x \in \mathbb{R}\ \wedge\ \sinh x = y\ :\equiv\ x = \ln\left(y + \sqrt{y^2 + 1}\right).$$

$$\mathrm{Arctanh}(y, x)\ :\equiv\ |y| < 1\ \wedge\ \tanh x = y\ :\equiv\ x = \tfrac{1}{2} \ln\left(\frac{1 + y}{1 - y}\right).$$

*Remark* (Negated quantified statement).

$$\neg\ \operatorname{Arctanh}(y,x) \iff |y| \geq 1 \ \vee\ \tanh x \neq y.$$

*Remark* (Derivation of arcsinh). Set $y = \sinh x = (e^x - e^{-x})/2$ and let $u = e^x > 0$. Then $y = (u - 1/u)/2$, so $u^2 - 2yu - 1 = 0$. The positive root is $u = y + \sqrt{y^2+1}$, giving $x = \ln(y + \sqrt{y^2+1})$.

*Remark* (Derivatives). $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2+1}$; $(\operatorname{arccosh} x)' = 1/\sqrt{x^2-1}$ for $x > 1$; $(\operatorname{arctanh} x)' = 1/(1-x^2)$ for $|x| < 1$. These follow from the inverse function theorem applied to sinh, cosh, and tanh respectively.

*Remark* (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Limit at Infinity
- Limit of a Quotient

*Remark* (Dependencies).

- Hyperbolic Functions
- Natural Logarithm
- Continuous Inverse Theorem
- Absolute Value on $\mathbb{R}$

## 1.2 The Logarithm

**Logarithmic Functions — Quick Reference**

| **Concept** | **Meaning** |
|---|---|
| Natural logarithm | Inverse of the exponential on $(0, \infty)$ |
| Fundamental logarithmic limit | $\ln(1+x)/x \to 1$ |
| Power dominates log | Logarithmic growth is below every positive power |

## The Logarithm

**Definition (Natural Logarithm)**

**Definition 1.2.1** (Natural Logarithm)**.** The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \;:=\; \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \;:=\; \int_1^x \frac{1}{t}\, dt.$$

*Remark* (Standard quantified statement)*.*

$$\forall x > 0 :\; e^{\ln x} = x. \qquad \forall y \in \mathbb{R} :\; \ln(e^y) = y.$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{NaturalLog}(x, y) \;:\equiv\; x > 0 \;\wedge\; e^y = x \;:\equiv\; x > 0 \;\wedge\; y = \int_1^x \frac{1}{t}\, dt.$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{FunctionLimit}\left(\frac{\ln(1+x)}{x},\, 0,\, 1\right) \;:\equiv\; \forall \varepsilon > 0,\; \exists \delta > 0,\; \forall x :\; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigg(\frac{\ln$$

*Remark* (Negated quantified statement)*.*

$$\neg\, \mathrm{NaturalLog}(x, y) \iff x \le 0 \;\vee\; e^y \ne x.$$

*Remark* (Two definitions, same function)*.* The inverse-of-exponential definition requires $e^x$ to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

*Remark* (Key properties)*.* (i) $\ln 1 = 0$; $\ln e = 1$.

(ii) **Functional equation:** $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$.

(iii) $\ln(x/y) = \ln x - \ln y$; $\ln(x^r) = r \ln x$ for $r \in \mathbb{R}$.

(iv) $\ln$ is strictly increasing on $(0, \infty)$.

(v) $\ln x \to +\infty$ as $x \to \infty$; $\ln x \to -\infty$ as $x \to 0^+$.

(vi) $(\ln x)' = 1/x$ for all $x > 0$.

*Remark* (General logarithm)*.* For $a > 0$, $a \ne 1$, the **logarithm base** $a$ is $\log_a x := \ln x / \ln a$. The common logarithm is $\log_{10}$; the binary logarithm is $\log_2$.

**Proposition (Fundamental Logarithmic Limit)**

**Proposition 1.2.2** (Fundamental Logarithmic Limit)**.**

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{\ln(1+x)}{x} - 1 \right| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\neg\left(\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1\right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta\ \wedge\ \left| \frac{\ln(1+x)}{x} - 1 \right| \geq \varepsilon_0.$$

*Remark* (Equivalence with exponential limit). The fundamental exponential limit $\lim(e^x - 1)/x = 1$ and the fundamental logarithmic limit $\lim \ln(1+x)/x = 1$ are equivalent: each follows from the other via the substitution $x \leftrightarrow e^x - 1$. Both express the first-order Taylor approximation of the respective function at 0.

*Remark* (Proof strategy). From the series: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (valid for $|x| < 1$). Dividing by $x$ gives $1 - x/2 + x^2/3 - \cdots \to 1$ as $x \to 0$ by continuity of power series.

*Remark* (Dependencies).

- Exponential Function
- Inverse Bijection
- Continuous Inverse Theorem

**Proposition (Every Power Dominates the Logarithm)**

**Proposition 1.2.3** (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x\to 0^+} x^r \ln x = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall r > 0 :\ \lim_{x\to\infty} x^{-r} \ln x = 0\ \wedge\ \lim_{x\to 0^+} x^r \ln x = 0.$$

*Remark* (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^{-r}\ln x,\ +\infty,\ 0)\ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}(x^{-r}\ln x, 0, \varepsilon).$$

*Remark* (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty}\frac{\ln x}{x^r}=0\right) \iff \exists\varepsilon_0>0,\ \forall M>0,\ \exists x>M:\ x^{-r}|\ln x|\geq\varepsilon_0.$$

*Remark* (Proof strategy). For the first: substitute $x=e^t$ to get $t/e^{rt}\to 0$ by exponential dominance. For the second: substitute $x=1/t$ and reduce to the first: $x^r\ln x=(-\ln t)/t^r\to 0$.

*Remark* (Significance). These limits establish the position of $\ln x$ in the growth hierarchy: $\ln x$ is dominated by every positive power. Combined with the exponential dominance result, the full ordering is $\ln x \ll x^r \ll e^x$ as $x\to\infty$ for any $r>0$. The second limit says that near $0^+$, the positive power $x^r$ wins over the logarithmic blow-up $|\ln x|\to\infty$. Used in integrability arguments near 0.

*Remark* (Dependencies).

- Natural Logarithm
- Fundamental Exponential Limit
- Function Limit

*Remark* (Dependencies).

- Natural Logarithm
- Power Function
- Exponential Dominates Every Power
- Limit at Infinity

## 1.3 Power Functions and the Exponential

**Power and Exponential Functions — Quick Reference**

| Concept | Meaning |
|---|---|
| Power function | Real powers $x^r$ on positive inputs |
| The number $e$ | Series and compound-interest limit definitions |
| Exponential function | Power series with $e^{x+y}=e^xe^y$ |
| Fundamental exponential limit | $(e^x-1)/x\to 1$ |
| Compound-interest limit | $(1+1/n)^n\to e$ |
| Growth hierarchy | $e^x$ dominates powers at infinity |

## Power Functions and the Exponential

**Definition (Power Function)**

**Definition 1.3.1** (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \;:=\; e^{r \ln x}.$$

*Remark* (Standard quantified statement)*.*

$$\forall r \in \mathbb{R},\ \forall x > 0:\ x^r = e^{r \ln x}.$$

For $r \in \mathbb{N}$ this reduces to the $r$-fold product $x \cdot x \cdots x$.

*Remark* (Definition predicate reading)*.*

$$\mathrm{RealPower}(x, r, y) \;:\equiv\; x > 0 \;\wedge\; y = e^{r \ln x}.$$

*Remark* (Laws of exponents)*.* For $x, y > 0$ and $r, s \in \mathbb{R}$:

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition $x^r = e^{r \ln x}$.

*Remark* (Interpretation)*.* Defining $x^r := e^{r \ln x}$ unifies all three cases — integer, rational, irrational exponents — into a single formula and makes differentiability of $x^r$ immediate from that of $e^x$ and $\ln x$ via the chain rule: $(x^r)' = r x^{r-1}$.

**Definition (The Number $e$)**

**Definition 1.3.2** (The Number $e$)**.** The **Euler number** is

$$e \;:=\; \sum_{n=0}^{\infty} \frac{1}{n!} \;=\; \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \;\approx\; 2.71828\ldots$$

Both expressions define the same real number.

*Remark* (Standard quantified statement)*.*

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \;\wedge\; e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n.$$

*Remark* (Interpretation)*.* The series $\sum 1/n!$ converges rapidly by comparison with a geometric series. The limit $(1 + 1/n)^n$ arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

*Remark* (Dependencies)*.*

- Integer Powers in a Field

- Exponential Function
- Natural Logarithm
- Multiplication on $\mathbb{R}$

*Remark* (Dependencies).

- Sequence
- Convergent Sequence

**Definition (Exponential Function)**

**Definition 1.3.3** (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x \;:=\; \exp(x) \;:=\; \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{convergent absolutely since } \sum \frac{|x|^n}{n!} < \infty.$$

*Remark* (Definition predicate reading).

$$\text{Exponential}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

*Remark* (Key properties). (i) $e^0 = 1;\; e^1 = e$.

(ii) **Functional equation:** $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$.

(iii) $e^x > 0$ for all $x$; $e^{-x} = 1/e^x$.

(iv) $e^x$ is strictly increasing and bijective $\mathbb{R} \to (0, \infty)$.

(v) $(e^x)' = e^x$ — the exponential is its own derivative.

*Remark* (General exponential). For $a > 0$, $a \neq 1$: $a^x := e^{x \ln a}$. Strictly increasing if $a > 1$, strictly decreasing if $0 < a < 1$. $(a^x)' = a^x \ln a$. Only $e^x$ is its own derivative.

**Proposition (Fundamental Exponential Limit)**

**Proposition 1.3.4** (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}\left( \frac{e^x - 1}{x},\ 0,\ 1 \right) \ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\left( \frac{e^x -}{x} \right.$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \to 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta\ \wedge\ \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

*Remark* (Interpretation). This is the statement $(e^x)'\big|_{x=0} = 1$. The series gives $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \to 1$ as $x \to 0$ by continuity of power series at their center. Combined with the functional equation, this implies $(e^x)' = e^x$ everywhere.

*Remark* (Dependencies).

- The Number $e$
- Integer Powers in a Field
- Sequence
- Convergent Sequence

**Proposition (Compound Interest Limit)**

**Proposition 1.3.5** (Compound Interest Limit)**.**

$$\lim_{x \to \infty} \left( 1 + \frac{1}{x} \right)^x = e.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists M > 0 :\ x > M \Rightarrow \left| \left( 1 + \frac{1}{x} \right)^x - e \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{LimitAtInfinity}\left( \left(1 + \tfrac{1}{x}\right)^x,\ +\infty,\ e \right) \ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}\left( \left(1 + \tfrac{1}{x}\right)^x, e, \varepsilon \right)$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \to \infty} \left(1 + \tfrac{1}{x}\right)^x = e \right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ \left| \left( 1 + \frac{1}{x} \right)^x - e \right| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Write $(1 + 1/x)^x = e^{x \ln(1+1/x)}$. Set $t = 1/x \to 0^+$ and use the fundamental logarithmic limit $\ln(1 + t)/t \to 1$. Then $x \ln(1 + 1/x) = \ln(1 + t)/t \to 1$, so $(1 + 1/x)^x = e^{x \ln(1+1/x)} \to e^1 = e$ by continuity of the exponential.

*Remark* (Dependencies).

- Exponential Function
- Function Limit
- Derivative at a Point

**Proposition (Exponential Dominates Every Power)**

**Proposition 1.3.6** (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{x^r}{e^x} = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall r > 0 : \lim_{x\to\infty} x^r e^{-x} = 0.$$

*Remark* (Definition predicate reading).

$$\text{LimitAtInfinity}(x^r e^{-x},\ +\infty,\ 0) \ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \text{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$$

*Remark* (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{x^r}{e^x} = 0\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ x^r e^{-x} \geq \varepsilon_0.$$

*Remark* (Proof strategy). Pick integer $n > r$. The series bound $e^x \geq x^n/n!$ gives $x^r/e^x \leq n!\, x^{r-n} \to 0$ since $r - n < 0$.

*Remark* (Growth hierarchy). For any $r > 0$:

$$\ln x \ \ll\ x^r \ \ll\ e^x \qquad \text{as } x \to \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

*Remark* (Dependencies).

- The Number $e$
- Exponential Function

- Natural Logarithm
- Fundamental Logarithmic Limit
- Limit at Infinity

*Remark* (Dependencies).

- Power Function
- Exponential Function
- Limit at Infinity
- Squeeze Theorem for Function Limits

## 1.4 Trigonometric Functions

**Trigonometric Functions — Quick Reference**

| Concept | Meaning |
|---|---|
| Sine and cosine | Defined analytically by power series |
| Pythagorean identity | $\sin^2 x + \cos^2 x = 1$ |
| Addition formulas | Angle addition identities from exponentials |
| Other trig functions | Tangent, cotangent, secant, cosecant |
| Inverse trig functions | Inverses on monotone restricted domains |
| Fundamental trig limit | $\sin x/x \to 1$ as $x \to 0$ |
| Second trig limit | $(1 - \cos x)/x^2 \to 1/2$ |

### Trigonometric Functions

**Definition (Sine and Cosine)**

**Definition 1.4.1** (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x \;:=\; \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \;=\; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x \;:=\; \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \;=\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} :\ \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

*Remark* (Definition predicate reading).

$$\mathrm{Sine}(x,y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \qquad \mathrm{Cosine}(x,y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

*Remark* (Key properties). (i) $\sin 0 = 0$; $\cos 0 = 1$.

(ii) $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even).

(iii) $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$.

(iv) $|\sin x| \le 1$; $|\cos x| \le 1$ for all $x$.

(v) $\sin^2 x + \cos^2 x = 1$ (Pythagorean identity).

(vi) Periodic: $\sin(x + 2\pi) = \sin x$; $\cos(x + 2\pi) = \cos x$.

*Remark* (Relationship to $e^{ix}$). Euler's formula: $e^{ix} = \cos x + i \sin x$, where $e^{ix}$ is defined by the complex exponential series. This gives $\cos x = \mathrm{Re}(e^{ix})$ and $\sin x = \mathrm{Im}(e^{ix})$. Euler's identity $e^{i\pi} + 1 = 0$ follows immediately.

**Proposition (Pythagorean Identity)**

**Proposition 1.4.2** (Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \sin^2 x + \cos^2 x = 1.$$

*Remark* (Definition predicate reading).

$$\mathrm{PythagoreanIdentity}(x) \;:\equiv\; \mathrm{Sine}(x,s) \;\wedge\; \mathrm{Cosine}(x,c) \;\wedge\; s^2 + c^2 = 1.$$

*Remark* (Negated quantified statement).

$$\neg\,(\forall x : \sin^2 x + \cos^2 x = 1) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

*Remark* (Proof strategy). Define $f(x) := \sin^2 x + \cos^2 x$. Then $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$, so $f$ is constant. Since $f(0) = 0 + 1 = 1$, the identity follows.

*Remark* (Dependencies).

- Real Numbers

- Integer Powers in a Field
- Sequence
- Convergent Sequence

**Proposition (Addition Formulas)**

**Proposition 1.4.3** (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sin(x+y) = \sin x \cos y + \cos x \sin y \ \wedge \ \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

*Remark* (Definition predicate reading).

$$\mathrm{TrigAddition}(x, y) \ :\equiv \ \sin(x+y) = \sin x \cos y + \cos x \sin y \wedge \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

*Remark* (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R} : \ \sin(x+y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

*Remark* (Proof strategy). From the Cauchy product: $e^{i(x+y)} = e^{ix}e^{iy}$. Taking real and imaginary parts yields both addition formulas simultaneously.

*Remark* (Derived identities).

$$\begin{aligned} \sin(2x) &= 2\sin x \cos x \\ \cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \\ \sin^2 x &= \tfrac{1}{2}(1 - \cos 2x) \\ \cos^2 x &= \tfrac{1}{2}(1 + \cos 2x) \end{aligned}$$

*Remark* (Dependencies).

- Sine and Cosine
- Derivative at a Point
- Zero Derivative Implies Constant

**Definition**

**Definition 1.4.4** (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R},\ \cos x \neq 0 :\ \tan x = \frac{\sin x}{\cos x}.$$

*Remark* (Definition predicate reading).

$$\mathrm{Tangent}(x, y)\ :\equiv\ \cos x \neq 0\ \wedge\ y = \frac{\sin x}{\cos x}.$$

*Remark* (Pythagorean identities for tan). Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$: $\cot^2 x + 1 = \csc^2 x$.

*Remark* (Dependencies).

- Sine and Cosine
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

**Definition**

**Definition 1.4.5** (Inverse Trigonometric Functions).

- $\arcsin : [-1, 1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1, 1] \to [0, \pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

*Remark* (Standard quantified statement).

$$\forall y \in [-1, 1] :\ \sin(\arcsin y) = y. \qquad \forall x \in [-\pi/2, \pi/2] :\ \arcsin(\sin x) = x.$$

*Remark* (Definition predicate reading).

$$\mathrm{Arctan}(y, x)\ :\equiv\ x \in (-\pi/2, \pi/2)\ \wedge\ \tan x = y.$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\sin x}{x},\ 0,\ 1\right)\ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\left(\frac{\sin x}{x},\right.$$

*Remark* (Negated quantified statement).

$$\neg\ \mathrm{Arctan}(y, x)\ \iff\ x \notin (-\pi/2, \pi/2)\ \vee\ \tan x \neq y.$$

*Remark* (Key limits of arctan).

$$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

arctan is bounded, strictly increasing, and continuous on $\mathbb{R}$; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.

*Remark* (Dependencies).

- Sine and Cosine
- $\mathbb{R}$ Is a Field

**Theorem (Fundamental Trigonometric Limit)**

**Theorem 1.4.6** (Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0}\frac{\sin x}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall\varepsilon > 0,\ \exists\delta > 0:\ 0 < |x| < \delta \Rightarrow \left|\frac{\sin x}{x} - 1\right| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\neg\left(\lim_{x\to 0}\frac{\sin x}{x} = 1\right) \iff \exists\varepsilon_0 > 0,\ \forall\delta > 0,\ \exists x:\ 0 < |x| < \delta\ \wedge\ \left|\frac{\sin x}{x} - 1\right| \geq \varepsilon_0.$$

*Remark* (Proof strategy — series). From the series: $\sin x/x = 1 - x^2/3! + x^4/5! - \cdots \to 1$ as $x \to 0$ by continuity of power series at their center.

*Remark* (Proof strategy — geometric squeeze). For $0 < x < \pi/2$, area comparison of two triangles and a circular sector gives $\cos x \leq \sin x/x \leq 1$. Since $\cos x \to 1$ as $x \to 0^+$, the Squeeze Theorem gives $\sin x/x \to 1$. Even symmetry extends this to $x \to 0^-$.

*Remark* (Significance). This is the statement $(\sin x)'\big|_{x=0} = 1$, which implies $(\sin x)' = \cos x$ everywhere. It underlies all trigonometric differentiation.

*Remark* (Dependencies).

- Sine and Cosine
- Tangent, Cotangent, Secant, Cosecant
- Continuous Inverse Theorem
- Closed Interval

**Proposition (Second Fundamental Trigonometric Limit)**

**Proposition 1.4.7** (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{1 - \cos x}{x^2},\ 0,\ \tfrac{1}{2}\right) :\equiv \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigg($$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge\ \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Use the half-angle identity $1 - \cos x = 2\sin^2(x/2)$:

$$\frac{1 - \cos x}{x^2} = \frac{2\sin^2(x/2)}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \to \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

*Remark* (Dependencies).

- Sine and Cosine
- Function Limit
- Squeeze Theorem for Function Limits

*Remark* (Dependencies).

- Sine and Cosine
- Trigonometric Addition Formulas
- Fundamental Trigonometric Limit
- Function Limit

# Chapter 2

# Continuity

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK II
Functions, Continuity, and Differentiation

CHAPTER 9
Continuity

## 2.1 Limits

**One-Sided Limits — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Right-hand limit | Approach the point from inputs larger than $c$. |
| Left-hand limit | Approach the point from inputs smaller than $c$. |
| Right sequential criterion | Test right limits by right-approaching sequences. |
| Left sequential criterion | Test left limits by left-approaching sequences. |
| Matching criterion | Two-sided limits exist exactly when side limits agree. |

**Limits of Functions — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Function limit | Punctured input control near a cluster point. |
| Sequential limit | All approaching sequences have the same output limit. |
| Neighbourhood limit | Neighbourhood formulation of the same condition. |
| Uniqueness | A function has at most one limit at a point. |
| Sequential criterion | Sequence convergence is the main working test. |
| Local boundedness | Finite limits bound the function near the point. |
| Squeeze theorem | Ordered comparison transfers limits. |
| Locality | Only behaviour near the approach point matters. |

**Algebra of Limits — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Sum rule | Limits add pointwise. |
| Difference rule | Limits subtract pointwise. |
| Scalar multiple rule | Scalars pass through limits. |
| Product rule | Products of finite limits multiply. |
| Quotient rule | Division is valid when the denominator limit is nonzero. |

## Limits of Functions

**Definition ($\varepsilon$-Closeness of a Function to a Value)**

**Definition 2.1.1** ($\varepsilon$-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function $f$ is **$\varepsilon$-close to** $L$ if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

*Remark* (Standard quantified statement).

$$f \ \varepsilon\text{-close to } L \iff \forall x \in X \ \big(|f(x) - L| \leq \varepsilon\big).$$

*Remark* (Interpretation). A global condition: every output lies within $\varepsilon$ of $L$, regardless of where in $X$ the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of $c$.

**Definition (Local $\delta$-Closeness Near a Point)**

**Definition 2.1.2** (Local $\delta$-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let $x_0$ be adherent to $X$, and let $\varepsilon, \delta > 0$. The function $f$ is **$\varepsilon$-close to** $L$ **near** $x_0$ **within** $\delta$ if

$$\forall x \in X \ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\big).$$

*Remark* (Standard quantified statement).

$$\forall x \in X : \ |x - x_0| < \delta \ \Rightarrow \ |f(x) - L| \leq \varepsilon.$$

*Remark* (Interpretation). A local condition: only outputs from the $\delta$-ball around $x_0$ need lie within $\varepsilon$ of $L$. The input condition here is Within — unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every $\varepsilon$ such a $\delta$ exists.

*Remark* (Dependencies).

- $\varepsilon$-Closeness of a Function to a Value

**Definition (Limit of a Function)**

**Definition 2.1.3** (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. A real number $L$ is the **limit of $f$ at** $c$, written $\lim_{x\to c} f(x) = L$, if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Go to uniqueness proof.*

*Remark* (Standard quantified statement)*.*

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta\ \Rightarrow\ |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading)*.* Define

$$\begin{aligned} \mathrm{OutputTol}(\varepsilon) &\iff \varepsilon > 0, \\ \mathrm{InputTol}(\delta) &\iff \delta > 0, \\ \mathrm{PuncturedNbhd}(x, c, \delta) &\iff 0 < |x - c| < \delta, \\ \mathrm{OutputNbhd}(y, L, \varepsilon) &\iff |y - L| < \varepsilon. \end{aligned}$$

Then

$$\mathrm{FunctionLimit}(f, A, c, L) \iff \forall \varepsilon \Big(\mathrm{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left(\mathrm{InputTol}(\delta) \wedge \forall x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta) \Rightarrow \mathrm{O}\right.\right.$$

*Remark* (Negated quantified statement)*.*

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading)*.*

$$\begin{aligned} \neg\, \mathrm{FunctionLimit}(f, A, c, L) &\iff \exists \varepsilon_0 \Big(\mathrm{OutputTol}(\varepsilon_0) \wedge \forall \delta \left(\mathrm{InputTol}(\delta) \Rightarrow \exists x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta)\right.\right. \\ &\iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big). \end{aligned}$$

*Remark* (Failure modes)*.*

- The proposed value $L$ is the wrong target: the function approaches some value $M \neq L$ near $c$.
- The function does not settle near any single value as $x$ approaches $c$ through $A$.
- The outputs repeatedly stay at least a fixed distance from $L$ no matter how tightly the inputs are restricted around $c$.

*Remark* (Failure mode decomposition)*.*

$$\neg\, \mathrm{FunctionLimit}(f, A, c, L) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \Big(\underbrace{0 < |x - c| < \delta}_{\text{arbitrarily close to } c} \wedge \underbrace{|f(x) - L| \geq \varepsilon_0}_{\text{fixed output gap from } L}\Big).$$

*Remark* (Interpretation). The quantifier structure is asymmetric by design. The output tolerance $\varepsilon$ is a *demand*: the caller specifies how precise the output must be. The input tolerance $\delta$ is a *response*: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because $f$ may not be defined at $c$, and because continuity — not limits — is the concept that includes $c$ itself. This is the deepest structural difference between the two definitions. Four critical features: (i) $f$ need not be defined at $c$; (ii) $c$ must be a cluster point of $A$ else the condition is vacuously satisfied by every $L$; (iii) $\delta$ may depend on $\varepsilon$; (iv) the condition is about all $x$ near $c$ simultaneously, not just one.

*Remark* (Dependencies).

- Cluster Point

**Definition (Sequential Definition of Limit)**

**Definition 2.1.4** (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to $L$.

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow f(x_n) \to L\big).$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}_{\text{seq}}(f, c, L) \ :\equiv \ \forall (x_n) \subseteq A \backslash \{c\} : \ \text{ConvergentSequence}(x_n, c) \ \Rightarrow \ \text{ConvergentSequence}(f(x_n),$$

*Remark* (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \ \wedge \ f(x_n) \not\to L\big).$$

*Remark* (Interpretation). The sequence $(x_n)$ must lie in $A \setminus \{c\}$: the term $x_n = c$ is excluded, mirroring the $0 <$ in the $\varepsilon$-$\delta$ form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The $\varepsilon$-$\delta$ form is better for constructing explicit $\delta$ functions.

*Remark* (Dependencies).

- Limit of a Function
- Cluster Point

**Definition (Neighbourhood Form of Limit)**

**Definition 2.1.5** (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. The limit $\lim_{x\to c} f(x) = L$ in the **neighbourhood sense** holds if for every $\varepsilon$-neighbourhood $V_\varepsilon(L)$ there exists a deleted $\delta$-neighbourhood $V^*_\delta(c)$ such that

$$x \in V^*_\delta(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in V_\delta^*(c) \cap A \; f(x) \in V_\varepsilon(L).$$

*Remark* (Interpretation). The neighbourhood form packages the same local limit condition in terms of sets around the input point and output value. The domain is still restricted to $A$, and the neighbourhood of $c$ is deleted so the value of $f$ at $c$ itself is irrelevant.

*Remark* (Dependencies).

- Limit of a Function
- $\varepsilon$-Neighbourhood
- Deleted $\varepsilon$-Neighbourhood

**Proposition**

**Proposition 2.1.6** (Neighbourhood Form of the Limit). *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. Then $\lim_{x \to c} f(x) = L$ if and only if for every $\varepsilon$-neighbourhood $V_\varepsilon(L)$ there exists a $\delta$-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\Big[\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \\ \iff \Big[\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in V_\delta^*(c) \cap A \; f(x) \in V_\varepsilon(L)\Big].$$

*Remark* (Interpretation). The equivalence is purely notational: $V_\delta^*(c) \cap A$ is exactly the set of $x \in A$ satisfying $0 < |x - c| < \delta$, and $f(x) \in V_\varepsilon(L)$ is exactly $\mathrm{OutputTolerance}(f(x), L, \varepsilon)$. The neighbourhood form makes the geometric content explicit: the image of every punctured $\delta$-slice of $A$ must fit inside the $\varepsilon$-ball around $L$. This is the form of continuity used in metric space topology.

*Remark* (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- $\varepsilon$-Neighbourhood
- Deleted $\varepsilon$-Neighbourhood

**Theorem**

**Theorem 2.1.7** (Uniqueness of Limits of Functions). *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. Then $f$ has at most one limit at $c$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \;\wedge\; \lim_{x\to c} f(x) = L' \;\Rightarrow\; L = L'.$$

*Remark* (Failure modes). Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

*Remark* (Failure mode decomposition). Non-uniqueness would require two values $L \neq L'$ each satisfying the limit definition. The $\varepsilon = |L - L'|/2$ argument shows any $x$ within $\min(\delta_1, \delta_2)$ of $c$ satisfies both $|f(x)-L| < \varepsilon$ and $|f(x)-L'| < \varepsilon$, but then $|L-L'| \leq |L-f(x)|+|f(x)-L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster point condition is essential: it guarantees such an $x$ exists.

*Remark* (Interpretation). The limit, when it exists, is uniquely determined by the local behavior of $f$ near $c$. Uniqueness is what licenses writing $\lim_{x\to c} f(x) = L$ as an equation rather than an assertion.

*Remark* (Dependencies).

- Limit of a Function

**Proposition**

**Proposition 2.1.8** (Sequential Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

*Remark* (Negated quantified statement).

$$\neg \lim_{x\to c} f(x) = L \quad\iff\quad \exists (x_n) \subseteq A \setminus \{c\} :\ x_n \to c \;\wedge\; f(x_n) \not\to L.$$

*Remark* (Interpretation). The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xRightarrow{\text{Sequential Criterion}} \text{algebra of function limits.}$$

Every sequence result — sum, product, quotient, squeeze — transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove $\lim_{x\to c} f(x) = L$, exhibit one sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$ and $f(x_n) \not\to L$.

*Remark* (Dependencies).

- Limit of a Function
- Sequential Definition of Limit

**Corollary**

**Corollary 2.1.9** (Three Equivalent Forms of the Function Limit)**.** *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. The $\varepsilon$-$\delta$ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*
*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ \iff &\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big]\\ \iff &\Big[\forall(x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].\end{aligned}$$

*Remark* (Interpretation)*.* All three encodings capture the same mathematical condition. The $\varepsilon$-$\delta$ form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

*Remark* (Dependencies)*.*

- Limit of a Function
- Neighbourhood Form of Limit
- Sequential Definition of Limit
- Neighbourhood Form of the Limit
- Sequential Criterion for Function Limits

**Theorem**

**Theorem 2.1.10** (Limit Implies Local Boundedness)**.** *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*
*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\lim_{x\to c} f(x) = L \ \Rightarrow\ \exists\delta > 0\ \exists M > 0\ \forall x \in A\ \big(0 < |x - c| < \delta \Rightarrow |f(x)| \le M\big).$$

*Remark* (Proof sketch)*.* Apply the definition with $\varepsilon = 1$: get $\delta > 0$ such that $|f(x) - L| < 1$ holds for all $x$ satisfying $0 < |x - c| < \delta$. The triangle inequality then gives $|f(x)| \le |f(x) - L| + |L| < 1 + |L|$. Set $M = 1 + |L|$.

*Remark* (Interpretation). A finite limit does more than control the output near $L$: it prevents the function from growing without bound near $c$. Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

*Remark* (Dependencies).

- Limit of a Function

**Theorem**

**Theorem 2.1.11** (Bounded Limits)**.** *Let $f : A \to \mathbb{R}$, let $c$ be a cluster point of $A$, and let $a, b \in \mathbb{R}$. If $a \le f(x) \le b$ for all $x \in A$ with $x \ne c$, and $\lim_{x\to c} f(x) = L$, then $a \le L \le b$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\big(\forall x \in A \setminus \{c\} : \ a \le f(x) \le b\big) \ \wedge \ \lim_{x\to c} f(x) = L \ \Rightarrow \ a \le L \le b.$$

*Remark* (Interpretation). If every nearby value of $f$ is trapped in $[a, b]$, the limit cannot escape outside $[a, b]$. This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

*Remark* (Dependencies).

- Limit of a Function

**Theorem**

**Theorem 2.1.12** (Squeeze Theorem for Function Limits)**.** *Let $f, g, h : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. Suppose $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A, \ 0 < |x - c| < \delta_0 \ \Rightarrow \ f(x) \le h(x) \le g(x).$$

*Then $\lim_{x\to c} h(x) = L$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L \ \wedge \ f \le h \le g \text{ near } c \ \Rightarrow \ \lim_{x\to c} h(x) = L.$$

*Remark* (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new $\varepsilon$-$\delta$ argument is needed. The condition $f \le h \le g$ is only required near $c$, not globally.

*Remark* (Dependencies).

- Limit of a Function

- Bounded Limits

**Theorem**

**Theorem 2.1.13** (Limits Are Local). *Let $E \subseteq X \subseteq \mathbb{R}$, let $x_0$ be a cluster point of $E$, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x \to x_0;\, x \in E} f(x) = L \iff \lim_{x \to x_0;\, x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \to x_0;\, x \in E} f(x) = L \iff \lim_{x \to x_0;\, x \in E \cap (x_0 - \delta,\, x_0 + \delta)} f(x) = L.$$

*Remark* (Interpretation). The existence and value of FunctionLimit$(f, x_0, L)$ depend only on the behavior of $f$ within any fixed neighbourhood of $x_0$. Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around $c$ without any loss of generality.

*Remark* (Dependencies).

- Limit of a Function

**Proposition**

**Proposition 2.1.14** (Limit of Absolute Value). *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x \to c} f(x) = L$, then $\lim_{x \to c} |f(x)| = |L|$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \to c} f(x) = L \Rightarrow \lim_{x \to c} |f(x)| = |L|.$$

*Remark* (Negated quantified statement).

$$\lim_{x \to c} |f(x)| = |L| \not\Rightarrow \lim_{x \to c} f(x) = L.$$

*Remark* (Failure modes). The absolute values may converge while the signs of $f(x)$ oscillate or approach the opposite sign from the proposed limiting value.

*Remark* (Proof strategy). The reverse triangle inequality $\big||f(x)| - |L|\big| \leq |f(x) - L|$ feeds the $\varepsilon$-$\delta$ definition directly: the same $\delta$ that delivers $|f(x) - L| < \varepsilon$ also delivers $\big||f(x)| - |L|\big| < \varepsilon$.

*Remark* (Interpretation). Taking absolute values can collapse sign information, so convergence of $f$ forces convergence of $|f|$, but convergence of $|f|$ alone does not recover the original limiting sign.

*Remark* (Dependencies).

- Limit of a Function

**Theorem**

**Theorem 2.1.15** (Composition of Limits)**.** *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let $c_1$ be a cluster point of $A$, and let $c_2 \in \mathbb{R}$ be a cluster point of $B$. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x\to c_1} f(x) = c_2 \ \wedge\ \lim_{y\to c_2} g(y) = L \ \wedge\ \big(c_2 \in B \Rightarrow g(c_2) = L\big) \ \Rightarrow\ \lim_{x\to c_1} g(f(x)) = L.$$

*Remark* (Failure modes). The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

*Remark* (Failure mode decomposition). The hypothesis $g(c_2) = L$ is essential. Without it the composition can fail: take $f \equiv c_2$ (constant) and $g(c_2) \neq L$. Then $g(f(x)) = g(c_2) \neq L$ for all $x$, so $\lim_{x\to c_1} g(f(x)) = L$ fails even though both component limits exist. The failure is that $f(x) = c_2$ is excluded from the punctured condition on $g$, but $f$ hits $c_2$ constantly.

*Remark* (Interpretation). Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis $g(c_2) = L$ bridges the gap: when $f(x) = c_2$, the output condition $g(f(x)) = g(c_2) = L$ is satisfied directly. This hypothesis is equivalent to requiring $g$ to be continuous at $c_2$ (within $B$).

*Remark* (Dependencies).

- Limit of a Function

**Theorem**

**Theorem 2.1.16** (Monotone Limit Theorem)**.** *Let $a < b$ and let $f : (a,b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing $f$ and every $x_0 \in (a,b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$f \text{ is monotone on } (a,b) \wedge \exists B > 0\ \forall x \in (a,b)\ (|f(x)| \leq B)\ \Rightarrow\ \forall x_0 \in (a,b) :\ \lim_{x\to x_0^-} f(x) = f(x_0^-) \wedge \lim_{x\to x_0^+} f($$

*Remark* (Interpretation). Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing $f$, the values $f(x)$ for $x < x_0$ form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities — never oscillatory ones.

*Remark* (Dependencies).

- Monotone Function
- Bounded Function
- Left-Hand Limit
- Right-Hand Limit

**Proposition**

**Proposition 2.1.17** (Cauchy Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \text{OutputTolerance}(\varepsilon),\ \exists\ \text{InputTolerance}(\delta),\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta\ \Rightarrow\ |f(x)-f(y)$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \big(\lim_{x\to c} f(x) = L\big) \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \Rightarrow |f(x)-f(y)|$$

*Remark* (Interpretation). The Cauchy criterion tests limit existence without knowing $L$. Two outputs from any sufficiently small punctured ball must lie within $\varepsilon$ of each other — the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of $\mathbb{R}$ in exactly the same way.

*Remark* (Dependencies).

- Limit of a Function

## Algebra of Limits

**Theorem**

**Theorem 2.1.18** (Sum Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f+g)(x) = L + M.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x \in A\ 0 < |x-c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon\Big] \\
&\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x \in A\ 0 < |x-c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon\Big] \\
&\quad \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f+g)(x) - (L+M)| < \varepsilon.
\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \implies \mathrm{FunctionLimit}(f + g, c, L + M).$$

*Remark* (Interpretation). Limits respect addition: sufficiently close values of $f$ and $g$ add to values of $f + g$ close to the sum of the two limits.

*Remark* (Dependencies).

- Limit of a Function
- Pointwise Sum of Functions

**Theorem**

**Theorem 2.1.19** (Difference Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f - g)(x) = L - M.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |(f - g)(x) - (L - M)| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \implies \mathrm{FunctionLimit}(f - g, c, L - M).$$

*Remark* (Interpretation). The difference rule is the sum rule applied to $f + (-g)$.

*Remark* (Dependencies).

- Limit of a Function
- Pointwise Difference of Functions

**Theorem**

**Theorem 2.1.20** (Scalar Multiple Rule for Function Limits)**.** *Let $f : A \to \mathbb{R}$, let $c$ be a cluster point of $A$, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |\alpha f(x) - \alpha L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \implies \text{FunctionLimit}(\alpha f, c, \alpha L).$$

*Remark* (Interpretation). Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

*Remark* (Dependencies).

- Limit of a Function
- Pointwise Scalar Multiple of a Function

**Theorem**

**Theorem 2.1.21** (Product Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c} (fg)(x) = LM.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x)g(x) - LM| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(fg, c, LM).$$

*Remark* (Interpretation). The product rule combines closeness of $f$ to $L$ with local boundedness of $g$ near $c$.

*Remark* (Dependencies).

- Limit of a Function
- Pointwise Product of Functions

**Theorem**

**Theorem 2.1.22** (Quotient Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let $c$ be a cluster point of $A$. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \wedge M \neq 0\\ &\quad \Rightarrow \exists \delta_0 > 0\ \forall x \in A\ 0 < |x-c| < \delta_0 \Rightarrow g(x) \neq 0,\\ &\quad \text{and} \quad \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| < \varepsilon.\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \wedge M \neq 0 \implies \mathrm{FunctionLimit}(f/g, c, L/M).$$

*Remark* (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

*Remark* (Dependencies).

- Limit of a Function
- Pointwise Quotient of Functions

## One-Sided Limits

**Definition (Right-Hand Limit)**

**Definition 2.1.23** (Right-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose $c$ is a cluster point of $A \cap (c, \infty)$. A real number $L$ is the **right-hand limit** of $f$ at $c$ if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \ :=\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading).

$$\neg\ \mathrm{RightLimit}(f, c, L).$$

*Remark* (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to $c$ from the right.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Interpretation). Only inputs to the right of $c$ are tested. The value of $f$ at $c$, if it is defined, is irrelevant to the right-hand limit.

*Remark* (Dependencies).

- Limit of a Function
- Cluster Point

**Definition (Left-Hand Limit)**

**Definition 2.1.24** (Left-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose $c$ is a cluster point of $A \cap (-\infty, c)$. A real number $L$ is the **left-hand limit** of $f$ at $c$ if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L)\ :=\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading).

$$\neg\ \mathrm{LeftLimit}(f, c, L).$$

*Remark* (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to $c$ from the left.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Interpretation). Only inputs to the left of $c$ are tested. This is the mirror image of the right-hand limit definition.

*Remark* (Dependencies).

- Limit of a Function
- Cluster Point

**Theorem**

**Theorem 2.1.25** (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose $c$ is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

*Remark* (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L.$$

*Remark* (Interpretation). Right-hand limits can be tested by sequences approaching $c$ from above. Every such sequence must force the corresponding output sequence toward $L$.

*Remark* (Dependencies).

- Right-Hand Limit
- Sequential Definition of Limit

**Corollary**

**Corollary 2.1.26** (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose $c$ is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x \to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*
*Go to proof.*

*Remark* (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

*Remark* (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L.$$

*Remark* (Interpretation). Left-hand limits can be tested by sequences approaching $c$ from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

*Remark* (Dependencies).

- Left-Hand Limit

- Sequential Definition of Limit

**Theorem**

**Theorem 2.1.27** (Two-Sided Limit via One-Sided Limits)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If $c$ is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x\to c} f(x) = L$ if and only if both $\lim_{x\to c^+} f(x) = L$ and $\lim_{x\to c^-} f(x) = L$.*

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\quad \iff \Big[\forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in A\ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\qquad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in A\ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon\Big].\end{aligned}$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{FunctionLimit}(f, c, L) \iff \mathrm{RightLimit}(f, c, L) \wedge \mathrm{LeftLimit}(f, c, L).$$

*Remark* (Negated quantified statement)*.*

$$\begin{aligned}&\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\\ &\quad \iff \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big]\\ &\qquad \vee \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big].\end{aligned}$$

*Remark* (Negation predicate reading)*.*

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

*Remark* (Failure modes)*.* A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

*Remark* (Failure mode decomposition)*.*

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

*Remark* (Interpretation)*.* A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

*Remark* (Dependencies)*.*

- Limit of a Function
- Right-Hand Limit
- Left-Hand Limit

## 2.2 Point Continuity

**Oscillation — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Oscillation on a set | Diameter of the image over a set. |
| Oscillation at a point | Limiting local oscillation near the point. |
| Zero oscillation criterion | Continuity is exactly vanishing point oscillation. |
| Discontinuity set | Discontinuities are detected by positive oscillation. |

**Discontinuity — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Point of discontinuity | Negation of continuity at a point. |
| Sequential discontinuity | A bad sequence witnesses failure. |
| Neighbourhood discontinuity | Neighbourhood form of the same failure. |
| Equivalence lemma | The discontinuity formulations agree. |
| Types of discontinuity | Removable, jump, and essential behavior. |

**Continuity — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Relative neighbourhood | Neighbourhoods are read inside the domain. |
| Point continuity | The unpunctured $\varepsilon$-$\delta$ condition. |
| Neighbourhood continuity | Images of nearby inputs stay near the value. |
| Sequential continuity | Convergent input sequences preserve limits. |
| Oscillation criterion | Continuity is equivalent to zero local oscillation. |

### Continuity

**Definition (Relative Neighborhood)**

**Definition 2.2.1** (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of $a$ in $E$ if and only if there exists $\delta > 0$ such that $U = E \cap (a-\delta, a+\delta)$. Equivalently, every neighborhood of $a$ in $E$ is a set of the form $U_E(a) := E \cap (a-\delta, a+\delta)$ for some $\delta > 0$.

*Remark* (Standard quantified statement).

$$\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.$$
$$U \text{ is a neighborhood of } a \text{ in } E \iff \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

*Remark* (Definition predicate reading).

$$\mathrm{RelativeNeighborhood}(U, E, a) := \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ fails to be a neighborhood of } a \text{ in } E \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).$$

*Remark* (Failure modes). Failure occurs precisely when no radius $\delta$ makes $U$ coincide with the relative interval slice: for every $\delta > 0$ either $U$ contains a point outside $E \cap (a-\delta, a+\delta)$ or the slice contains a point missing from $U$.

*Remark* (Failure mode decomposition).

$$\forall \delta > 0\ \underbrace{\big(\exists x \in U :\ x \notin E \cap (a-\delta, a+\delta)\big)}_{\text{extraneous point}} \vee \underbrace{\big(\exists y \in E \cap (a-\delta, a+\delta) :\ y \notin U\big)}_{\text{missing point}}.$$

*Remark* (Interpretation). Relative neighborhoods capture the local behavior of $E$ around $a$ by intersecting $E$ with an ordinary open interval centered at $a$. The construction restricts the ambient notion of openness to the geometry of $E$, so it is suited to relative topologies. Failure simply means that $U$ never matches the precise slice of $E$ at any scale, typically because $U$ omits some nearby points of $E$ or includes points that fall outside $E$ near $a$.

*Remark* (Dependencies).

- Subset
- Epsilon Neighbourhood
- Order on $\mathbb{R}$

**Definition (Continuity at a Point)**

**Definition 2.2.2** (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function $f$ is **continuous at** $c$ if and only if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A : \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, A, c) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is not continuous at } c \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).$$

*Remark* (Failure modes). Continuity at $c$ fails precisely when there exists a persistent output gap $\varepsilon_0 > 0$ such that no matter how small the chosen neighborhood of $c$ is, a point of $A$ within that neighborhood keeps the function values at least $\varepsilon_0$ away from $f(c)$. The single failure branch records this inescapable oscillation near $c$.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{persistent gap}}\quad \underbrace{\forall \delta > 0}_{\text{every neighborhood}}\quad \underbrace{\exists x \in A}_{\text{nearby witness}}\ :\ |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0.$$

*Remark* (Interpretation). Continuity at $c$ means that the output $f(x)$ can be kept arbitrarily close to $f(c)$ by taking $x$ sufficiently close to $c$ while staying in the domain $A$. The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on $c$, so nearby inputs cannot force the outputs to settle near $f(c)$. This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at $c$.

*Remark* (Dependencies).

- Definition: Limit of a Function

**Definition (Continuity at a Point — Neighbourhood Form)**

**Definition 2.2.3** (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function $f$ is continuous at $c$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0 :\\ &\qquad f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, c) := \forall \varepsilon > 0\ \exists \delta > 0 :\ f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

*Remark* (Negated quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A. \\
&\neg\big(f \text{ is continuous at } c\big) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 : \\
&\qquad f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c) - \varepsilon_0, f(c) + \varepsilon_0).
\end{aligned}
$$

*Remark* (Negation predicate reading).

$$\text{DiscontinuousAt}(f, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c) - \varepsilon_0, f(c) + \varepsilon_0).$$

*Remark* (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about $c$. This manifests by finding points of $A$ arbitrarily close to $c$ whose images either exceed the upper endpoint $f(c) + \varepsilon_0$ or fall below the lower endpoint $f(c) - \varepsilon_0$.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \geq f(c) + \varepsilon_0\Big)}_{\text{overshoot}} \vee \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \leq f(c) - \varepsilon_0}_{\text{undershoot}}$$

*Remark* (Interpretation). Continuity at $c$ demands that every output tolerance around $f(c)$ has a matching neighborhood around $c$ whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from $A$ approaches $c$ while their images either spike above or dip below $f(c)$ by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above $c$ must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs $c$.

**Definition (Continuity at a Point — Sequential Form)**

**Definition 2.2.4** (Continuity at a Point — Sequential Form)**.** Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function $f$ is continuous at $c$ if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in $A$, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

*Remark* (Standard quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}. \\
&f \text{ is continuous at } c \iff \forall (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = f(c)\Big).
\end{aligned}
$$

*Remark* (Definition predicate reading).

$$\text{ContinuousAt}(f, c) :\!= \forall x_n\ \Big(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n), f$$

*Remark* (Negated quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}. \\
&f \text{ is not continuous at } c \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \wedge \lim_{n\to\infty} f(x_n) \neq f(c)\Big).
\end{aligned}
$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \exists x_n \Big(\mathrm{Sequence}(x_n, A) \wedge \mathrm{ConvergentSequence}(x_n, c) \wedge \neg\,\mathrm{ConvergentSequence}(f(x_n$$

*Remark* (Failure modes). Continuity at $c$ fails sequentially when there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in $A$ with $x_n \to c$ but $f(x_n)$ does not converge to $f(c)$. This manifests in two distinct ways: either $f(x_n)$ converges to some limit different from $f(c)$, or $f(x_n)$ fails to converge entirely (oscillating without settling or diverging to infinity).

*Remark* (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A : \underbrace{\lim_{n\to\infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n\to\infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

*Remark* (Interpretation). The sequential characterization asserts that information about $f$ near $c$ is completely encoded by its action on limits of sequences taken within the domain. Any approach to $c$ through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that $f$ cannot introduce jumps or oscillations that persist along every infinitesimal path toward $c$.

*Remark* (Dependencies).

Definition 2.2.2.

## Characterization of Discontinuity

**Definition (Point of Discontinuity)**

**Definition 2.2.5** (Point of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $a \in A$. The function $f$ is discontinuous at $a$ if and only if $f$ is not continuous at $a$.

*Remark* (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A.$$
$$f \text{ is discontinuous at } a \iff \neg\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon\big)\Big).$$

*Remark* (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f,a) := \neg\,\mathrm{ContinuousAt}(f,a).$$

*Remark* (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A.$$
$$f \text{ is discontinuous at } a \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x-a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon\big).$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f,a) \iff \mathrm{ContinuousAt}(f,a) \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x)-f(a)| <$$

*Remark* (Failure modes). Discontinuity arises either because the nearby values of $f$ do not approach any single real number as $x \to a$, or because a well-defined limit exists but disagrees with the actual value $f(a)$.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg\exists L \in \mathbb{R}\ \text{FunctionLimit}(f, A, a, L)}_{\text{no limit exists}} \vee \underbrace{\big(\exists L \in \mathbb{R}\ \text{FunctionLimit}(f, A, a, L) \wedge L \neq f(a)\big)}_{\text{limit-value mismatch}}.$$

*Remark* (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

*Remark* (Dependencies).

Definition 2.2.2.

**Definition (Sequential Characterization of Discontinuity)**

**Definition 2.2.6** (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function $f$ is discontinuous at $c$ if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

*Remark* (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A\ \big[x_n \to c\ \wedge\ f(x_n) \not\to f(c)\big].$$

*Remark* (Definition predicate reading).

$$\text{DiscontinuousAt}(f, c) :\equiv \exists (x_n) \subseteq A :\ x_n \to c\ \wedge\ \neg\big(f(x_n) \to f(c)\big).$$

*Remark* (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).$$

*Remark* (Negation predicate reading).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \text{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).\blacksquare$$

*Remark* (Failure modes). The characterization fails — meaning $f$ is NOT sequentially discontinuous — when every sequence $(x_n)$ in $A$ converging to $c$ has $f(x_n) \to f(c)$. No sequential witness exists.

*Remark* (Failure mode decomposition).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c)}_{\text{every approach preserves the limit}}.$$

*Remark* (Interpretation). A discontinuity at $c$ can be certified by a single sequence approaching $c$ whose images refuse to settle at $f(c)$.

*Remark* (Dependencies).

Definition 2.2.4; Definition 2.2.5.

**Definition (Neighbourhood Characterization of Discontinuity)**

**Definition 2.2.7** (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function $f$ is discontinuous at $c$ if and only if there exists a neighborhood $V$ of $f(c)$ such that for every neighborhood $U$ of $c$ relative to $A$ there exists $x \in U \cap A$ with $f(x) \notin V$.

*Remark* (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A\ \big(f(x) \notin V\big).$$

*Remark* (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, c) := \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V.$$

*Remark* (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V\big).$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \mathrm{ContinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V$$

*Remark* (Failure modes)*.* The neighbourhood discontinuity characterization fails — meaning $f$ is actually continuous — when every output neighbourhood $V$ of $f(c)$ can be matched by some input neighbourhood $U$ such that $f$ maps all of $U \cap A$ into $V$. No perpetually-missed output neighbourhood exists.

*Remark* (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} :\ \underbrace{\forall x \in U \cap A :\ f(x) \in V}_{\text{all outputs inside tolerance}}.$$

*Remark* (Interpretation)*.* Discontinuity means a fixed output neighborhood $V$ is perpetually missed no matter how tightly the input is restricted around $c$.

*Remark* (Dependencies).

Definition 2.2.1.

**Lemma (Equivalent Formulations of Discontinuity at a Point)**

**Lemma 2.2.8** (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *$c$ is a point of discontinuity in the $\varepsilon$-$\delta$ sense.*
2. *$c$ is a sequential point of discontinuity.*
3. *$c$ is a neighbourhood point of discontinuity.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$
\begin{aligned}
&(1)\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A:\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
\iff\ &(2)\ \exists (x_n) \subseteq A:\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\
\iff\ &(3)\ \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A:\ f(x) \notin V.
\end{aligned}
$$

*Remark* (Predicate reading).

$$
\begin{aligned}
\text{DiscontinuousAt}(f, c) &\iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A:\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
&\iff \exists (x_n) \subseteq A:\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\
&\iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A:\ f(x) \notin V.
\end{aligned}
$$

*Remark* (Failure modes). The equivalence cannot fail for functions at limit points of $A$: the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where $c$ is not a limit point of $A$, the punctured-neighbourhood conditions are vacuously satisfied by every $L$, and the equivalence becomes trivial rather than informative.

*Remark* (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

*Remark* (Dependencies).

- Point of Discontinuity
- Sequential Characterization of Discontinuity
- Neighbourhood Characterization of Discontinuity

**Definition (Types of Discontinuity)**

**Definition 2.2.9** (Types of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with $f$ discontinuous at $x_0$.

(i) *Removable* iff $\exists L \in \mathbb{R}:\ \lim_{x \to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R}:\ \lim_{x \to x_0^-} f(x) = L_-$, $\lim_{x \to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

*Remark* (Standard quantified statement).

$$
\begin{aligned}
&\text{Removable} \iff \exists L \in \mathbb{R}\ \Big(\lim_{x \to x_0} f(x) = L \wedge L \neq f(x_0)\Big). \\
&\text{Jump} \iff \exists L_-, L_+ \in \mathbb{R}\ \Big(\lim_{x \to x_0^-} f(x) = L_- \wedge \lim_{x \to x_0^+} f(x) = L_+ \wedge L_- \neq L_+\Big). \\
&\text{Essential} \iff \neg \exists L \in \mathbb{R}\ \Big(\lim_{x \to x_0} f(x) = L \wedge L \neq f(x_0)\Big).
\end{aligned}
$$

*Remark* (Definition predicate reading).

$$
\begin{aligned}
\text{RemovableDiscontinuity}(f, x_0) &:= \exists L \in \mathbb{R}\ \big(\text{FunctionLimit}(f, A, x_0, L) \wedge L \neq f(x_0)\big). \\
\text{JumpDiscontinuity}(f, x_0) &:= \exists L_-, L_+ \in \mathbb{R}\ \big(\text{LeftLimit}(f, A, x_0, L_-) \wedge \text{RightLimit}(f, A, x_0, L_+) \wedge L_- \neq \\
\text{EssentialDiscontinuity}(f, x_0) &:= \neg\, \text{RemovableDiscontinuity}(f, x_0).
\end{aligned}
$$

*Remark* (Negated quantified statement).

$$\begin{aligned} \neg\text{Removable} &\iff \forall L \in \mathbb{R}\, \big(\lim_{x\to x_0} f(x) \neq L \vee L = f(x_0)\big), \\ \neg\text{Jump} &\iff \forall L_-, L_+ \in \mathbb{R}\, \big(\lim_{x\to x_0^-} f(x) \neq L_- \vee \lim_{x\to x_0^+} f(x) \neq L_+ \vee L_- = L_+\big), \\ \neg\text{Essential} &\iff \text{Removable}. \end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned} \neg\,\text{RemovableDiscontinuity}(f, x_0) &\iff \forall L \in \mathbb{R}\, \big(\neg\,\text{FunctionLimit}(f, A, x_0, L) \vee L = f(x_0)\big). \\ \neg\,\text{JumpDiscontinuity}(f, x_0) &\iff \forall L_-, L_+ \in \mathbb{R}\, \big(\neg\,\text{LeftLimit}(f, A, x_0, L_-) \vee \neg\,\text{RightLimit}(f, A, x_0, L_+) \\ \neg\,\text{EssentialDiscontinuity}(f, x_0) &\iff \text{RemovableDiscontinuity}(f, x_0). \end{aligned}$$

*Remark* (Failure modes).

- Removable discontinuity fails when no bilateral limit exists at $x_0$, or when the bilateral limit exists but equals $f(x_0)$ (making the function actually continuous).
- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from $f(x_0)$).

*Remark* (Failure mode decomposition).

$$\begin{aligned} \neg\,\text{RemovableDiscontinuity}(f, x_0) &\iff \underbrace{\neg\exists L \in \mathbb{R}\ \lim_{x\to x_0} f(x) = L}_{\text{no bilateral limit exists}} \vee \underbrace{\lim_{x\to x_0} f(x) = f(x_0)}_{\text{limit equals function value}}. \\ \neg\,\text{JumpDiscontinuity}(f, x_0) &\iff \underbrace{\neg\exists L_- \in \mathbb{R}\ \lim_{x\to x_0^-} f(x) = L_-}_{\text{no left limit}} \vee \underbrace{\neg\exists L_+ \in \mathbb{R}\ \lim_{x\to x_0^+} f(x) = L_+}_{\text{no right limit}} \\ &\quad \vee \underbrace{\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x)}_{\text{one-sided limits agree}}. \end{aligned}$$

*Remark* (Interpretation). Removable discontinuities can be healed by redefining $f(x_0)$. Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

*Remark* (Dependencies).

- Point of Discontinuity
- Limit of a Function
- Left-Hand Limit
- Right-Hand Limit

## Oscillation

**Definition (Oscillation on a Set)**

**Definition 2.2.10** (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of $f$ on $E$ is the extended real number

$$\omega(f; E) := \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \operatorname{diam}(f(E)).$$

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].\\ &\omega(f; E) = \omega \iff \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

*Remark* (Definition predicate reading).

$$\operatorname{OscillationOn}(f, E, \omega) := \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].\\ &\omega(f; E) \neq \omega \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \operatorname{OscillationOn}(f, E, \omega) \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

*Remark* (Failure modes)*.* The proposed value $\omega$ fails either because it is not an upper bound for pairwise differences of function values on $E$, or because it is an upper bound but not the least such upper bound.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists x_1, x_2 \in E :\ |f(x_1) - f(x_2)| > \omega}_{\text{not an upper bound}} \vee \underbrace{\exists \eta < \omega\ \forall x_1, x_2 \in E :\ |f(x_1) - f(x_2)| \leq \eta}_{\text{not least}}.$$

*Remark* (Interpretation)*.* Oscillation on a set records the spread of the image — the diameter of $f(E)$. Small oscillation means all values of $f$ on $E$ lie close together. Zero oscillation means $f$ is constant on $E$.

*Remark* (Dependencies).

- Function
- Subset
- Supremum
- Absolute Value on $\mathbb{R}$

**Definition (Oscillation at a Point)**

**Definition 2.2.11** (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of $f$ at $x_0$ is the extended real number

$$\omega(f; x_0) = \lim_{\delta \to 0^+} \omega\bigl(f;\, U_\delta(x_0) \cap E\bigr),$$

with $\omega(f; A)$ denoting the oscillation of $f$ on the set $A$. Because the function $\delta \mapsto \omega\bigl(f;\, U_\delta(x_0) \cap E\bigr)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\bigl[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\bigr]\\
&\iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall r\ (0 < r < \delta \Rightarrow |\omega(f; U_r(x_0) \cap E) - \omega| < \varepsilon).
\end{aligned}$$

*Remark* (Definition predicate reading).

$$\operatorname{OscillationAt}(f, x_0, \omega) :\equiv \lim_{\delta \to 0^+} \omega\bigl(f;\, U_\delta(x_0) \cap E\bigr) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\neg\bigl[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\bigr]\\
&\iff \Bigl[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Bigr]\\
&\qquad \vee \Bigl(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)\Bigr).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \operatorname{OscillationAt}(f, x_0, \omega) \iff \Bigl[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Bigr] \vee \Bigl(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E$$

*Remark* (Failure modes). The definition fails in two distinct ways: either the point $x_0$ has a punctured neighborhood missing $E$, so there are no arbitrarily small samples to measure, or the local oscillation values near $x_0$ refuse to stabilize, preventing the limit from existing.

*Remark* (Failure mode decomposition).

$$\neg \operatorname{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing}_{\text{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E)}_{\text{oscillation limit does not exist}}$$

*Remark* (Interpretation). Oscillation at $x_0$ measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around $x_0$. The shrinking neighborhoods $U_r(x_0) \cap E$ supply finer and finer samples, and the oscillation on each such set records the spread of $f$ there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when $f$ becomes arbitrarily close to a single value, i.e., when $f$ is continuous at $x_0$. Failure occurs either because $x_0$ has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

*Remark* (Dependencies).

Definition 2.2.10

**Theorem (Continuity Characterised by Oscillation)**

**Theorem 2.2.12** (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then $f$ is continuous at $c$ if and only if $\omega(f;c) = 0$. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A.\\ &\big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\big)\\ &\Longleftrightarrow\\ &\left(\inf_{\delta > 0} \sup\big\{|f(x) - f(y)| : x, y \in A,\ |x - c| < \delta,\ |y - c| < \delta\big\} = 0\right).\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \Longleftrightarrow \mathrm{OscillationAt}(f, c, 0).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A.\\ &\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big) \wedge \Big(\inf_{\delta > 0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x - c| < \delta,\\ &\qquad \vee\\ &\Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)\Big) \wedge \Big(\inf_{\delta > 0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x - c| < \delta\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\big(\mathrm{ContinuousAt}(f, c) \Longleftrightarrow \mathrm{OscillationAt}(f, c, 0)\big).$$

*Remark* (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either $f$ remains continuous at $c$ while the oscillation retains a positive lower bound, or $\omega(f;c)$ vanishes even though $f$ does not satisfy the $\varepsilon$-$\delta$ continuity condition at $c$.

*Remark* (Failure mode decomposition).

$$\underbrace{\mathrm{ContinuousAt}(f, c) \wedge \neg\, \mathrm{OscillationAt}(f, c, 0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg\, \mathrm{ContinuousAt}(f, c) \wedge \mathrm{OscillationAt}(f, c, 0)}_{\text{Zero oscillation without continuity}}.$$

*Remark* (Interpretation). The theorem states that measuring the oscillation of $f$ near $c$ captures exactly the same local regularity as the classical $\varepsilon$-$\delta$ definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing $f(x)$ to approach $f(c)$. Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of $c$.

*Remark* (Dependencies).

Definition 2.2.2, Definition 2.2.11.

**Proposition**

**Proposition 2.2.13** (Discontinuity Set via Oscillation)**.** *Let $f : E \to \mathbb{R}$. Then*

$$D(f) := \{x \in E : f \textit{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}. \\
&\forall x \in E : \ x \in D(f) \iff \omega(f; x) > 0 \iff \exists n \in \mathbb{N} : \ \omega(f; x) \geq \tfrac{1}{n}.
\end{aligned}$$

*Remark* (Predicate reading).

$$\begin{aligned}
x \in D(f) &\iff \mathrm{DiscontinuousAt}(f, x) \\
&\iff \exists \omega \ (\mathrm{OscillationAt}(f, x, \omega) \wedge \omega > 0) \\
&\iff \exists n \in \mathbb{N} \ \exists \omega \ (\mathrm{OscillationAt}(f, x, \omega) \wedge \omega \geq \tfrac{1}{n}).
\end{aligned}$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}. \\
&\neg\Big[\forall x \in E \ \big(x \in D(f) \iff \omega(f; x) > 0 \iff \exists n \in \mathbb{N} \ \omega(f; x) \geq \tfrac{1}{n}\big)\Big] \\
&\iff \exists x \in E \ \Big((x \in D(f) \wedge \omega(f; x) = 0) \vee (x \notin D(f) \wedge \omega(f; x) > 0) \\
&\qquad\qquad \vee (\omega(f; x) > 0 \wedge \forall n \in \mathbb{N} \ \omega(f; x) < \tfrac{1}{n})\Big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned}
\exists x \in E \ \Big(&\exists \omega \ (\mathrm{DiscontinuousAt}(f, x) \wedge \mathrm{OscillationAt}(f, x, \omega) \wedge \omega = 0) \ \vee \\
&\exists \omega \ (\neg \mathrm{DiscontinuousAt}(f, x) \wedge \mathrm{OscillationAt}(f, x, \omega) \wedge \omega > 0) \ \vee \\
&\exists \omega \ (\mathrm{OscillationAt}(f, x, \omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N} \ \omega < \tfrac{1}{n})\Big).
\end{aligned}$$

*Remark* (Failure modes)*.* The proposition fails precisely when some $x \in E$ is marked as discontinuous although its oscillation vanishes, or when $x$ is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts $\omega(f; x) > 0$ into the countable union description.

*Remark* (Failure mode decomposition).

$$\exists x \in E \begin{cases} x \in D(f) \wedge \omega(f;x) = 0 & \text{(Type I)}, \\ x \notin D(f) \wedge \omega(f;x) > 0 & \text{(Type II)}, \\ \omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \frac{1}{n} & \text{(Type III)}. \end{cases}$$

*Remark* (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces $\omega(f;x)$ to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set $\{x \in E : \omega(f;x) \geq 1/n\}$ is closed, the discontinuity set $D(f)$ is an $F_\sigma$, a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of $D(f)$ controls integrability.

*Remark* (Dependencies).

Definition 2.2.5; Definition 2.2.11.

## 2.3 Global Theorems

**Global Theorems — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Bounded on a set | The output set is bounded in $\mathbb{R}$. |
| Boundedness theorem | Continuous functions on $[a,b]$ are bounded. |
| Absolute extrema | Global maximum and minimum values on a set. |
| Extreme value theorem | Continuous functions on $[a,b]$ attain extrema. |
| Intermediate value theorem | Continuous functions do not skip values. |
| Image interval theorem | $f([a,b])$ is a closed bounded interval. |
| Darboux property | Interval values are preserved without gaps. |

### Intermediate Value Theorem

**Definition (Bounded on a Set)**

**Definition 2.3.1** (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function $f$ is bounded on $A$ if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.\\ &f \text{ is bounded on } A \iff \exists M > 0\ \forall x \in A\ (|f(x)| \leq M).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{BoundedOn}(f, A) :\!= \exists M > 0\ \forall x \in A\ (|f(x)| \leq M).$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \forall x \in [a,b]\ (f(x) \neq c) \implies \Big[\big(\forall x \in [a,b]\ f(x) > c\big) \vee \big(\forall x \in [a,b]\ f(x) < c\big)\Big].$$ ∎

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.\\ &\neg(f \text{ bounded on } A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f, A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

*Remark* (Failure modes). Failure occurs precisely when every proposed positive bound $M$ is exceeded by some value of $|f(x)|$ with $x \in A$.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}}\quad \underbrace{\exists x \in A\ (|f(x)| > M)}_{\text{value exceeding } M}.$$

*Remark* (Interpretation). The definition packages the informal idea that the values of $f$ stay within a fixed vertical strip over $A$. The witness $M$ serves as a uniform ceiling for the magnitudes $|f(x)|$, so once such an $M$ is known the graph of $f$ never escapes the band $[-M, M]$. Failure means that the outputs of $f$ grow without limit along some sequence of points inside $A$, so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

*Remark* (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

**Theorem (Boundedness Theorem)**

**Theorem 2.3.2** (Boundedness Theorem)**.** *Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a,b]$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ and } f : [a,b] \to \mathbb{R}.\\ &\Big(\forall c \in [a,b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a,b]\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)\\ &\qquad \implies \exists M > 0\ \forall x \in [a,b]\ (|f(x)| \leq M).\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \implies \mathrm{BoundedOn}(f,[a,b]).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a<b \text{ and } f:[a,b]\to\mathbb{R}.\\
&\Big(\forall c\in[a,b]\ \forall\varepsilon>0\ \exists\delta>0\ \forall x\in[a,b]\big(|x-c|<\delta \Rightarrow |f(x)-f(c)|<\varepsilon\big)\Big)\\
&\qquad\wedge\Big(\forall M>0\ \exists x\in[a,b]\ (|f(x)|>M)\Big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \neg\,\mathrm{BoundedOn}(f,[a,b]).$$

*Remark* (Failure modes). Each component hypothesis is indispensable: without closedness (for instance $f(x)=1/x$ on $(0,1]$) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example $f(x)=x$ on $[0,\infty)$) linear growth forces unbounded values; without continuity (take $f(0)=0$ and $f(x)=1/x$ for $x\in(0,1]$) the discontinuity at 0 again creates arbitrarily large magnitudes.

*Remark* (Failure mode decomposition).

$$\neg\big[\mathrm{ClosedInterval}(a,b)\wedge\mathrm{BoundedSet}([a,b])\wedge\mathrm{ContinuousOn}(f,[a,b])\big] = \underbrace{\neg\,\mathrm{ClosedInterval}(a,b)}_{\text{missing endpoint}} \vee \underbrace{\neg\,\mathrm{Bounde}}_{\text{unbound}}$$

*Remark* (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval $[a,b]$ can be covered by finitely many neighborhoods in which the oscillation of $f$ is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

*Remark* (Dependencies).

Definition 2.2.2; Definition 2.3.1.

**Definition (Absolute Maximum and Absolute Minimum)**

**Definition 2.3.3** (Absolute Maximum and Absolute Minimum). Let $A\subseteq\mathbb{R}$ and let $f\colon A\to\mathbb{R}$. A point $x^*\in A$ is an absolute maximum of $f$ on $A$ if and only if $f(x^*)\ge f(x)$ for every $x\in A$. A point $x_*\in A$ is an absolute minimum of $f$ on $A$ if and only if $f(x_*)\le f(x)$ for every $x\in A$.

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A\subseteq\mathbb{R},\ f\colon A\to\mathbb{R},\ x^*,x_*\in A.\\
&x^* \text{ is an absolute maximum of } f \text{ on } A \iff \forall x\in A\ (f(x)\le f(x^*)).\\
&x_* \text{ is an absolute minimum of } f \text{ on } A \iff \forall x\in A\ (f(x_*)\le f(x)).
\end{aligned}$$

*Remark* (Definition predicate reading).

$$(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A)) \iff (x^* \in A) \wedge \forall x \in A\ (f(x) \leq f(x^*)),$$
$$(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A)) \iff (x_* \in A) \wedge \forall x \in A\ (f(x_*) \leq f(x)).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in \mathbb{R}.\\
&x^* \text{ fails to be an absolute maximum of } f \text{ on } A\\
&\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big).\\
&x_* \text{ fails to be an absolute minimum of } f \text{ on } A\\
&\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] \iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big),$$
$$\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] \iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).$$

*Remark* (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain $A$, or because some $x \in A$ produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in $A$ with a strictly smaller value.

*Remark* (Failure mode decomposition).

$$\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] = \underbrace{(x^* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x) > f(x^*))\big)}_{\text{better value}},$$
$$\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] = \underbrace{(x_* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x_*) > f(x))\big)}_{\text{better value}}.$$

*Remark* (Interpretation). An absolute maximum identifies the highest value that $f$ attains on its domain $A$, together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to $A$, so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above $A$, while the absolute minimum sits at the lowest point.

*Remark* (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

- Maximum
- Minimum

**Theorem (Extreme Value Theorem)**

**Theorem 2.3.4** (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, $f$ attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}$$
$$\exists x_m, x_M \in [a, b]\ \forall x \in [a, b]\ (f(x_m) \leq f(x) \leq f(x_M)).$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b]\ \big(\mathrm{Maximum}(f(x_M), f([a, b])) \wedge \mathrm{Minimum}(f(x_m), f([a, b]))\big).$$

*Remark* (Failure modes). The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg(a < b)}_{\text{invalid interval}} \vee \underbrace{\neg\big(f : [a, b] \to \mathbb{R}\big)}_{\text{wrong function type}} \vee \underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}}.$$

*Remark* (Interpretation). Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

*Remark* (Dependencies).

Uses Definition 2.3.3 and Theorem 2.3.2.

**Theorem (Location of Roots)**

**Theorem 2.3.5** (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } f : [a, b] \to \mathbb{R} \text{ continuous.}$$
$$\big(f(a) < 0 < f(b) \vee f(a) > 0 > f(b)\big) \Rightarrow \exists c \in (a, b)\ \big(f(c) = 0\big).$$

*Remark* (Failure modes). The root conclusion can fail if $f$ is not continuous on $[a, b]$, or if the endpoint values do not have opposite signs.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \vee \underbrace{\big(f(a)f(b) \geq 0\big)}_{\text{no sign change}}.$$

*Remark* (Interpretation). Continuity forces the image of the interval $[a, b]$ to contain every intermediate value between $f(a)$ and $f(b)$. A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when $f$ is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the $x$-axis must intersect the axis somewhere in between.

*Remark* (Dependencies).

Bolzano's Intermediate Value Theorem.

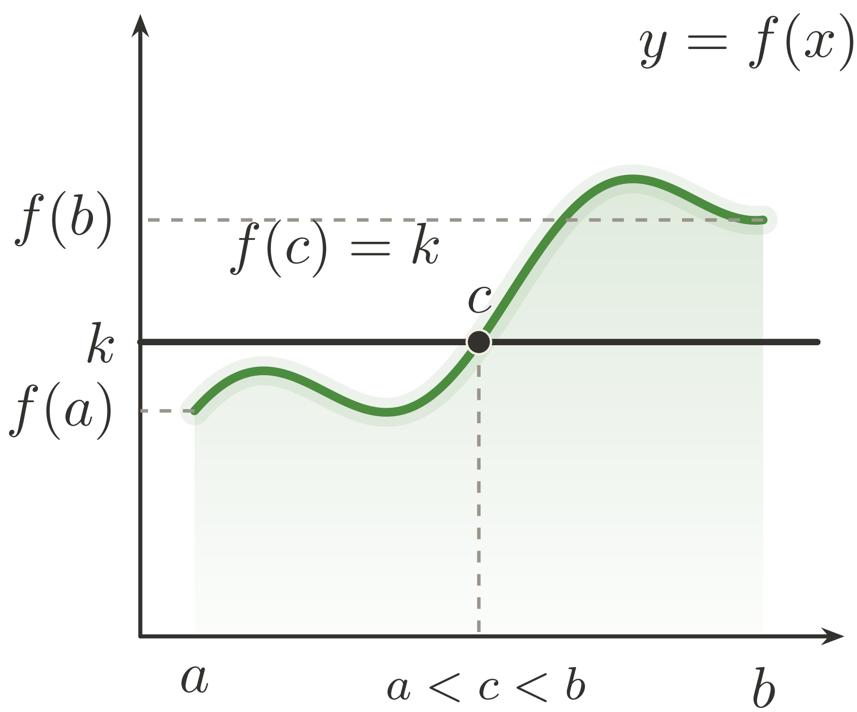


Figure 2.1: Intermediate Value Theorem: a continuous path hits every intermediate height.

**Theorem (Bolzano's Intermediate Value Theorem)**

**Theorem 2.3.6** (Bolzano's Intermediate Value Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be continuous on $I$, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that* $\min\{a, b\} < c < \max\{a, b\}$ *and $f(c) = k$. Go to proof.*

*Remark* (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall a, b \in I\ \forall k \in \mathbb{R}\ \Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in$$

*Remark* (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level $k$ is not strictly between the two attained endpoint values.

*Remark* (Failure mode decomposition).

$$\begin{aligned} &\underbrace{\exists x, y \in I\ \exists z \in \mathbb{R}\ \big(x < z < y \wedge z \notin I\big)}_{\text{not an interval}} \\ \vee\ &\underbrace{\exists c \in I\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \\ \vee\ &\underbrace{\neg\big(f(a) < k < f(b)\big)}_{\text{level not between endpoint values}} . \end{aligned}$$

*Remark* (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between $f(a)$ and $f(b)$. The conclusion provides an interior point $c$ whose image equals the prescribed level $k$, ensuring root-finding for $f - k$. If the hypothesis fails, either $f$ is not continuous or the level $k$ does not lie between the endpoint values, and no such interior point need exist.

*Remark* (Dependencies).

- Function
- Subset
- Continuity at a Point
- Order on $\mathbb{R}$
- Maximum
- Minimum

**Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)**

**Theorem 2.3.7** (Image of Closed Bounded Interval Is Closed Bounded Interval)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \forall f : [a, b] \to \mathbb{R} :$$
$$\Big(a < b \wedge \forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big) \implies f([a, b]) = [\min_{[a,b]} f$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \implies f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

*Remark* (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on $[a,b]$, with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image $f([a,b])$ is a closed interval with endpoints given by the absolute extremal values of $f$ on $[a,b]$, showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

*Remark* (Dependencies).

Extreme Value Theorem, Bolzano's Intermediate Value Theorem

**Theorem (Preservation of Intervals)**

**Theorem 2.3.8** (Preservation of Intervals). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R} :\\
&\quad \Big[\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\Big] \implies \forall y_1, y_2 \in f(I)\ \forall y \in \mathbb{R}\ \big(y_1 <
\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f, I) \implies \mathrm{Interval}(f(I)).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\exists I \subseteq \mathbb{R}\ \exists f : I \to \mathbb{R}\ \exists y_1, y_2 \in f(I)\ \exists y \in \mathbb{R} :\\
&\quad \big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\\
&\quad \wedge y_1 < y < y_2 \wedge y \notin f(I).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\mathrm{Interval}(I) \wedge \mathrm{ContinuousOn}(f, I) \wedge \neg\,\mathrm{Interval}(f(I)).$$

*Remark* (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when $I$ is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

*Remark* (Dependencies).

thm:bolzano-intermediate-value

**Theorem (Darboux Property)**

**Theorem 2.3.9** (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \Rightarrow \Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

*Remark* (Negated quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \Big(\exists u \in [a, b]\ f(u) \leq c\Big) \wedge \Big(\exists v \in [a, b]\ f(v) \geq c\Big).$$

*Remark* (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \neg\Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

*Remark* (Contrapositive quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\exists x \in [a, b]\ f(x) \leq c\Big) \wedge \Big(\exists y \in [a, b]\ f(y) \geq c\Big) \Rightarrow \exists z \in [a, b]\ f(z) = c.$$

*Remark* (Contrapositive predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{MeetsBothSides}(f, [a, b], c) \Longrightarrow \exists z \in [a, b]\ (f(z) = c).$$

*Remark* (Interpretation). A continuous function on a compact interval cannot oscillate across a level c without actually attaining that level. Stated contrapositive fashion, if c is never taken then the entire graph lies strictly above or strictly below the horizontal line y=c. The only way this guarantee can fail is when the function produces values on both sides of c, in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting c forces the image to stay in one of the open half-lines determined by c.

*Remark* (Dependencies).

Bolzano's Intermediate Value Theorem.

## 2.4 Uniform Continuity

**Uniform Continuity — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Uniform continuity | One $\delta(\varepsilon)$ works for all pairs in the domain. |
| Sequential test | Close inputs with separated outputs witness failure. |
| Heine–Cantor | Continuity on $[a, b]$ upgrades to uniform continuity. |
| Cauchy preservation | Uniform continuity preserves Cauchy sequences. |
| Lipschitz | A global slope bound gives a uniform modulus. |
| Lipschitz $\Rightarrow$ uniform | Lipschitz control implies uniform continuity. |
| Bi-Lipschitz maps | Distances are controlled above and below. |

### Uniform Continuity

**Definition (Uniform Continuity)**

**Definition 2.4.1** (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function $f$ is uniformly continuous on $A$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R})\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).\end{aligned}$$

*Remark* (Definition predicate reading)*.*

$$\text{UniformlyContinuous}(f, A) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

*Remark* (Negated quantified statement)*.*

$$\begin{aligned}&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R})\\ &\qquad \vee \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).\end{aligned}$$

*Remark* (Negation predicate reading)*.*

$$\neg\,\text{UniformlyContinuous}(f, A) \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

*Remark* (Failure modes)*.* The definition fails exactly when a single positive tolerance $\varepsilon_0$ exposes pairs of inputs arbitrarily close together whose function values remain at least $\varepsilon_0$ apart, so no universal $\delta$ works for that $\varepsilon_0$.

*Remark* (Failure mode decomposition)*.*

$$\neg\,\text{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon_0)}_{\text{bad witness pair}}.$$

*Remark* (Interpretation). Uniform continuity demands that a single proximity scale $\delta$ answers any prescribed accuracy $\varepsilon$ simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of $\delta$ because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length $\delta$ horizontally and $\varepsilon$ vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

*Remark* (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

**Theorem (Algebra of Uniform Continuity on a General Domain)**

**Theorem 2.4.2** (Algebra of Uniform Continuity on a General Domain)**.** *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on $A$, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and $cf$ are uniformly continuous on $A$.*

*(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product $fg$ is not uniformly continuous on $A$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\Big[\big(\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\big)$$
$$\wedge \big(\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\big)\Big] \Rightarrow \Big[\forall \varepsilon > 0\ \exists \delta_1 > 0\ \forall x, y \in A\ (|x -$$
$$\wedge\ \forall \varepsilon > 0\ \exists \delta_2 > 0\ \forall x, y \in A\ (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon)$$
$$\wedge\ \forall \varepsilon > 0\ \exists \delta_3 > 0\ \forall x, y \in A\ (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon)\Big],$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\ \Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\Big]$$
$$\wedge\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ \big(|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0\big).$$

*Remark* (Predicate reading).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\big(\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A)\big) \Longrightarrow \big(\text{UniformlyContinuous}(f + g, A) \wedge \text{U}$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\quad \text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A) \wedge \neg\,\text{UniformlyContin}$$

*Remark* (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

*Remark* (Dependencies).

- Uniform Continuity
- Linear Combination of Real-Valued Functions
- Pointwise Difference of Functions

**Theorem (Algebra of Uniform Continuity on a Bounded Domain)**

**Theorem 2.4.3** (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on $A$.*

*(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on $A$.*

*(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on $A$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R} \text{ be bounded and } f, g : A \to \mathbb{R}.\\
&\big[\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\Rightarrow \forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon),\\
&\big[\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\wedge \big[\exists k > 0\, \forall x \in A\, (|g(x)| \geq k)\big]\\
&\Rightarrow \forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
\end{aligned}
$$

*Remark* (Predicate reading).

$$
\begin{aligned}
&\text{Assume } A \subseteq \mathbb{R} \text{ is bounded and } f, g : A \to \mathbb{R}.\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(fg, A),\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \exists k > 0\, \forall x \in A\, (|g(x)| \geq k)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(f/g, A).
\end{aligned}
$$

*Remark* (Failure modes)*.* The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded, $f$ is not uniformly continuous on $A$, or $g$ is not uniformly continuous on $A$. The quotient conclusion can also fail if the denominator is not bounded away from zero on $A$.

*Remark* (Failure mode decomposition)*.*

$$\underbrace{\neg\,\mathrm{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(f, A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(g, A)}_{\text{second factor not uniformly continuous}}$$
$$\vee \underbrace{\neg\exists k > 0\ \forall x \in A :\ |g(x)| \geq k}_{\text{denominator not bounded away from zero}}.$$

*Remark* (Interpretation)*.* Bounded subsets of $\mathbb{R}$ prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for $f$ and $g$ can be combined to control $fg$. The quotient is uniformly continuous whenever $g$ never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for $|g|$ are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

*Remark* (Dependencies)*.*

Definition 2.4.1.

**Proposition (Sequential Characterization of Uniform Continuity)**

**Proposition 2.4.4** (Sequential Characterization of Uniform Continuity)**.** *Let* $A \subseteq \mathbb{R}$ *and let* $f : A \to \mathbb{R}$*. Then* $f$ *is uniformly continuous on* $A$ *if and only if for every pair of sequences* $(x_n)$ *and* $(y_n)$ *in* $A$ *with* $\lim_{n\to\infty}(x_n - y_n) = 0$ *we have* $\lim_{n\to\infty}(f(x_n) - f(y_n)) = 0$*. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ \big(|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon\big)\Big] \iff \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n$$

*Remark* (Predicate reading)*.*

$$\mathrm{UniformlyContinuous}(f, A) \iff \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big).$$

*Remark* (Negated quantified statement)*.*

$$\begin{aligned}
&\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} :\\
&\Big(\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\qquad \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big)\\
&\vee \Big(\big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\big] \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned}
&\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} : \\
&\quad \Big(\mathrm{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big) \\
&\quad \vee \Big(\neg\, \mathrm{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)\Big).
\end{aligned}$$

*Remark* (Failure modes). The equivalence can fail in exactly two ways: either $f$ is uniformly continuous while some pair of asymptotically coincident sequences in $A$ yields outputs that do not converge together, or every such pair of sequences does converge together while $f$ nonetheless fails to be uniformly continuous on $A$.

*Remark* (Failure mode decomposition).

$$\begin{aligned}
&\underbrace{\mathrm{UniformlyContinuous}(f, A)}_{\text{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)}_{\text{sequential test fails}} \\
&\vee\ \underbrace{\neg\, \mathrm{UniformlyContinuous}(f, A)}_{\text{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)}_{\text{sequential test holds}}.
\end{aligned}$$

*Remark* (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of $A$ are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in $A$ but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

*Remark* (Dependencies).

Definition 2.4.1

**Theorem (Heine–Cantor Theorem)**

**Theorem 2.4.5** (Heine–Cantor Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then $f$ is uniformly continuous on $[a, b]$. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}. \\
&\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big] \\
&\Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in [a, b]\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).
\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Rightarrow \mathrm{UniformlyContinuous}(f, [a, b]).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.\\ &\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]\\ &\wedge \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in [a, b]\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big].\end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \neg\, \text{UniformlyContinuous}(f, [a, b]).$$

*Remark* (Interpretation). Continuity on the compact interval $[a, b]$ forces a single modulus of continuity that works simultaneously at every point, so oscillations of $f$ cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

*Remark* (Dependencies).

Definition 2.2.2; Definition 2.4.1.

**Proposition (Uniform Continuity Preserves Cauchy Sequences)**

**Proposition 2.4.6** (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let $(x_n)$ be a sequence in $S$. If $f$ is uniformly continuous on $S$ and $(x_n)$ is a Cauchy sequence in $S$, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } S \subseteq \mathbb{R},\ f : S \to \mathbb{R},\ (x_n)_{n\in\mathbb{N}} \subseteq S.\\ &\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in S\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]\\ &\wedge \Big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_n - x_m| < \varepsilon)\Big]\\ &\Rightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|f(x_n) - f(x_m)| < \varepsilon).\end{aligned}$$

*Remark* (Predicate reading).

$$\text{UniformlyContinuous}(f, S) \wedge \text{CauchySequence}(x_n) \;\Rightarrow\; \text{CauchySequence}(f(x_n)).$$

*Remark* (Interpretation). Uniform continuity controls the change in $f(x)$ solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when $f$ is merely continuous and the sequence approaches a boundary point where continuity loses uniform control, producing large oscillations in the image sequence.

Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by $(x_n)$, ensuring that eventually all image terms lie in an arbitrarily small common interval.

*Remark* (Dependencies).

Definition 2.4.1; Definition 6.11.1.

**Definition (Lipschitz Condition)**

**Definition 2.4.7** (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function $f$ satisfies the Lipschitz condition on $A$ with constant $K$ if and only if $|f(x) - f(u)| \le K|x - u|$ for every $x, u \in A$.

*Remark* (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \wedge (K > 0) \\ \implies \forall x, u \in A \ \big(|f(x) - f(u)| \le K|x - u|\big).$$

*Remark* (Definition predicate reading).

$$\mathrm{Lipschitz}(f, A, K) \coloneqq (K > 0) \wedge \forall x, u \in A \ \big(|f(x) - f(u)| \le K|x - u|\big).$$

*Remark* (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0) \\ \vee \exists x, u \in A \ \big(|f(x) - f(u)| > K|x - u|\big).$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{Lipschitz}(f, A, K) \iff \neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0) \vee \exists x, u \in A \ \big(|f(x) - f(u)| > K|x - u|\big).$$

*Remark* (Failure modes)*.* The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in $A$ whose images separate faster than $K$ times the separation of the points themselves.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg(A \subseteq \mathbb{R})}_{\text{bad domain}} \vee \underbrace{\neg(f : A \to \mathbb{R})}_{\text{bad function type}} \vee \underbrace{K \le 0}_{\text{bad constant}} \vee \underbrace{\exists x, u \in A : \ |f(x) - f(u)| > K|x - u|}_{\text{excessive separation of outputs}}.$$

*Remark* (Interpretation)*.* The Lipschitz condition requires a single constant $K$ that simultaneously bounds the rate at which $f$ can stretch distances across the entire set $A$. It encapsulates a global slope cap: every secant line through the graph has absolute slope at most $K$. Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by $K$, so nearby points cannot be blown apart faster than $K$ times their original spacing.

*Remark* (Dependencies).

- Function

**Proposition (Lipschitz Implies Uniform Continuity)**

**Proposition 2.4.8** (Lipschitz Implies Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then $f$ is uniformly continuous on $A$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}, \text{ and suppose } \exists K \geq 0\ \forall x, y \in A\ (|f(x) - f(y)| \leq K|x - y|).\\ &\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).\end{aligned}$$

*Remark* (Predicate reading).

$$\big[\exists K \geq 0\ \ \mathrm{Lipschitz}(f, A, K)\big] \Rightarrow \mathrm{UniformlyContinuous}(f, A).$$

*Remark* (Contrapositive quantified statement). If $f$ fails to be uniformly continuous on $A$, then $f$ is not Lipschitz on $A$ with any constant.

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}.\\ &\Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big]\\ &\Longrightarrow \forall K \geq 0\ \exists x, y \in A\ (|f(x) - f(y)| > K|x - y|).\end{aligned}$$

*Remark* (Contrapositive predicate reading).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) \implies \neg \exists K \geq 0 :\ \mathrm{Lipschitz}(f, A, K).$$

*Remark* (Interpretation). A Lipschitz bound controls the variation of $f$ by the distance between inputs with a single constant $K$, so the same $K$ yields a usable $\delta = \varepsilon/K$ for every pair of points at once, proving uniform continuity. When $K = 0$ the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

*Remark* (Strict implication note). The implication Lipschitz $\Rightarrow$ uniform continuity is strict. The canonical counterexample is $f(x) = \sqrt{x}$ on $[0, 1]$, which is uniformly continuous by the Heine–Cantor theorem but satisfies $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \to \infty$ as $x \to 0^+$, so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable $K$ for the Lipschitz condition; only an explicit Lipschitz bound does.

*Remark* (Dependencies).

Definition 2.4.7, Definition 2.4.1.

Theorem (Algebra of Lipschitz Functions)

**Theorem 2.4.9** (Algebra of Lipschitz Functions)**.** *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

*(a) For every $c \in \mathbb{R}$, the function $cf$ is Lipschitz on $A$ with constant $|c|K_f$.*

*(b) The functions $f + g$ and $f - g$ are Lipschitz on $A$ with constant $K_f + K_g$.*

*(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on $A$ with constant $K_h K_f$.*

*(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then $fg$ is Lipschitz on $A$ with constant $M_g K_f + M_f K_g$.*

*(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if $g$ is Lipschitz on $A$ with constant $K_g$, then $f/g$ is Lipschitz on $A$ with constant $K_f/k + M_f K_g/k^2$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f, g : A \to \mathbb{R},\ K_f, K_g \geq 0,\\
&\forall x, y \in A:\ |f(x) - f(y)| \leq K_f|x - y| \text{ and } |g(x) - g(y)| \leq K_g|x - y|.\\
&\text{(a) } \forall c \in \mathbb{R},\ \forall x, y \in A:\ |cf(x) - cf(y)| \leq |c|K_f|x - y|.\\
&\text{(b) } \forall x, y \in A:\ |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|.\\
&\text{(c) If } B \subseteq \mathbb{R},\ f(A) \subseteq B,\ h : B \to \mathbb{R},\ \forall u, v \in B:\ |h(u) - h(v)| \leq K_h|u - v|,\\
&\qquad \text{then } \forall x, y \in A:\ |h(f(x)) - h(f(y))| \leq K_h K_f|x - y|.\\
&\text{(d) If } |f(x)| \leq M_f,\ |g(x)| \leq M_g\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad |f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|.\\
&\text{(e) If } |f(x)| \leq M_f,\ |g(x)| \geq k > 0\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad \left|\frac{f(x)}{g(x)} - \frac{f(y)}{g(y)}\right| \leq \left(\frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)|x - y|.
\end{aligned}
$$

*Remark* (Predicate reading).

$$
\begin{aligned}
&\mathrm{Lipschitz}(f, A, K_f) \wedge \mathrm{Lipschitz}(g, A, K_g)\\
&\Longrightarrow \Big[\forall c \in \mathbb{R} : \mathrm{Lipschitz}(cf, A, |c|K_f)\Big]\\
&\qquad \wedge \mathrm{Lipschitz}(f + g, A, K_f + K_g) \wedge \mathrm{Lipschitz}(f - g, A, K_f + K_g)\\
&\qquad \wedge \Big(\mathrm{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \Rightarrow \mathrm{Lipschitz}(h \circ f, A, K_h K_f)\Big)\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \leq M_g) \Rightarrow \mathrm{Lipschitz}(fg, A, M_g K_f + M_f K_g)\Big]\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \geq k > 0) \Rightarrow \mathrm{Lipschitz}\Big(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\Big)\Big].
\end{aligned}
$$

*Remark* (Interpretation)*.* Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

*Remark* (Dependencies)*.*

Definition 2.4.7, Definition 2.3.1

**Definition (Bi-Lipschitz Map)**

**Definition 2.4.10** (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map $f$ is bi-Lipschitz on $A$ if and only if there exist constants $0 < \alpha \le K$ such that for every $x, u \in A$,

$$\alpha|x-u| \le |f(x)-f(u)| \le K|x-u|.$$

Any such pair $(\alpha, K)$ is called a bi-Lipschitz constant pair for $f$ on $A$.

*Remark* (Standard quantified statement)*.*

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \\ \implies \exists \alpha, K \in \mathbb{R}\left(0 < \alpha \le K \wedge \forall x, u \in A : \ \alpha|x-u| \le |f(x)-f(u)| \le K|x-u|\right).$$

*Remark* (Definition predicate reading)*.*

$$\operatorname{BiLipschitz}(f, A, \alpha, K) \coloneqq (0 < \alpha \le K) \wedge \forall x, u \in A \ \left(\alpha|x-u| \le |f(x)-f(u)| \le K|x-u|\right).$$

*Remark* (Negated quantified statement)*.*

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \\ \vee \forall \alpha, K \in \mathbb{R}\Big(0 < \alpha \le K \Rightarrow \exists x, u \in A : \ |f(x)-f(u)| < \alpha|x-u| \vee |f(x) -$$

*Remark* (Negation predicate reading)*.*

$$\neg \operatorname{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \le K) \vee \exists x, u \in A \ \left(|f(x)-f(u)| < \alpha|x-u| \vee |f(x)-f(u)| > K|x -\right.$$

*Remark* (Failure modes)*.* The property fails either because $f$ collapses some pair of points faster than the allowed lower constant $\alpha$, or because it stretches some pair more than the upper constant $K$ permits. Both distortions prevent a uniform two-sided control of distances.

*Remark* (Failure mode decomposition)*.*

$$\exists x, u \in A : \quad \underbrace{|f(x)-f(u)| < \alpha|x-u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x)-f(u)| > K|x-u|}_{\text{upper-distortion failure}}.$$

*Remark* (Interpretation). A bi-Lipschitz map distorts every distance in $A$ by factors lying between $\alpha$ and $K$, so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

*Remark* (Dependencies).

- Function

**Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)**

**Theorem 2.4.11** (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|$ for all $x, y \in A$. Then $f$ is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u-v| \le |f^{-1}(u) - f^{-1}(v)| \le \frac{1}{\alpha}|u-v| \quad \text{for all } u, v \in f(A).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ \alpha, K > 0.\\
&\Big[\forall x, y \in A\ \big(\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|\big)\Big]\\
&\implies \Big[\forall x, y \in A\ (f(x) = f(y) \Rightarrow x = y)\\
&\qquad \wedge \exists g : f(A) \to A\ \big(\forall u \in f(A)\ (f(g(u)) = u)\big)\\
&\qquad \wedge \forall u, v \in f(A)\ \Big(\frac{1}{K}|u-v| \le |g(u)-g(v)| \le \frac{1}{\alpha}|u-v|\Big)\Big].
\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{BiLipschitz}(f, A, \alpha, K) \Rightarrow \big(\mathrm{Injective}(f) \wedge \mathrm{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)\big).$$

*Remark* (Interpretation). A bi-Lipschitz bound simultaneously controls the expansion and contraction of $f$, so no two points of $A$ can share the same image and the inverse is automatically a well-defined function on $f(A)$. The inequalities for $f^{-1}$ follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

*Remark* (Dependencies).

Definition 2.4.7; Definition 2.4.10.

## 2.5 Monotone Functions

**Monotone Functions — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| One-sided limits | Monotone functions have finite or infinite side limits. |
| Continuity criterion | Continuity means both side limits equal the value. |
| Jump | The size of the monotone discontinuity. |
| First-kind gaps | Monotone discontinuities are jumps. |
| Countability | Only countably many jumps can occur. |
| Continuous inverse | Strict monotonicity gives a continuous inverse. |

**Limits Superior and Inferior — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Function $\limsup$ and $\liminf$ | Tail suprema become punctured-neighbourhood suprema. |
| Order relation | Always $\limsup \geq \liminf$. |
| Limit criterion | A finite limit exists exactly when they agree. |

### Monotone Functions and Continuity

**Theorem (One-Sided Limits of Monotone Functions)**

**Theorem 2.5.1** (One-Sided Limits of Monotone Functions). *Let $I \subseteq \mathbb{R}$ be an interval, let $c$ be interior to $I$, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- \coloneqq \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ \coloneqq \inf\{f(x) : x \in I,\ c < x\}.$$

*Then both one-sided limits exist and satisfy*

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&I \subseteq \mathbb{R} \wedge \big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge f : I \to \mathbb{R} \\
&\wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \leq f(y))\big) \wedge c \in I \wedge \exists a, b \in I\ (a < c < b) \\
&\wedge L_- = \sup\{f(x) : x \in I,\ x < c\} \wedge L_+ = \inf\{f(x) : x \in I,\ c < x\} \\
&\Longrightarrow \\
&\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in I\ \big((0 < c - x < \delta_- \wedge x < c) \Rightarrow |f(x) - L_-| < \varepsilon\big) \\
&\wedge \forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in I\ \big((0 < x - c < \delta_+ \wedge c < x) \Rightarrow |f(x) - L_+| < \varepsilon\big).
\end{aligned}$$

*Remark* (Predicate reading).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ increasing, and $c$ interior to $I$.

$$\mathrm{LeftLimit}\Big(f, c, \sup\{f(x) : x \in I,\ x < c\}\Big) \wedge \mathrm{RightLimit}\Big(f, c, \inf\{f(x) : x \in I,\ x > c\}\Big).$$

*Remark* (Interpretation). For an increasing function, the values just to the left of $c$ form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when $c$ is not an interior point or $f$ is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

*Remark* (Dependencies).

- Completeness of $\mathbb{R}$
- Monotone Function
- Supremum
- Infimum

**Corollary (Continuity Criterion for Monotone Functions)**

**Corollary 2.5.2** (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let $c$ be an interior point of $I$. Then $f$ is continuous at $c$ if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$. Go to proof.*

*Remark* (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and $c$ an interior point of $I$.

$$\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) = f(c) = \lim_{x\to c^+} f(x)\right).$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \big(\mathrm{LeftLimit}(f, c, f(c)) \wedge \mathrm{RightLimit}(f, c, f(c))\big).$$

*Remark* (Negated quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and $c$ an interior point of $I$.

$$\neg\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) < f(c)\right) \vee \left(f(c) < \lim_{x\to c^+} f(x)\right).$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \big(\exists L^-\,[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)]\big) \vee \big(\exists L^+\,[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) < L$$

*Remark* (Failure modes). Continuity can fail only through a jump: either the graph arrives at $c$ from the left strictly below $f(c)$, or it departs to the right strictly above $f(c)$. Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

*Remark* (Failure mode decomposition).

$$\neg\,\mathrm{ContinuousAt}(f,c) = \underbrace{\big(\exists L^-\,[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)]\big)}_{\text{left jump at } c} \vee \underbrace{\big(\exists L^+\,[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) < L^+]}_{\text{right jump at } c}$$

*Remark* (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at $c$ demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through $c$. Any failure manifests as a jump discontinuity: the graph either leaps upward at $c$ or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

*Remark* (Dependencies).

Definition 2.2.2; Theorem 2.5.1.

**Definition (Jump of a Function)**

**Definition 2.5.3** (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let $c$ be an interior point of $I$. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of $f$ at $c$ is

$$j_f(c) := f(c^+) - f(c^-).$$

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R}\\
&\Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \exists a, b \in I\ (a < c < b)\\
&\quad \wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to c^-} f(x) = L_- \wedge \lim_{x\to c^+} f(x) = L_+\big)\Big]\\
&\quad \Rightarrow \big(j = j_f(c) \iff j = f(c^+) - f(c^-)\big).
\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{JumpOfFunction}(f,I,c,j) := \mathrm{Interval}(I) \wedge \mathrm{MonotoneIncreasing}(f,I) \wedge \mathrm{InteriorPoint}(c,I) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\mathrm{Le}$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} :\\
&\neg\big(j = j_f(c)\big) \iff \neg\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big)\\
&\qquad \vee \neg\big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \vee \neg\big(\exists a, b \in I\ (a < c < b)\big)\\
&\qquad \vee \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to c^-} f(x) \ne L_- \vee \lim_{x\to c^+} f(x) \ne L_+ \vee j \ne L_+ - L_-\big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{JumpOfFunction}(f, I, c, j) \iff \neg\,\mathrm{Interval}(I) \vee \neg\,\mathrm{MonotoneIncreasing}(f, I) \vee \neg\,\mathrm{InteriorPoint}(c, I) \vee \forall L_-, L$$

*Remark* (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg\big(f(c^-)\text{ exists}\big)}_{\text{missing left limit}} \vee \underbrace{\neg\big(f(c^+)\text{ exists}\big)}_{\text{missing right limit}} \vee \underbrace{j \neq f(c^+) - f(c^-)}_{\text{incorrect jump magnitude}}.$$

*Remark* (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

*Remark* (Dependencies).

Theorem 2.5.1.

**Proposition (Discontinuities of a Monotone Function Are of First Kind)**

**Proposition 2.5.4** (Discontinuities of a Monotone Function Are of First Kind)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of $I$ strictly less and strictly greater than $a$. If $f$ is discontinuous at $a$, then both one-sided limits $\lim_{x\uparrow a} f(x)$ and $\lim_{x\downarrow a} f(x)$ exist in $\mathbb{R}$. Go to proof.*

*Remark* (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone, and $a \in I$ interior to $I$.

$$\big[f \text{ is discontinuous at } a\big] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim\limits_{x\uparrow a} f(x) = L^-, \\ \lim\limits_{x\downarrow a} f(x) = L^+. \end{cases}$$

*Remark* (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \wedge \mathrm{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R}\ \big(\mathrm{LeftLimit}(f, a, L^-) \wedge \mathrm{RightLimit}(f, a, L$$

*Remark* (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

*Remark* (Dependencies).

Definition 2.2.5

**Corollary (Jump Intervals Are Disjoint)**

**Corollary 2.5.5** (Jump Intervals Are Disjoint)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of $f$, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.Go to proof.*

*Remark* (Standard quantified statement).

$$\forall a, b \in I \left[a \neq b \wedge \left(\exists \varepsilon_a > 0\ \forall \delta > 0\ \exists x \in I\ (|x-a| < \delta \wedge |f(x)-f(a)| \geq \varepsilon_a)\right) \wedge \left(\exists \varepsilon_b > 0\ \forall \delta > 0\ \exists y \in I\ (|y-b| <\right.\right.$$

*Remark* (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \mathrm{DiscontinuousAt}(f, a), \\ \mathrm{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \varnothing.$$

*Remark* (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

*Remark* (Dependencies).

Definition 2.5.3; Proposition 2.5.4

**Theorem (Discontinuities of a Monotone Function Are Countable)**

**Theorem 2.5.6** (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } I \subseteq \mathbb{R} \text{ be an interval and } f : I \to \mathbb{R} \text{ be monotone.} \\ &\#\{c \in I : f \text{ fails to be continuous at } c\} \leq \aleph_0. \end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \implies \#\{c \in I : \mathrm{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

*Remark* (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap $(f(c^-), f(c^+))$. Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into $\mathbb{Q}$, forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

*Remark* (Dependencies).

Theorem 2.5.1 and Corollary 2.5.5.

Corollary (Monotone Function Is Continuous on a Dense Set)

**Corollary 2.5.7** (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f$ is continuous at $x\}$ is dense in $[a, b]$. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be monotone.}\\ &\forall u, v \in [a, b]\ (u < v \Rightarrow \exists c \in (u, v) \text{ such that } f \text{ is continuous at } c).\end{aligned}$$

*Remark* (Predicate reading)*.*

$$\text{MonotoneFunction}(f, [a, b]) \implies \text{DenseSubset}\big(\{x \in [a, b] : \text{ContinuousAt}(f, x)\}, [a, b]\big).$$

*Remark* (Negated quantified statement)*.*

$$\exists u, v \in [a, b]\ \Big(u < v \wedge \forall c \in (u, v)\ \exists \varepsilon_c > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_c\big)\Big).$$

*Remark* (Negation predicate reading)*.*

$$\exists u, v \in [a, b]\ \Big(u < v \wedge \forall c \in (u, v)\ \neg\,\text{ContinuousAt}(f, c)\Big).$$

*Remark* (Interpretation)*.* A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

*Remark* (Dependencies)*.*

Discontinuities of a Monotone Function Are Countable

Proposition (Continuous Injective Is Strictly Monotone)

**Proposition 2.5.8** (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then $f$ is injective if and only if $f$ is strictly monotone on $[a, b]$. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous on } [a, b].\\ &\big[\forall x, y \in [a, b]\ (f(x) = f(y) \Rightarrow x = y)\big] \iff \big[\forall x, y \in [a, b]\ (x < y \Rightarrow f(x) < f(y))\big]\\ &\qquad\qquad \vee \big[\forall x, y \in [a, b]\ (x < y \Rightarrow f(y) < f(x))\big].\end{aligned}$$

*Remark* (Predicate reading)*.*

$$\text{ContinuousOn}(f, [a, b]) \implies \big(\text{Injective}(f) \iff \text{StrictlyMonotone}(f, [a, b])\big).$$

*Remark* (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

*Remark* (Dependencies).

Bolzano's Intermediate Value Theorem.

**Theorem (Continuous Inverse Theorem)**

**Theorem 2.5.9** (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on $I$. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$. Go to proof.*

*Remark* (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

$$\Big[(\forall x, y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big]$$
$$\Longrightarrow \Big[\exists g : f(I) \to I \text{ such that } (\forall x \in I)\, g(f(x)) = x \wedge (\forall y \in f(I))\, f(g(y)) = y$$
$$\wedge\ (\forall u, v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I)\ \forall \varepsilon > 0\ \exists \eta > 0\ \forall u \in f(I)\big(|u - y| < \eta \Rightarrow |g(u) - g$$

*Remark* (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of $f^{-1}$ is the reflection of the graph of $f$ across the diagonal line $y = x$, so the smoothness and strict increase or decrease are visibly preserved by symmetry.

*Remark* (Dependencies).

- Bolzano's Intermediate Value Theorem
- Strictly Increasing Function
- Strictly Decreasing Function
- Continuity at a Point

## Limits Superior and Inferior of Functions

**Definition (Limits Superior and Inferior of a Function)**

**Definition 2.5.10** (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let $x_0$ be an accumulation point of $E$. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

*Remark* (Standard quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$\limsup\limits_{x\to x_0} f(x) = \inf\limits_{\delta>0} \sup U(\delta),$
$\liminf\limits_{x\to x_0} f(x) = \sup\limits_{\delta>0} \inf U(\delta).$

*Remark* (Definition predicate reading)*.*

$$\mathrm{FunctionLimsup}(L, f, E, x_0) :\equiv L = \inf_{\delta>0} \sup U(\delta),$$
$$\mathrm{FunctionLiminf}(l, f, E, x_0) :\equiv l = \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

*Remark* (Negated quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$\neg\Big[\limsup\limits_{x\to x_0} f(x) = \inf\limits_{\delta>0} \sup U(\delta) \ \wedge\ \liminf\limits_{x\to x_0} f(x) = \sup\limits_{\delta>0} \inf U(\delta)\Big]$
$\iff \limsup\limits_{x\to x_0} f(x) \neq \inf\limits_{\delta>0} \sup U(\delta) \vee \liminf\limits_{x\to x_0} f(x) \neq \sup\limits_{\delta>0} \inf U(\delta).$

*Remark* (Negation predicate reading)*.*

$$\neg\big(\mathrm{FunctionLimsup}(L, f, E, x_0) \wedge \mathrm{FunctionLiminf}(l, f, E, x_0)\big)$$
$$\iff L \neq \inf_{\delta>0} \sup U(\delta) \vee l \neq \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

*Remark* (Failure modes)*.* The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists\delta > 0 :\ U(\delta) = \varnothing}_{\text{missing punctured samples}} \vee \underbrace{\left(\inf_{\delta>0} \sup U(\delta) = +\infty \wedge L < +\infty\right)}_{\text{upper envelope blow-up}} \vee \underbrace{\left(\sup_{\delta>0} \inf U(\delta) = -\infty \wedge l > -\infty\right)}_{\text{lower envelope collapse}}.$$

*Remark* (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to $x_0$, and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

*Remark* (Dependencies).

- Function
- Supremum
- Infimum
- Epsilon Neighbourhood

**Proposition**

**Proposition 2.5.11** ($\limsup \geq \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $x_0$ be a limit point of $A$, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\forall A \subseteq \mathbb{R}\ \forall f : A \to \mathbb{R}\ \forall x_0 \in \mathbb{R} :\\ &\qquad \Big(\forall \eta > 0\ \exists x \in A \setminus \{x_0\}\ (|x - x_0| < \eta)\Big) \implies \limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).\end{aligned}$$

*Remark* (Interpretation). The limsup records the largest cluster value that $f$ attains near $x_0$, while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

*Remark* (Dependencies).

Definition (Limits Superior and Inferior of a Function).

**Proposition**

**Proposition 2.5.12** (Limit Exists iff $\limsup = \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let $x_0$ be a limit point of $A$ with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals $L$ if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ (\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A \setminus \{x_0\}\ (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon))\\ \iff \quad \left(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\right).\end{gathered}$$

*Remark* (Predicate reading)*.*

$$\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\big).$$

*Remark* (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)\Big]\\ \iff \quad \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).\end{gathered}$$

*Remark* (Negation predicate reading)*.*

$$\neg\,\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).$$

*Remark* (Failure modes)*.* The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach $L$ in the epsilon-delta sense, or one of the two extremal envelopes near $x_0$ does not equal $L$.

*Remark* (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)}_{\text{limit condition fails}} \vee \underbrace{\limsup_{x\to x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \vee \underbrace{\liminf_{x\to x_0} f(x) \neq}_{\text{lower envelope mism}}$$

*Remark* (Interpretation)*.* The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the limsup exceeds the liminf, as happens for $f(x) = \sin(1/x)$ near $x_0 = 0$, the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

*Remark* (Dependencies)*.*

Definition 2.5.10

## 2.6 Limits at Infinity

**Limits at Infinity — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Infinite adherent points | Unboundedness makes $\pm\infty$ available as approach points. |
| Limit at infinity | Replace $0 < \|x - c\| < \delta$ by sufficiently large $\|x\|$. |
| Sequential criterion | Limits at infinity are tested along escaping sequences. |
| Limit laws | Algebraic limit rules carry over unchanged. |

### Limits at Infinity

**Definition (Infinite Adherent Points)**

**Definition 2.6.1** (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to $X$ if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to $X$ if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

*Remark* (Standard quantified statement).

$$\begin{aligned} +\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ -\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\begin{aligned} \mathrm{PlusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ \mathrm{MinusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

*Remark* (Negated quantified statement).

$$\begin{aligned} \neg(+\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg(-\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned} \neg\,\mathrm{PlusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg\,\mathrm{MinusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

*Remark* (Failure modes). Infinite adherence fails at $+\infty$ exactly when $X$ is bounded above, and fails at $-\infty$ exactly when $X$ is bounded below.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \le M)}_{\text{finite upper bound}} \qquad \underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M)}_{\text{finite lower bound}}.$$

*Remark* (Interpretation). Both finite and infinite adherence are instances of adherence in $\overline{\mathbb{R}}$: finite adherence uses balls; infinite adherence uses rays.

**Definition (Limit at Infinity)**

**Definition 2.6.2** (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of $f$ at $+\infty$ equals $L$ if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

*Remark* (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\lim_{x\to+\infty} f(x) = L \iff \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\\ &\qquad (x > M \Rightarrow |f(x) - L| < \varepsilon).\end{aligned}$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{LimitAtInfinity}(f, +\infty, L) :\equiv \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\ (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

*Remark* (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\neg\big(\lim_{x\to+\infty} f(x) = L\big) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\\ &\qquad (x > M \wedge |f(x) - L| \geq \varepsilon_0).\end{aligned}$$

*Remark* (Negation predicate reading)*.*

$$\neg\,\mathrm{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M \wedge |f(x) - L| \geq \varepsilon_0).$$

*Remark* (Failure modes)*.* The definition fails only when there is a fixed tolerance $\varepsilon_0$ that the function cannot satisfy on any tail of $X$, meaning every attempt to choose $M$ leaves some point $x > M$ with $|f(x) - L| \geq \varepsilon_0$.

*Remark* (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{fixed tolerance}}\ \underbrace{\forall M \in \mathbb{R}}_{\text{no tail succeeds}}\ \underbrace{\exists x \in X\ (x > M \wedge |f(x) - L| \geq \varepsilon_0)}_{\text{far-point violation}}.$$

*Remark* (Interpretation)*.* The limit at $+\infty$ demands that eventually the function values stay within any prescribed accuracy of $L$. Intuitively, once $x$ is large enough, the graph of $f$ must lie in the horizontal band of height $\varepsilon$ around $L$. Failure occurs when some positive tolerance is violated infinitely often by points escaping to $+\infty$. This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of $f$ with respect to the ideal point at infinity.

*Remark* (Dependencies)*.*

Definition 2.6.1.

## 2.7 Approximation

**Approximation — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Step approximation | Uniform approximation by step functions. |
| Piecewise linear function | Linear pieces joined across a partition. |
| PL approximation | Uniform approximation by polygonal functions. |
| Weierstrass | Uniform polynomial approximation on compact intervals. |
| Bernstein polynomial | Constructive polynomial approximants on $[0, 1]$. |
| Bernstein convergence | Bernstein polynomials converge uniformly. |

### Approximation of Continuous Functions

**Theorem (Step Function Approximation)**

**Theorem 2.7.1** (Step Function Approximation)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.\\ \exists s_\varepsilon : [a, b] \to \mathbb{R} \text{ step function } \forall x \in [a, b]\ (|f(x) - s_\varepsilon(x)| < \varepsilon).\end{gathered}$$

*Remark* (Predicate reading)*.*

$$\mathrm{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) :\equiv \mathrm{StepFunction}(s_\varepsilon, [a, b]) \wedge \forall x \in [a, b]\ (|f(x) - s_\varepsilon(x)| < \varepsilon).$$

*Remark* (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.\\ \forall s : [a, b] \to \mathbb{R} \text{ step function } \exists x \in [a, b]\ (|f(x) - s(x)| \geq \varepsilon).\end{gathered}$$

*Remark* (Negation predicate reading)*.*

$$\neg\exists s_\varepsilon\ \mathrm{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) \iff \forall s : [a, b] \to \mathbb{R}\ \Big(\mathrm{StepFunction}(s, [a, b]) \Rightarrow \exists x \in [a, b]\ (|f(x) - s$$

*Remark* (Failure modes)*.* The statement fails precisely when a continuous $f$ and tolerance $\varepsilon$ are fixed but every step function leaves an error of at least $\varepsilon$ at some point in $[a, b]$.

*Remark* (Failure mode decomposition)*.*

$$\underbrace{\forall s : [a, b] \to \mathbb{R} \text{ step function}}_{\text{all approximants}}\ \underbrace{\exists x \in [a, b]\ (|f(x) - s(x)| \geq \varepsilon)}_{\text{witnessed large deviation}}.$$

*Remark* (Interpretation). Uniform continuity on $[a, b]$ guarantees that $f$ does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track $f$ within any prescribed uniform error. The typical construction samples $f$ at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least $\varepsilon$, contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

*Remark* (Dependencies).

Heine–Cantor Theorem.

**Definition (Piecewise Linear Function)**

**Definition 2.7.2** (Piecewise Linear Function). Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function $g$ is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1},x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\ &g \text{ is piecewise linear}\\ &\iff \exists m \in \mathbb{N} \Bigl(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a, b] \; \bigl[a = x_0 < x_1 < \cdots < x_m = b\\ &\qquad\qquad \wedge \forall k \in \{1, \ldots, m\} \; g|_{[x_{k-1},x_k]} \text{ is affine}\bigr]\Bigr).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{PiecewiseLinear}(g, [a, b]) :\Longleftrightarrow \exists m \in \mathbb{N} \Bigl(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a, b] \; \bigl[a = x_0 < \cdots < x_m = b \wedge \forall k \in \{1, \ldots, m\}$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\ &g \text{ is not piecewise linear}\\ &\iff \forall m \in \mathbb{N} \Bigl(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \; \bigl[a = x_0 < \cdots < x_m = b\\ &\qquad\qquad \Rightarrow \exists k \in \{1, \ldots, m\} \; g|_{[x_{k-1},x_k]} \text{ is not affine}\bigr]\Bigr).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\, \mathrm{PiecewiseLinear}(g, [a, b]) \iff \forall m \in \mathbb{N} \Bigl(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \; \bigl[a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in \{1$$

*Remark* (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of $[a, b]$ is available, or because for every such partition at least one subinterval has a restriction of $g$ that is not affine.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N}\ (m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a,b]\ (a = x_0 < \cdots < x_m = b \Rightarrow \cdots))}_{\text{no valid finite affine subdivision}} \vee \underbrace{\exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is not}}_{\text{non-affine segment}}$$

*Remark* (Interpretation). A piecewise linear function on $[a,b]$ has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

*Remark* (Dependencies).

- Function
- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

**Theorem**

**Theorem 2.7.3** (Piecewise Linear Approximation). *Let $f\colon [a,b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a,b] \to \mathbb{R}$ such that $\sup_{x \in [a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } f\colon [a,b] \to \mathbb{R},\ f \text{ continuous},\ \varepsilon > 0.\\ &\exists \text{ continuous piecewise linear } \ell_\varepsilon\colon [a,b] \to \mathbb{R} : \sup_{x \in [a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon.\end{aligned}$$

*Remark* (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding $\varepsilon$ on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

*Remark* (Dependencies).

- Heine–Cantor Theorem
- Piecewise Linear Function
- Continuity at a Point

**Theorem (Weierstrass Approximation Theorem)**

**Theorem 2.7.4** (Weierstrass Approximation Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}\\ &\forall \varepsilon > 0\ \exists p \in \mathbb{R}[x]\ (\|f - p\|_{\infty,[a,b]} < \varepsilon).\end{aligned}$$

*Remark* (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Longrightarrow \forall \varepsilon > 0\ \exists p \in \mathbb{R}[x]\ \big(\|f - p\|_{\infty,[a,b]} < \varepsilon\big).$$

*Remark* (Interpretation). Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

*Remark* (Dependencies).

Depends on thm:bernstein-approximation.

**Definition (Bernstein Polynomial)**

**Definition 2.7.5** (Bernstein Polynomial)**.** Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the $n$th Bernstein polynomial of $f$ if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}.$$

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } n \in \mathbb{N},\ f : [0, 1] \to \mathbb{R},\ B_n : [0, 1] \to \mathbb{R}.\\ &B_n \text{ is the } n\text{th Bernstein polynomial of } f\\ &\Longleftrightarrow \forall x \in [0, 1]\ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1 - x)^{n-k}\Big).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{BernsteinPolynomial}(B_n, f, n) :\equiv \Big(B_n : [0, 1] \to \mathbb{R}\Big) \wedge \forall x \in [0, 1]\ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}\Big).$$

*Remark* (Interpretation). The nth Bernstein polynomial samples f at the uniformly spaced nodes k/n, weights those samples by the binomial probabilities of order n, and produces for each x an average that depends polynomially on x. This construction smooths f into a polynomial that agrees with f at the endpoints and approximates f uniformly as n grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

*Remark* (Dependencies).

- Function
- Natural Numbers
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

**Theorem (Bernstein Approximation Theorem)**

**Theorem 2.7.6** (Bernstein Approximation Theorem). *Let $f : [0, 1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let $B_n$ denote the nth Bernstein polynomial of $f$. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

*Remark* (Standard quantified statement).

Let $f : [0, 1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let $B_n$ be the $n$th Bernstein polynomial of $f$. [redacted]
$\forall \varepsilon > 0\ \exists n_\varepsilon \in \mathbb{N}\ \forall n \geq n_\varepsilon :\ \|f - B_n\|_\infty < \varepsilon.$

*Remark* (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on $[0, 1]$ converge uniformly to the original function, so the entire graph of $f$ can be approximated to any prescribed sup-norm accuracy by explicit degree-$n$ polynomials once $n$ is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of $f$, which forces the uniform error to vanish as $n \to \infty$. This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when $f$ is not continuous on $[0, 1]$, in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each $B_n$ is a convex combination of the samples $f(k/n)$, so the polygons formed by these samples become finer and eventually hug the graph of $f$ everywhere on the interval.

*Remark* (Dependencies).

Definition (Continuity at a Point), Definition (Bernstein Polynomial).

## 2.8 Gauge

**Gauges and Tagged Partitions — Quick Reference**

| Concept/Statement | Meaning/Role |
|---|---|
| Interval partition | Finite subdivision of a compact interval. |
| Tagged partition | Each subinterval carries a selected tag. |
| Gauge | A position-dependent positive tolerance. |
| $\delta$-fine partition | Each subinterval lies within its tag's gauge window. |
| Cousin's theorem | Every gauge on $[a, b]$ admits a fine tagged partition. |
| Mesh | The largest subinterval length. |
| Common refinement | A partition refining two given partitions. |

### Gauges and Tagged Partitions

**Definition (Partition)**

**Definition 2.8.1** (Partition)**.** Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of $I$ if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide $I$.

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I. \\
&P \text{ is a partition of } I \\
&\iff \big(x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \dots, n\}\ (x_{i-1} < x_i)\big).
\end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{PartitionOfInterval}(P, I) :\equiv \exists n \in \mathbb{N}\ \Big(n \geq 1 \wedge P = \{x_0, \dots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, \dots,$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I. \\
&P \text{ is not a partition of } I \\
&\iff \big((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \dots, n\}\ (x_{i-1} \geq x_i)\big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\text{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N}\ \Big(n \geq 1 \Rightarrow \big(P \neq \{x_0, \dots, x_n\} \vee P \nsubseteq I \vee x_0 \neq \min I \vee x_n \neq \max I \vee \exists i$$

*Remark* (Failure modes)*.* The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

*Remark* (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint missing}} \quad \vee \ \underbrace{\exists i \in \{1, \ldots, n\} \ (x_{i-1} \geq x_i)}_{\text{nodes not strictly increasing}}.$$

*Remark* (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals $[x_{i-1}, x_i]$ tile $[a, b]$ without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

*Remark* (Dependencies).

- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

**Definition (Tagged Partition)**

**Definition 2.8.2** (Tagged Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. Equivalently, a tagged partition is exactly a partition $P$ of $[a, b]$ together with one tag chosen inside every subinterval of $P$.

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is a tagged partition of } [a, b] \\
&\iff \big(a = x_0 < \cdots < x_n = b \wedge \forall i \in \{1, \ldots, n\} \ (t_i \in [x_{i-1}, x_i])\big).
\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) :\equiv \Big(\mathrm{Partition}(P, [a, b]) \wedge \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n} \wedge \forall i \ (t_i \in [x_{i-1}, x_i])\Big),$$

where $\mathrm{Partition}(P, [a, b])$ abbreviates $a = x_0 < \cdots < x_n = b$ for $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$.

*Remark* (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is not a tagged partition of } [a, b] \\
&\iff \Big(x_0 \neq a \vee x_n \neq b \vee \exists j \in \{1, \ldots, n\} \ (x_{j-1} \geq x_j) \vee \exists i \in \{1, \ldots, n\} \ (t_i \notin [x_{i-1}, x_i])\Big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b]) \iff (x_0\neq a)\vee(x_n\neq b)\vee\Big(\exists j\ (x_{j-1}\geq x_j)\Big)\vee\Big(\exists i\ (t_i\notin[x_{i-1},x_i])\Big).$$

*Remark* (Failure modes)*.* A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

*Remark* (Failure mode decomposition).

$$\underbrace{x_0\neq a}_{\text{left endpoint defect}}\quad\vee\quad\underbrace{x_n\neq b}_{\text{right endpoint defect}}\quad\vee\ \underbrace{\exists j\ (x_{j-1}\geq x_j)}_{\text{ordering defect}}\ \vee\ \underbrace{\exists i\ (t_i\notin[x_{i-1},x_i])}_{\text{tag misplaced}}.$$

*Remark* (Interpretation)*.* A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

*Remark* (Dependencies).

Definition 2.8.1.

**Definition (Gauge)**

**Definition 2.8.3** (Gauge)**.** Let $a,b\in\mathbb{R}$ with $a<b$ and set $I=[a,b]$. A gauge on $I$ is a function $\delta: I\to(0,\infty)$.

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } a,b\in\mathbb{R} \text{ with } a<b,\ \text{put } I=[a,b],\ \text{and let } \delta: I\to\mathbb{R}.\\ &\delta \text{ is a gauge on } I \iff \forall x\in I\ (\delta(x)>0).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\mathrm{Gauge}(\delta,I) := \big(\delta: I\to\mathbb{R}\big)\wedge\forall x\in I\ (\delta(x)>0).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } a,b\in\mathbb{R} \text{ with } a<b,\ \text{put } I=[a,b],\ \text{and let } \delta: I\to\mathbb{R}.\\ &\delta \text{ fails to be a gauge on } I \iff \exists x\in I\ (\delta(x)\leq 0).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{Gauge}(\delta,I) \iff \neg\big(\delta: I\to\mathbb{R}\big)\vee\exists x\in I\ (\delta(x)\leq 0).$$

*Remark* (Failure modes)*.* A purported gauge fails either because it is not a function from $I$ to $\mathbb{R}$, or because it takes on a nonpositive value somewhere in $I$.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg\big(\delta : I \to \mathbb{R}\big)}_{\text{wrong function type}} \lor \underbrace{\exists x \in I \ (\delta(x) \le 0)}_{\text{nonpositive gauge value}}$$

*Remark* (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

*Remark* (Dependencies).

- Function
- Closed Interval
- Order on $\mathbb{R}$

**Definition ($\delta$-Fine Partition)**

**Definition 2.8.4** ($\delta$-Fine Partition)**.** Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is $\delta$-fine if and only if for every $i \in \{1, \dots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

*Remark* (Standard quantified statement).

Let $[a, b]$ be a closed interval, $\delta : [a, b] \to (0, \infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ a tagged partition of $[a, b]$. $\dot{\mathcal{P}}$ is $\delta$-fine $\iff \forall i \in \{1, \dots, n\} \ \forall x \in [x_{i-1}, x_i] \ (|x - t_i| < \delta(t_i))$.

*Remark* (Definition predicate reading).

$$\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) :\!= \forall i \in \{1, \dots, n\} \ \forall x \in [x_{i-1}, x_i] \ (|x - t_i| < \delta(t_i)).$$

*Remark* (Negated quantified statement).

$$\exists i \in \{1, \dots, n\} \ \exists x \in [x_{i-1}, x_i] \ \big(|x - t_i| \ge \delta(t_i)\big).$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \iff \exists i \in \{1, \dots, n\} \ \exists x \in [x_{i-1}, x_i] \ (|x - t_i| \ge \delta(t_i)).$$

*Remark* (Failure modes). A tagged partition fails to be $\delta$-fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius $\delta(t_i)$, so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists i \in \{1, \dots, n\} \ \exists x \in [x_{i-1}, x_i]}_{\text{choose offending subinterval and point}} \ \underbrace{(|x - t_i| \ge \delta(t_i))}_{\text{distance not controlled by } \delta} .$$

*Remark* (Interpretation). A $\delta$-fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

*Remark* (Dependencies).

Definition 2.8.3; Definition 2.8.2

**Lemma**

**Lemma 2.8.5** (Every Point Is Covered by a Tag)**.** *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is $\delta$-fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \ldots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } a < b,\ \delta : [a, b] \to (0, \infty),\ \text{and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \text{ be } \delta\text{-fine on } [a, b].$$
$$\forall x \in [a, b]\ \exists i \in \{1, \ldots, n\}\ (|x - t_i| < \delta(t_i)).$$

*Remark* (Interpretation). A $\delta$-fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

*Remark* (Dependencies).

Definition 2.8.2, Definition 2.8.4.

**Theorem (Cousin's Theorem)**

**Theorem 2.8.6** (Cousin's Theorem)**.** *Every gauge $\delta : [a, b] \to (0, \infty)$ on a closed interval $[a, b]$ admits a $\delta$-fine tagged partition of $[a, b]$. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a, b] \to (0, \infty).\\
&\exists m \in \mathbb{N}\ \exists x_0, \ldots, x_m\ \exists t_1, \ldots, t_m\\
&\qquad \big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j \in \{1, \ldots, m\}\ (t_j \in [x_{j-1}, x_j])\\
&\qquad \wedge \forall j \in \{1, \ldots, m\}\ [x_{j-1}, x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\big).
\end{aligned}$$

*Remark* (Predicate reading).

$$\forall \delta : [a, b] \to (0, \infty) \quad \exists \dot{\mathcal{P}} \Big( \mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \wedge \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \Big).$$

*Remark* (Negated quantified statement).

$$
\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a,b] \to (0,\infty).\\
&\neg\Big(\text{there exists a } \delta\text{-fine tagged partition of } [a,b]\Big)\\
&\iff \forall m \in \mathbb{N}\ \forall x_0, \ldots, x_m\ \forall t_1, \ldots, t_m\\
&\qquad \Big[\big(a = x_0 < x_1 < \cdots < x_m = b \land \forall j\ (t_j \in [x_{j-1}, x_j])\big)\\
&\qquad \Rightarrow \exists j \in \{1, \ldots, m\}\ [x_{j-1}, x_j] \nsubseteq (t_j - \delta(t_j), t_j + \delta(t_j))\Big].
\end{aligned}
$$

*Remark* (Negation predicate reading).

$$\exists \delta : [a,b] \to (0,\infty)\ \forall \dot{\mathcal{P}}\ \Big(\text{TaggedPartition}(\dot{\mathcal{P}}, [a,b]) \Rightarrow \neg\,\text{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).$$

*Remark* (Failure modes). Failure would mean that a positive gauge $\delta$ defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall \dot{\mathcal{P}}\ \ \text{TaggedPartition}(\dot{\mathcal{P}}, [a,b])}_{\text{all tagged partitions}} \quad \Rightarrow \quad \underbrace{\neg\,\text{DeltaFine}(\dot{\mathcal{P}}, \delta)}_{\text{some interval escapes its gauge ball}} \quad .$$

*Remark* (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

*Remark* (Dependencies).

Definition 2.8.3, Definition 2.8.2, Definition 2.8.4.

*Remark* (Gauge proofs of the four classical theorems). Each proof follows the same pattern: define a gauge $\delta$ from the continuity hypothesis, invoke Cousin's theorem for a $\delta$-fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

**Boundedness.** At each $t$, continuity gives $\delta(t)$ with $\text{Within}(x, t, \delta(t)) \Rightarrow |f(x) - f(t)| \le 1$. A $\delta$-fine partition with tags $t_1, \ldots, t_n$ gives $K = \max_i |f(t_i)|$; the Covering Lemma yields $|f(x)| \le 1 + K$ for all $x$.

**EVT.** Let $M = \sup f(I)$; suppose $f(x) < M$ everywhere. At each $t$, take $\delta(t)$ with $\text{Within}(x, t, \delta(t)) \Rightarrow f(x) < \frac{1}{2}(M + f(t))$. A $\delta$-fine partition produces $\widetilde{M} = \frac{1}{2}(M + \max_i f(t_i)) < M$ bounding $f$ everywhere — contradiction.

**Location of Roots.** Assume $f(t) \neq 0$ for all $t$. At each $t$, take $\delta(t)$ preserving $\operatorname{sgn}(f(t))$. A $\delta$-fine partition forces consistent sign from $f(a) < 0$ to $f(b) > 0$ — contradiction.

**Heine–Cantor.** Given $\varepsilon > 0$, at each $t$ let $\delta(t)$ satisfy $\text{Within}(x, t, 2\delta(t)) \Rightarrow |f(x) - f(t)| \le \varepsilon/2$. A $\delta$-fine partition yields $\delta_\varepsilon = \min_i \delta(t_i) > 0$. For $|x - u| < \delta_\varepsilon$: find $t_i$ covering $x$; then $|u - t_i| \le |u - x| + |x - t_i| < 2\delta(t_i)$; so $|f(x) - f(u)| \le \varepsilon$.

## Tagged Partitions and Mesh

**Definition (Mesh of a Partition)**

**Definition 2.8.7** (Mesh of a Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. A real number $m$ is the *mesh* (or *norm*) of $P$ if and only if

$$m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

The mesh of $P$ is denoted by $\|P\|$ and is the unique $m$ satisfying this equivalence.

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \dots, x_n\} \text{ be a partition of } [a, b].\\ &m \text{ is the mesh of } P \iff m = \max_{1 \le i \le n} (x_i - x_{i-1}).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{NormOfPartition}(P, m) :\equiv m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \dots, x_n\} \text{ be a partition of } [a, b].\\ &\neg\big(m \text{ is the mesh of } P\big) \iff \Big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\Big) \lor \Big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\Big).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\text{NormOfPartition}(P, m) \iff \Big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\Big) \lor \Big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\Big).$$

*Remark* (Failure modes). Failure occurs either when $m$ underestimates the largest subinterval of $P$ or when $m$ overshoots that largest subinterval.

*Remark* (Failure mode decomposition).

$$\neg\,\text{NormOfPartition}(P, m) = \underbrace{\Big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\Big)}_{\text{Underestimate}} \lor \underbrace{\Big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\Big)}_{\text{Overestimate}}.$$

*Remark* (Interpretation). The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

*Remark* (Dependencies).

Definition 2.8.1.

**Definition (Refinement)**

**Definition 2.8.8** (Refinement)**.** Let $[a, b] \subseteq \mathbb{R}$ and let $P$ and $Q$ be partitions of $[a, b]$. Then $Q$ is a refinement of $P$ if and only if every point of $P$ is also a point of $Q$; equivalently, $P \subseteq Q$.

*Remark* (Standard quantified statement)*.*

$$\forall x \ (x \in P \Rightarrow x \in Q).$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{Refinement}(Q, P) \coloneqq P \subseteq Q.$$

*Remark* (Negated quantified statement)*.*

$$\exists x \ (x \in P \wedge x \notin Q).$$

*Remark* (Negation predicate reading)*.*

$$\neg \, \mathrm{Refinement}(Q, P) \iff \exists x \in P \ (x \notin Q).$$

*Remark* (Failure modes)*.* Failure occurs precisely when some point listed in P fails to appear in Q, meaning Q omits at least one of the partition points it was supposed to inherit from P.

*Remark* (Failure mode decomposition)*.*

$$\underbrace{\exists x \in P \ (x \notin Q)}_{\text{point of } P \text{ omitted by } Q} .$$

*Remark* (Interpretation)*.* Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition $P \subseteq Q$ guarantees that every subinterval defined by $P$ is still represented when one passes to $Q$, allowing comparisons of sums or integrals computed on different meshes. Failure means that $Q$ discards at least one point of $P$, so information tied to that breakpoint would be lost when transitioning between partitions.

*Remark* (Dependencies)*.*

Definition 2.8.1.

**Definition (Common Refinement)**

**Definition 2.8.9** (Common Refinement)**.** Let $a < b$ and let $P$ and $Q$ be partitions of $[a, b]$. Their common refinement is the partition $R$ obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

*Remark* (Standard quantified statement)*.*

$$\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } P, Q \text{ be partitions of } [a, b].$$
$$\forall R \text{ partition of } [a, b] : \ R \text{ is the common refinement of } P \text{ and } Q \iff R = P \cup Q.$$

*Remark* (Definition predicate reading).

$$\mathrm{CommonRefinement}(R, P, Q) :\!= (R = P \cup Q).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
&a < b \wedge P, Q \subseteq [a, b] \wedge R \subseteq [a, b] \\
&\implies \neg\big(\forall x \in \mathbb{R}\ (x \in R \iff x \in P \cup Q)\big) \\
&\iff \big(\exists x \in R\ (x \notin P \cup Q)\big) \vee \big(\exists x \in P \cup Q\ (x \notin R)\big).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg\,\mathrm{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

*Remark* (Failure modes). The claim that a partition $R$ is the common refinement of $P$ and $Q$ fails exactly when $R$ either contains a node not present in $P \cup Q$ or omits a node that $P \cup Q$ contributes. These are the only two structural ways in which $R$ can differ from $P \cup Q$.

*Remark* (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\text{extra node}} \vee \underbrace{\exists x \in (P \cup Q) \setminus R}_{\text{missing node}}.$$

*Remark* (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in P or Q, and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

*Remark* (Dependencies).

Definition 2.8.1

# Chapter 3

# Differentiation



## 3.1 The Tangent Problem

**Secant and Tangent — Quick Reference**

| Concept | Meaning |
|---|---|
| Secant line | Line through two points of a graph |
| Tangent line | Limiting local line determined by the derivative |

*Remark* (Section guide). Before the derivative is a formula it is a question: what straight line best matches a curve at a single point? The route in is indirect: choose a second point, draw the computable secant line, and let that point slide toward the first. If the secants settle toward a limiting line, that line is the tangent and its slope is the derivative.

**Definition (Secant Line)**

**Definition 3.1.1** (Secant Line)**.** Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \tfrac{f(x_2)-f(x_1)}{x_2-x_1}).$$

*Remark* (Interpretation). With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y/\Delta x$ — the average rate of change of $f$ over the interval $[x_1, x_2]$. This is an exact algebraic identity; no limit is involved. As $x_2 \to x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as $x_2$ approaches $x_1$, and the limit position is the tangent line.

**Definition (Difference Quotient)**

**Definition 3.1.2** (Difference Quotient)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of $f$ from $c$ to $x$ is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

*Remark* (Interpretation)*.* The difference quotient is the slope of the secant line through $(c, f(c))$ and $(x, f(x))$. It is the quantity whose limiting value, when it exists as $x \to c$, becomes the derivative.

*Remark* (Dependencies)*.*

- Secant Line

**Definition (Tangent Line)**

**Definition 3.1.3** (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that $f$ is differentiable at $c$. The *tangent line* to the graph of $f$ at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

*Remark* (Standard quantified statement)*.*

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

*Remark* (Interpretation)*.* The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \to c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ — the best first-order approximation to $f$ near $c$. The error $f(x) - \ell(x) = o(x - c)$ as $x \to c$, meaning it vanishes faster than the displacement.

*Remark* (Dependencies)*.*

- Definition: Secant Line
- Definition: Derivative at a Point

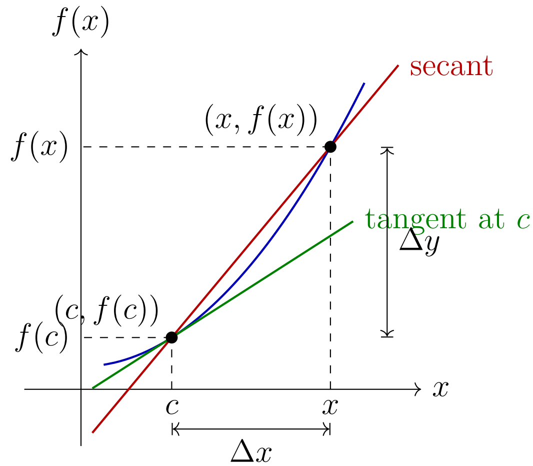


Figure 3.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y/\Delta x$. As $x \to c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

## 3.2 The Derivative

**Derivative Definition — Quick Reference**

| **Concept** | **Meaning** |
|---|---|
| Derivative at a point | Epsilon–delta limiting secant slope |
| Example: derivative of $x^2$ | The definition in a removable-singularity computation |
| Topological derivative | Neighbourhood formulation of the same limit |
| Sequential derivative | Sequence test for differentiability |
| Equivalent definitions | Epsilon–delta, topological, and sequential forms agree |
| $h$-form | Difference quotient written with $x = c + h$ |
| Differentiability implies continuity | Differentiability is stronger than continuity |
| Example: one corner | Continuous but not differentiable at one point |
| Example: sawtooth | Continuous with infinitely many isolated corners |
| Example: Weierstrass function | Continuous and nowhere differentiable |
| Uniqueness | The derivative value is single-valued |
| Identity derivative | Base case $D(x) = 1$ |
| Constant derivative | Base case $D(C) = 0$ |

*Remark* (Section guide)*.* The derivative is the number the secant slopes approach, but the word "limit" has three useful faces. The epsilon–delta form is the contract; the topological form strips it to neighbourhoods; the sequential form is the disproving tool, since one bad approach sequence is enough.

Definition (Derivative at a Point)

**Definition 3.2.1** (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \operatorname{int}(D)$. We say $f$ is *differentiable at* $c$ with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D\Big(0 < |x-c| < \delta \implies \left|\frac{f(x)-f(c)}{x-c} - L\right| < \varepsilon\Big).$$

*Remark* (Standard quantified statement)*.*

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D\Big(0 < |x-c| < \delta \implies \left|\frac{f(x)-f(c)}{x-c} - L\right| < \varepsilon\Big).$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{DerivativeAt}(f, c, L) \iff \mathrm{FunctionLimit}\left(\lambda x.\, \frac{f(x)-f(c)}{x-c},\ D,\ c,\ L\right).$$

The predicate $\mathrm{DerivativeAt}(f, c, L)$ asserts that $L$ is the value of the limit of the difference quotient of $f$ at $c$.

*Remark* (Negated quantified statement)*.*

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in D\Big(0 < |x-c| < \delta\ \wedge\ \left|\frac{f(x)-f(c)}{x-c} - L\right| \geq \varepsilon\Big).$$

*Remark* (Negation predicate reading)*.*

$$\neg\,\mathrm{DerivativeAt}(f, c, L) \iff \neg\,\mathrm{FunctionLimit}\left(\lambda x.\, \frac{f(x)-f(c)}{x-c},\ D,\ c,\ L\right).$$

*Remark* (Failure modes)*.*

- The difference quotient has no finite limit at $c$ (e.g. $|f(x) - f(c)|/|x-c| \to \infty$).
- The difference quotient oscillates without settling near any single value.
- The proposed value $L$ is the wrong candidate.

*Remark* (Interpretation)*.* The derivative is a limit of slopes of secant lines. The $\varepsilon$-$\delta$ form demands that the difference quotient can be made arbitrarily close to $L$ by taking $x$ sufficiently close to $c$. The input condition $0 < |x-c|$ is punctured: $x = c$ is excluded because $(f(c) - f(c))/(c-c)$ is $0/0$.

*Remark* (Dependencies)*.*

- Limit of a Function

*Example* (Derivative of $x^2$ from the Definition)*.* Let $f(x) = x^2$ and fix $c \in \mathbb{R}$. For $x \neq c$,

$$\frac{f(x)-f(c)}{x-c} = \frac{x^2-c^2}{x-c} = x + c.$$

The singularity has been removed by exact algebra, so the limit as $x \to c$ is $2c$. Hence $f'(c) = 2c$. Read through Carathéodory, this is the factor $\varphi(x) = x + c$, continuous at $c$ with $\varphi(c) = 2c$.

**Definition (Topological Definition of Derivative at a Point)**

**Definition 3.2.2** (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \operatorname{int}(D)$. $f$ is differentiable at $c$ with derivative $L$ if for every neighbourhood $V$ of $L$, there exists a neighbourhood $U$ of $c$ such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L\; \exists U \text{ neighbourhood of } c\; \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \forall V \ni L\; \exists U \ni c\; \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L\; \forall U \text{ neighbourhood of } c\; \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

*Remark* (Negation predicate reading).

$$\neg\, \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \exists V \ni L\; \forall U \ni c\; \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

*Remark* (Failure modes)*.* The topological derivative fails when a fixed output neighbourhood of $L$ cannot be entered by the difference quotient no matter how tightly the input is restricted around $c$. Geometric reading: no punctured neighbourhood around $c$ maps all secant slopes into any prescribed window around $L$.

*Remark* (Interpretation)*.* The topological form replaces $\varepsilon$-balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

*Remark* (Dependencies).

- Function
- Relative Neighborhood
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

**Definition (Sequential Definition of Derivative at a Point)**

**Definition 3.2.3** (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of $A$. We say that $f$ is differentiable at $c$ with derivative $L$ if

$$\forall (x_n) \subseteq A \setminus \{c\} \; \big(x_n \to c \Rightarrow \tfrac{f(x_n) - f(c)}{x_n - c} \to L\big).$$

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\Rightarrow\; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

*Remark* (Definition predicate reading).

$$\text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\Rightarrow\; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

*Remark* (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\wedge\; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

*Remark* (Negation predicate reading).

$$\neg\,\text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\wedge\; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

*Remark* (Failure modes). Failure occurs when a single approach sequence $(x_n) \to c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than $L$ (wrong slope candidate). A single such sequence certifies non-differentiability.

*Remark* (Failure mode decomposition).

$$\underbrace{x_n \to c}_{\text{approach condition}} \;\wedge\; \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\to L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

*Remark* (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \to c$ along which the difference quotient fails to converge to any single value.

*Remark* (Dependencies).

- Sequence
- Convergent Sequence
- Derivative at a Point
- Limit of a Function

**Theorem (Equivalence of Definitions of the Derivative at a Point)**

**Theorem 3.2.4** (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of $A$. The following are equivalent:*

1. *$f$ is differentiable at $c$ in the $\varepsilon$-$\delta$ sense with derivative $L$.*
2. *$f$ is differentiable at $c$ in the topological sense with derivative $L$.*
3. *$f$ is differentiable at $c$ in the sequential sense with derivative $L$.*

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\text{(i)} \iff \text{(ii)} \iff \text{(iii)}.$$

*Remark* (Predicate reading)*.*

$$\text{DerivativeAt}(f, c, L) \iff \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \text{DerivativeAt}_{\text{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of $f$ at $c$.

*Remark* (Negated quantified statement)*.*

$$\begin{aligned}\neg\text{(i)} &\iff \neg\text{(ii)} \iff \neg\text{(iii)} \\ &\iff \exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\wedge\; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.\end{aligned}$$

*Remark* (Failure modes)*.* The equivalence cannot produce contradictory verdicts: if $c$ is a limit point of $A$, all three forms agree. Degenerate case: if $c$ is not a limit point of $A$, the punctured-ball and sequence conditions are vacuously satisfied by every candidate $L$ simultaneously, making the equivalence trivially true for all $L$ rather than characterizing a unique derivative.

*Remark* (Interpretation)*.* The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

*Remark* (Dependencies)*.*

- Derivative at a Point
- Topological Definition
- Sequential Definition

**Proposition (Derivative: $h$-Form Equivalence)**

**Proposition 3.2.5** (Derivative: $h$-Form Equivalence)**.** *Let $f : A \to \mathbb{R}$ and $c \in \operatorname{int}(A)$. Then $f$ is differentiable at $c$ with derivative $L$ if and only if*

$$\lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = L.$$

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\text{DerivativeAt}(f, c, L) \iff \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = L.$$

*Remark* (Negated quantified statement)*.*

$$\neg\,\text{DerivativeAt}(f, c, L) \iff \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

*Remark* (Failure modes)*.* The $h$-form fails in two branches: (i) The incremental ratio $(f(c+h) - f(c))/h$ has no finite limit as $h \to 0$ (corner, cusp, or vertical tangent at $c$). (ii) The ratio converges to a finite value different from $L$ (proposed $L$ is the wrong slope).

*Remark* (Interpretation)*.* The $h$-form substitutes $x = c + h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

*Remark* (Dependencies)*.*

- Derivative at a Point

*Remark* (Discrete and continuous change)*.* The forward difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of the derivative. The $h$-form above is its continuous descendant: replace the fixed step 1 by a shrinking step $h$, divide by $h$, and pass to the limit.

The kinship matters, but so does the divergence. The discrete product rule has an unavoidable shift:

$$\Delta(fg)(n) = (\Delta f)(n)g(n) + f(n+1)(\Delta g)(n).$$

The continuous product rule loses this shift because $f(x+h) \to f(x)$ as $h \to 0$. Likewise, ordinary powers are not the clean basis for $\Delta$; falling factorials are:

$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$

Passing to the limit is therefore not just a sharpening of a quotient. It restores symmetries that the discrete operator must carry as shifts.

### 3.2.1 First Consequences

**Theorem (Differentiable Implies Continuous)**

**Theorem 3.2.6** (Differentiable Implies Continuous)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If $f$ is differentiable at $c$, then $f$ is continuous at $c$.*

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\text{DerivableAt}(f, c) \;\Rightarrow\; \text{ContinuousAt}(f, c).$$

*Remark* (Predicate reading)*.*

$$\text{DifferentiableAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

*Remark* (Failure modes)*.* The implication can only fail in the reverse direction: continuity at $c$ does not force differentiability at $c$.

*Remark* (Failure mode decomposition)*.* The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

*Remark* (Contrapositive quantified statement)*.*

$$\neg\text{ContinuousAt}(f, c) \;\Rightarrow\; \neg\text{DifferentiableAt}(f, c).$$

*Remark* (Contrapositive predicate reading)*.*

$$\neg\, \text{ContinuousAt}(f, c) \implies \neg\, \text{DifferentiableAt}(f, c).$$

*Remark* (Interpretation)*.* Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \to c$.

*Remark* (Dependencies)*.*

- Derivative at a Point
- Continuity at a Point

*Remark* (Converse failure)*.* The theorem runs one way only. Differentiability forces continuity, but continuity does not force differentiability. The negation of the converse is

$$\exists f\ \exists c:\ \text{ContinuousAt}(f, c) \wedge \neg\, \text{DifferentiableAt}(f, c).$$

The examples below escalate the failure: one corner, infinitely many isolated corners, and no differentiability anywhere.

*Example* (One Corner: $|x|$)*.* The function $f(x) = |x|$ is continuous on $\mathbb{R}$, but it is not differentiable at 0. Its one-sided derivatives are $-1$ and $+1$, so the two one-sided slopes do not agree. Thus a single corner is enough to separate continuity from differentiability.

*Example* (Finitely Many Corners: A Sawtooth). On a compact interval, a triangular sawtooth made from finitely many line segments can be continuous while failing to be differentiable at each corner. For example,

$$s(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 \le x \le 2, \end{cases}$$

is continuous on $[0, 2]$, but the left derivative at 1 is 1 and the right derivative is $-1$. Adding more teeth gives any finite number of corner failures without breaking continuity.

*Example* (The Weierstrass Function). For $0 < a < 1$ and an odd integer $b$ satisfying $ab > 1 + \frac{3}{2}\pi$, the Weierstrass function

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

is continuous on $\mathbb{R}$, because the series converges uniformly by the Weierstrass $M$-test. Classically, under the displayed condition, it is differentiable at no point. Thus continuity may hold everywhere while differentiability holds nowhere.

*Remark* (Proof status). The continuity of $W$ follows from uniform convergence once the function-series machinery is available. The nowhere-differentiability proof is a classical result of Weierstrass and is deferred to the function-series development; it is used here only as a counterexample to the converse of Differentiable Implies Continuous.

*Remark* (Forward flag). Brownian motion has sample paths that are almost surely continuous everywhere and differentiable nowhere. This is the stochastic analogue of the Weierstrass phenomenon and one reason ordinary calculus must be replaced by the Itô integral in stochastic analysis.

**Theorem (Uniqueness of the Derivative)**

**Theorem 3.2.7** (Uniqueness of the Derivative)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If $f$ is differentiable at $c$, its derivative is unique.*

*Go to proof. Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ \text{DerivativeAt}(f, c, L') \ \Rightarrow \ L = L'.$$

*Remark* (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ \text{DerivativeAt}(f, c, L') \implies L = L'.$$

*Remark* (Failure modes). The uniqueness proof assumes $L \neq L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies $\delta$ making the difference quotient within $\varepsilon$ of both $L$ and $L'$ simultaneously, forcing $|L - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster-point hypothesis on $c$ is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

*Remark* (Contrapositive quantified statement).

$$L \neq L' \;\Rightarrow\; \neg\big(\mathrm{DerivativeAt}(f, c, L) \;\wedge\; \mathrm{DerivativeAt}(f, c, L')\big).$$

Two distinct candidates cannot both be the derivative of $f$ at $c$.

*Remark* (Contrapositive predicate reading).

$$L \neq L' \implies \neg\,\mathrm{DerivativeAt}(f, c, L) \;\vee\; \neg\,\mathrm{DerivativeAt}(f, c, L').$$

*Remark* (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

*Remark* (Dependencies).

- Derivative at a Point

**Proposition (Derivative of the Identity)**

**Proposition 3.2.8** (Derivative of the Identity)**.** *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall c \in \mathbb{R} : \; \frac{d}{dx}[x]\Big|_{x=c} = 1.$$

*Remark* (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

*Remark* (Dependencies).

- Derivative at a Point
- Function

**Proposition (Derivative of a Constant)**

**Proposition 3.2.9** (Derivative of a Constant)**.** *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall k \in \mathbb{R}\, \forall c \in \mathbb{R} : \; \frac{d}{dx}[k]\Big|_{x=c} = 0.$$

*Remark* (Interpretation). A constant function has zero change across every secant line, so its instantaneous rate of change is zero everywhere. Together with the derivative of the identity, this is one of the base cases for the algebraic computation rules.

*Remark* (Dependencies).

- Derivative at a Point

## 3.3 One-Sided Derivatives

**One-Sided Derivatives — Quick Reference**

| Concept | Meaning |
|---|---|
| Left-hand derivative | Limiting slope approached from the left |
| Right-hand derivative | Limiting slope approached from the right |
| Two-sided criterion | Differentiability iff both one-sided derivatives agree |
| Example: $\lvert x\rvert$ at 0 | A continuous corner with unequal one-sided slopes |

*Remark* (Section guide). A corner is a place where a curve has a left answer and a right answer and refuses to choose. One-sided derivatives interrogate each side alone, and their agreement criterion is the fastest way to certify many failures of differentiability.

**Definition (Left-Hand Derivative)**

**Definition 3.3.1** (Left-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that $c$ is a left-hand limit point of $I$. We say that $f$ has a *left-hand derivative* at $c$ if the limit

$$\lim_{x\to c^-} \frac{f(x)-f(c)}{x-c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x\to c^-} \frac{f(x)-f(c)}{x-c}.$$

*Remark* (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \Big( -(\delta) < x-c < 0 \implies \left| \frac{f(x)-f(c)}{x-c} - L \right| < \varepsilon \Big).$$

In this case $f'_-(c) = L$.

*Remark* (Definition predicate reading).

$$\mathrm{LeftDerivativeAt}(f,c,L) \iff \mathrm{LeftLimit}\left( \lambda x.\ \frac{f(x)-f(c)}{x-c},\ c,\ L \right).$$

*Remark* (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \Big( -\delta < x-c < 0\ \wedge\ \left| \frac{f(x)-f(c)}{x-c} - L \right| \geq \varepsilon \Big).$$

*Remark* (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\,\mathrm{LeftDerivativeAt}(f,c,L).$$

No finite $L$ is the left-hand limit of the difference quotient.

*Remark* (Interpretation)*.* The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as $x$ approaches $c$ *strictly from the left*. It measures the instantaneous rate of change of $f$ at $c$ as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

*Remark* (Dependencies)*.*

- Definition: Derivative at a Point

**Definition (Right-Hand Derivative)**

**Definition 3.3.2** (Right-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that $c$ is a right-hand limit point of $I$. We say that $f$ has a *right-hand derivative* at $c$ if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

*Remark* (Standard quantified statement)*.*

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \left(0 < x - c < \delta \implies \left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon\right).$$

In this case $f'_+(c) = L$.

*Remark* (Definition predicate reading)*.*

$$\mathrm{RightDerivativeAt}(f, c, L) \iff \mathrm{RightLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ c,\ L\right).$$

*Remark* (Negated quantified statement)*.*

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \left(0 < x - c < \delta\ \wedge\ \left|\frac{f(x) - f(c)}{x - c} - L\right| \geq \varepsilon\right).$$

*Remark* (Negation predicate reading)*.*

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{RightDerivativeAt}(f, c, L).$$

*Remark* (Interpretation)*.* The right-hand derivative $f'_+(c)$ is the instantaneous rate of change seen from inputs larger than $c$. It is the endpoint derivative available at the left endpoint of a closed interval, and it is one half of the two-sided derivative test at interior points.

*Remark* (Dependencies)*.*

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

**Theorem**

**Theorem 3.3.3** (Differentiability and One-Sided Derivatives)**.** *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then $f$ is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \iff \Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for some } L \in \mathbb{R}.$$

*Remark* (Negated quantified statement).

$$\neg\,\mathrm{DifferentiableAt}(f, c) \iff \neg\Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

*Remark* (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show $f$ is not differentiable at $c$, compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so $f$ is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior $(a, b)$ in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

*Remark* (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

*Example* ($|x|$ Has No Derivative at 0). For $f(x) = |x|$ at $c = 0$, the one-sided difference quotients are

$$\frac{|x| - 0}{x - 0} = \frac{|x|}{x} = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Thus $f'_-(0) = -1$ and $f'_+(0) = 1$. The two values disagree, so the two-sided derivative does not exist at 0, even though $f$ is continuous there.

## 3.4 Carathéodory and the Chain Rule

**Carathéodory and Chain Rule — Quick Reference**

| Concept | Role |
|---|---|
| Carathéodory factorization | Continuous slope factor for differentiability |
| Carathéodory characterization | Differentiability iff the slope factor extends continuously |
| Chain Rule | Derivative of a composition |
| Leibniz Rule | $n$-th derivative of a product |
| Faa di Bruno | $n$-th derivative of a composition |
| Faa di Bruno at $n = 2$ | Coefficient check for the second derivative |
| Faa di Bruno at $n = 3$ | Coefficient check for the third derivative |

### 3.4.1 Carathéodory Characterization

*Remark* (Section guide). Carathéodory replaces division by factorisation: $f(x) - f(c) = (x - c)\varphi(x)$, with $\varphi$ continuous at $c$. Differentiability becomes continuity of the slope factor, and continuity composes. That is why the chain rule becomes clean rather than a fight with vanishing denominators.

**Definition (Carathéodory Factorization)**

**Definition 3.4.1** (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that $f$ admits a *Carathéodory factorization* at $c$ if there exists a function $\phi : I \to \mathbb{R}$, continuous at $c$, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, $f$ is differentiable at $c$ and $f'(c) = \phi(c)$.

*Remark* (Standard quantified statement).

$$\exists \phi : I \to \mathbb{R}\ \big(\mathrm{ContinuousAt}(\phi, c)\ \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).$$

*Remark* (Definition predicate reading). The existence of a Carathéodory factorization at $c$ is equivalent to $\mathrm{DifferentiableAt}(f, c)$ by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of $f$ at $c$ is precisely the statement that this piecewise-defined $\phi$ is continuous at $c$.

*Remark* (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at $c$ — with a function $\phi$ that is genuinely continuous at $c$. The factorization $f(x) - f(c) = (x - c)\phi(x)$ holds everywhere on $I$, including at $x = c$

where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

**Carathéodory vs. Lipschitz.** Both control the increment $f(x) - f(c)$ by the factor $(x - c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to $c$; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x)-f(c)| \leq L|x-c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if $\phi$ is continuous at $c$ it is bounded near $c$, so $|f(x)-f(c)| = |\phi(x)||x-c| \leq K|x-c|$ for some $K$ near $c$. The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values $\pm 1$ on either side of 0 with no continuous extension.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuous at a Point

**Theorem**

**Theorem 3.4.2** (Carathéodory Characterization of Differentiability)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

*(i) $f$ is differentiable at $c$.*

*(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at $c$, such that*

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

*In this case, $\phi(c) = f'(c)$.* *Go to proof. Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \text{DifferentiableAt}(f, c) \iff \exists \phi : I \to \mathbb{R} \, \big(&\text{ContinuousAt}(\phi, c) \\ &\wedge \ \forall x \in I \, [f(x) = f(c) + (x - c)\phi(x)]\big). \end{aligned}$$

In this case $\phi(c) = f'(c)$.

*Remark* (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define $\phi$ as the difference quotient extended by $f'(c)$ at $c$ and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from $\phi$ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of $f$ and $g$ yields one automatically.

*Remark* (Dependencies).

- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

**Theorem**

**Theorem 3.4.3** (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If $f$ is differentiable at $c$ and $g$ is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at $c$, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, f(c)) \implies (g \circ f)'(c) = g'(f(c))f'(c).$$

*Remark* (Interpretation). The derivative of a composition is the derivative of the outside function evaluated at the inside value, multiplied by the derivative of the inside function.

*Remark* (Interpretation). Let $\phi_f$ and $\phi_g$ be the Carathéodory slope factors for $f$ at $c$ and $g$ at $f(c)$. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x))(f(x) - f(c)) = \phi_g(f(x))\phi_f(x)(x - c).$$

The new slope factor is

$$\phi_{g\circ f}(x) = \phi_g(f(x))\phi_f(x),$$

which is continuous at $c$ and has value $g'(f(c))f'(c)$. This proof never divides by $f(x) - f(c)$, so it avoids the failure in the informal $\Delta g/\Delta f \cdot \Delta f/\Delta x$ argument when $f(x) = f(c)$ near $c$.

*Remark* (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

**Theorem (Leibniz Rule)**

**Theorem 3.4.4** (Leibniz Rule). *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \to \mathbb{R}$ are $n$-times differentiable on a neighbourhood of $x$. Then $fg$ is $n$-times differentiable at $x$, and*

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x)g^{(n-k)}(x).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{Diff}^n(f,x) \wedge \mathrm{Diff}^n(g,x) \Longrightarrow (fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

*Remark* (Interpretation). The binomial pattern is not an accident. Each of the $n$ differentiations lands on either $f$ or $g$, and $\binom{n}{k}$ counts the ways to send exactly $k$ differentiations to $f$. At $n = 1$, this is the ordinary product rule.

*Remark* (Dependencies).

- Theorem: Product Rule
- Definition: Higher Derivatives

**Theorem (Faa di Bruno's Formula)**

**Theorem 3.4.5** (Faa di Bruno's Formula). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \to J$ is $n$-times differentiable on a neighbourhood of $x$, and suppose $f : J \to \mathbb{R}$ is $n$-times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is $n$-times differentiable at $x$, and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^{n} m k_m = n} \frac{n!}{\prod_{m=1}^{n} k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^{n} \left(g^{(m)}(x)\right)^{k_m},$$

*where $k_m \geq 0$ and $k = \sum_{m=1}^{n} k_m$. Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{Diff}^n(g,x) \wedge \mathrm{Diff}^n(f,g(x)) \Longrightarrow (f \circ g)^{(n)}(x) = \sum_{\sum m k_m = n} \frac{n!}{\prod_m k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_m \left(g^{(m)}(x)\right)^{k_m}.$$

*Remark* (Interpretation). The chain rule gives the first derivative of a composition. Repeated differentiation is deeper: product-rule terms and chain-rule terms interlace, and the bookkeeping is governed by integer partitions. Faa di Bruno's formula is the all-orders chain rule.

*Remark* (Hypotheses). Smoothness is more than this formula needs. The $n$-th order statement requires $n$ derivatives of $g$ near $x$ and $n$ derivatives of $f$ near $g(x)$. At $n = 1$, the formula is exactly the chain rule.

*Remark* (Structural reading). The tuple $(k_1, \ldots, k_n)$ records an integer partition of $n$: $k_m$ is the number of parts of size $m$. The coefficient

$$\frac{n!}{\prod_m k_m!\,(m!)^{k_m}}$$

counts set partitions with that block-size profile. The outer derivative $f^{(k)}$ is differentiated once per block, while each block of size $m$ contributes one factor $g^{(m)}$.

*Remark* (Normalization). The formula uses the bare-product normalization: the coefficient contains $(m!)^{k_m}$, and the product contains $\big(g^{(m)}(x)\big)^{k_m}$. Equivalently, one may put $g^{(m)}(x)/m!$ inside the product and remove $(m!)^{k_m}$ from the denominator. Mixing these two conventions gives the wrong coefficients.

*Remark* (Cross-volume callback). The constraint $\sum_m mk_m = n$ is precisely the multiplicity form of an integer partition of $n$. Thus the number of summands is $p(n)$, the partition function from the discrete and number-system material, now appearing as bookkeeping for iterated differentiation.

*Remark* (Scope). This theorem is enrichment, not a critical dependency for the integration, measure, or manifold arcs. It is included because it is the complete all-orders chain rule and a useful callback to partitions.

*Remark* (Dependencies).

- Theorem: Chain Rule
- Theorem: Leibniz Rule
- Definition: Higher Derivatives
- Definition: Class $C^k$

*Example* (Faa di Bruno at $n = 2$). The partitions of 2 are represented by $(k_1, k_2) = (2, 0)$ and $(0, 1)$. The first gives $f''(g)(g')^2$, and the second gives $f'(g)g''$. Hence

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g''.$$

The coefficient on $f'(g)g''$ is 1, which is the quickest check that the normalization has not been mixed.

*Example* (Faa di Bruno at $n = 3$). The partitions of 3 produce the three terms

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The middle coefficient 3 counts the three ways to choose which differentiation contributes the second-derivative block of $g$.

## 3.5 The Shape Questions

**Shape Questions — Quick Reference**

| Concept | Meaning |
|---|---|
| Increasing at a point | Local right-up, left-down behavior |
| Decreasing at a point | Local right-down, left-up behavior |
| Relative minimum | Locally smallest function value |
| Relative maximum | Locally largest function value |
| Relative extremum | Relative minimum or maximum |
| Convex function | Graph lies below its chords |
| Concave function | Graph lies above its chords |
| Inflection point | Local change in bending behavior |

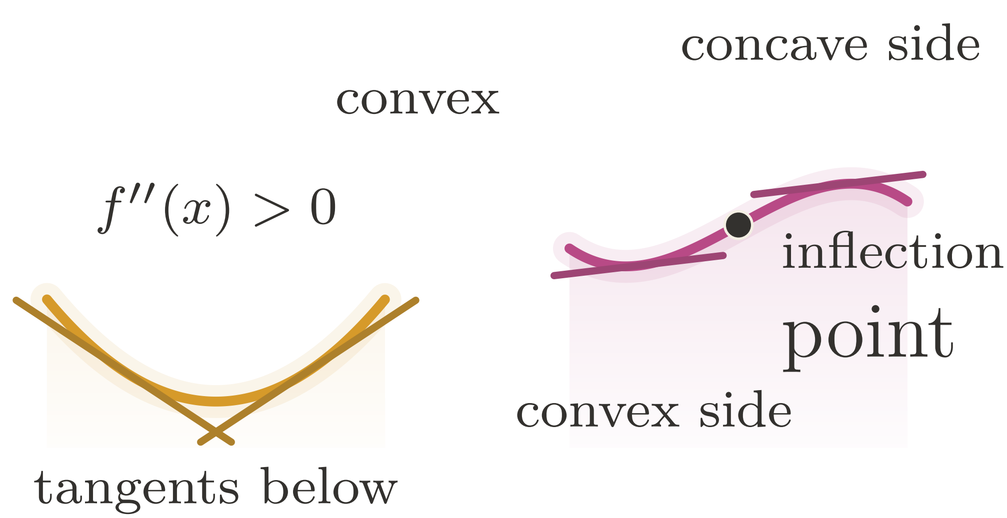


Figure 3.2: Convexity, concavity, and the change of bending at an inflection point.

**Definition (Convex Function)**

**Definition 3.5.1** (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is *convex* on $I$ if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

*Remark* (Standard quantified statement)*.*

$$\forall x, y \in I \ \forall \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

*Remark* (Interpretation)*.* Convexity is defined by chords, not derivatives. The graph lies below the line segment joining any two of its points. Derivative tests are later tools for detecting this geometric condition, not the definition of the condition.

*Remark* (Dependencies).

- Closed Interval

**Definition (Concave Function)**

**Definition 3.5.2** (Concave Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is *concave* on $I$ if $-f$ is convex on $I$.

*Remark* (Standard quantified statement).

$$\mathrm{ConcaveOn}(f, I) \iff \mathrm{ConvexOn}(-f, I).$$

*Remark* (Interpretation). Concavity is the downward-bending counterpart of convexity. Equivalently, the graph of a concave function lies above each chord joining two of its points.

*Remark* (Dependencies).

- Convex Function

**Definition (Inflection Point)**

**Definition 3.5.3** (Inflection Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be an interior point. The point $c$ is an *inflection point* of $f$ if $f$ changes convexity at $c$: there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that $f$ is convex on one side of $c$ and concave on the other side.

*Remark* (Standard quantified statement).

$$\mathrm{InflectionPoint}(f, c) \iff f \text{ changes convexity at } c.$$

*Remark* (Interpretation). An inflection point is a change in the direction of bending. It is not merely a point where $f''(c) = 0$; that equation is only a necessary condition under additional hypotheses.

*Remark* (Dependencies).

- Convex Function
- Concave Function

### 3.5.1 Interior Extrema

**Interior Extrema — Quick Reference**

| Statement | Role |
|---|---|
| Relative minimum | Local lower comparison point |
| Relative maximum | Local upper comparison point |
| Relative extremum | Umbrella term for local optima |
| Interior Extremum Theorem | Interior differentiable extremum has zero derivative |
| Necessary condition | Extrema occur at critical behavior |
| Critical-point corollary | Derivative zero or derivative undefined |

**Definition (Relative Minimum)**

**Definition 3.5.4** (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that $f$ has a *relative minimum* at $c$ if there exists $\delta > 0$ such that

$$f(c) \le f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then $f$ has a *strict relative minimum* at $c$.

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \; \big(|x-c| < \delta \implies f(c) \le f(x)\big).$$

*Remark* (Definition predicate reading).

$$\mathrm{LocalMinimum}(f, c) \iff \exists \delta > 0 \; \forall x \in A \; \big(\mathrm{Within}(x, c, \delta) \implies f(c) \le f(x)\big).$$

*Remark* (Interpretation). $f$ has a relative minimum at $c$ if $f(c)$ is the smallest value of $f$ in some open neighbourhood of $c$ within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of $f$ globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

*Remark* (Dependencies).

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

**Definition (Relative Maximum)**

**Definition 3.5.5** (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that $f$ has a *relative maximum* at $c$ if there exists $\delta > 0$ such that

$$f(x) \le f(c) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then $f$ has a *strict relative maximum* at $c$.

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \; \big(|x - c| < \delta \implies f(x) \le f(c)\big).$$

*Remark* (Definition predicate reading).

$$\mathrm{LocalMaximum}(f, c) \iff \exists \delta > 0 \; \forall x \in A \; \big(\mathrm{Within}(x, c, \delta) \implies f(x) \le f(c)\big).$$

*Remark* (Interpretation). A relative maximum is a point whose function value is at least as large as all nearby function values in the domain.

*Remark* (Dependencies).

- Definition: Relative Minimum

**Definition (Relative Extremum)**

**Definition 3.5.6** (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that $f$ has a *relative extremum* at $c$ if $f$ has either a relative minimum at $c$ or a relative maximum at $c$.

*Remark* (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

*Remark* (Definition predicate reading).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

*Remark* (Interpretation). Relative extrema are the local turning candidates of a graph: nearby values lie all on one side of $f(c)$.

*Remark* (Dependencies).

- Definition: Relative Minimum
- Definition: Relative Maximum

**Theorem**

**Theorem 3.5.7** (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If $f$ has a relative extremum at $c$ and $f$ is differentiable at $c$, then*

$$f'(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \land \mathrm{DifferentiableAt}(f, c) \Rightarrow \mathrm{DerivativeAt}(f, c, 0).$$

*Remark* (Contrapositive quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\mathrm{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then $c$ is not a relative extremum.

*Remark* (Contrapositive predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\mathrm{LocalMinimum}(f, c) \ \wedge \ \neg\,\mathrm{LocalMaximum}(f, c).$$

*Remark* (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points — where $f'(c) = 0$ or $f'(c)$ does not exist — are candidates for extrema, not guarantees.

*Remark* (Dependencies).

- Definition: Relative Extremum
- Definition: Derivative at a Point
- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

**Theorem**

**Theorem 3.5.8** (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If $f$ has a relative extremum at $c$, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \ \Rightarrow \ \neg\,\mathrm{DifferentiableAt}(f, c) \ \vee \ \mathrm{DerivativeAt}(f, c, 0).$$

*Remark* (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at $c$, or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where $f'$ is zero and points where $f'$ does not exist.

*Remark* (Dependencies).

- Theorem: Interior Extremum Theorem

**Corollary**

**Corollary 3.5.9** (Necessary Condition for an Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If $f$ has a relative extremum at $c$, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\mathrm{LocalExtremum}(f, c) \;\Rightarrow\; \mathrm{DerivativeAt}(f, c, 0) \;\vee\; \neg\,\mathrm{DifferentiableAt}(f, c).$$

*Remark* (Interpretation)*.* An extremum can occur only at a critical point: either the derivative vanishes or the derivative fails to exist.

*Remark* (Dependencies)*.*

- Theorem: Necessary Condition for an Interior Extremum

## 3.6 The Mean Value Theorems

**Mean Value Theorems — Quick Reference**

| **Statement** | **Role** |
|---|---|
| Rolle's Theorem | Equal endpoint values force a horizontal tangent |
| Mean Value Theorem | Some tangent slope equals the secant slope |
| Cauchy Mean Value Theorem | Compares two functions' rates of change |
| Derivative bound implies Lipschitz | Turns a slope bound into a global modulus |

*Remark* (Section guide)*.* Rolle's theorem says that a differentiable curve returning to the same height must level off somewhere between. The Mean Value Theorem tilts that picture: between any two points, some interior tangent runs parallel to the chord. Cauchy's version compares two functions at once and later powers L'Hôpital's rule.

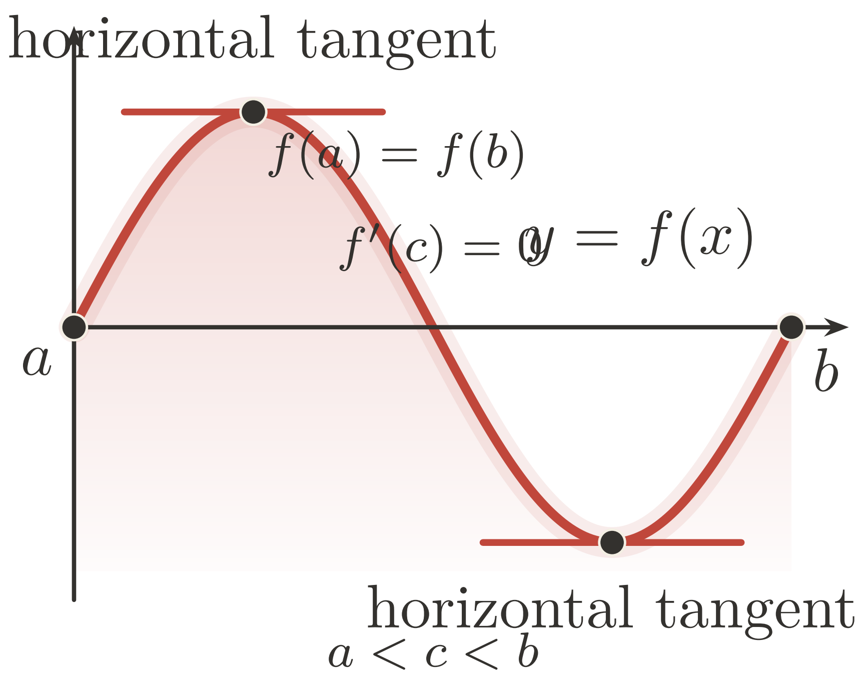


Figure 3.3: Rolle's Theorem: equal endpoint heights force a horizontal tangent.

**Theorem (Rolle's Theorem)**

**Theorem 3.6.1** (Rolle's Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on $(a, b)$, and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b) \ \text{DifferentiableAt}(f, x) \ \wedge \ f(a) = f(b) \\ &\Rightarrow \ \exists c \in (a, b) \ \text{DerivativeAt}(f, c, 0).\end{aligned}$$

*Remark* (Predicate reading).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \text{DifferentiableOn}(f, (a, b)) \ \wedge \ f(a) = f(b) \\ &\implies \exists c \in (a, b) \ : \ \text{DerivativeAt}(f, c, 0).\end{aligned}$$

*Remark* (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \text{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but $f$ has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.

- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and $f$ is continuous, but $f$ has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.

- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then $f$ is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

*Remark* (Contrapositive quantified statement).

$\big(\forall c \in (a, b) \;:\; f'(c) \neq 0\big) \;\Rightarrow\; f$ not continuous on $[a, b] \vee f$ not differentiable on $(a, b) \vee f(a) \neq f(b)$. $\blacksquare$

*Remark* (Contrapositive predicate reading).

$$\begin{aligned}&\big(\forall c \in (a, b) \;:\; \neg\,\mathrm{DerivativeAt}(f, c, 0)\big) \\ &\quad \Longrightarrow \neg\,\mathrm{ContinuousOn}(f, [a, b]) \;\vee\; \neg\,\mathrm{DifferentiableOn}(f, (a, b)) \;\vee\; f(a) \neq f(b).\end{aligned}$$

*Remark* (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then $f$ is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

*Remark* (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

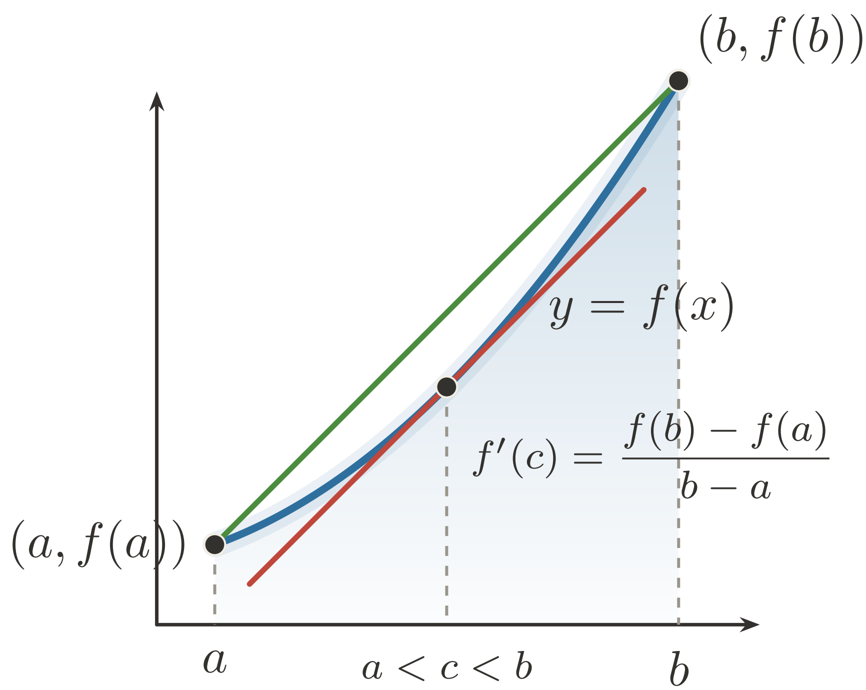


Figure 3.4: Mean Value Theorem: an interior tangent matches the secant slope.

**Theorem (Mean Value Theorem)**

**Theorem 3.6.2** (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on $(a, b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\begin{aligned} &\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b) \ \mathrm{DifferentiableAt}(f, x) \\ &\quad \Rightarrow \ \exists c \in (a, b) \ \mathrm{DerivativeAt}\Big(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\Big). \end{aligned}$$

*Remark* (Predicate reading)*.*

$$\begin{aligned} &\mathrm{ContinuousOn}(f, [a, b]) \ \wedge \ \mathrm{DifferentiableOn}(f, (a, b)) \\ &\quad \Longrightarrow \ \exists c \in (a, b) \ : \ \mathrm{DerivativeAt}\Big(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\Big). \end{aligned}$$

*Remark* (Failure modes)*.* Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then $f$ has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then $f$ is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere $f$ is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some $c$ exists but gives no formula for it.

*Remark* (Contrapositive quantified statement)*.*

$$\left(\forall c \in (a, b) \ : \ f'(c) \neq \frac{f(b) - f(a)}{b - a}\right) \ \Rightarrow \ f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

*Remark* (Contrapositive predicate reading)*.*

$$\begin{aligned} &\Big(\forall c \in (a, b) \ : \ \neg\,\mathrm{DerivativeAt}\Big(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\Big)\Big) \\ &\quad \Longrightarrow \ \neg\,\mathrm{ContinuousOn}(f, [a, b]) \ \vee \ \neg\,\mathrm{DifferentiableOn}(f, (a, b)). \end{aligned}$$

*Remark* (Interpretation)*.* The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point $c$ is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which satisfies

$g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some $c$ exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

*Remark* (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

**Theorem (Cauchy Mean Value Theorem)**

**Theorem 3.6.3** (Cauchy Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on $(a, b)$. Then there exists $c \in (a, b)$ such that*

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

*If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{ContinuousOn}(g, [a, b]) \wedge \mathrm{DifferentiableOn}(f, (a, b)) \wedge \mathrm{DifferentiableOn}(g, (a, b))$$
$$\implies \exists c \in (a, b)\ \big[(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)\big].$$

*Remark* (Predicate reading)*.* CauchyMVT$(f, g, [a, b])$ asserts the existence of an interior point where the velocity vector $(g'(c), f'(c))$ is parallel to the chord vector $(g(b) - g(a), f(b) - f(a))$. Taking $g(x) = x$ recovers the ordinary Mean Value Theorem.

*Remark* (Negated quantified statement).

$$\forall c \in (a, b) :\ \big(f(b) - f(a)\big)g'(c) \neq \big(g(b) - g(a)\big)f'(c).$$

*Remark* (Failure modes)*.* The continuity and differentiability hypotheses fail for the same reasons they fail in the ordinary MVT: jumps break endpoint-to-interior control, and corners leave no derivative to match. The quotient form has the additional nonvanishing hypothesis $g' \neq 0$ on $(a, b)$; without it the cross-multiplied conclusion may still hold, but the displayed quotient can be undefined.

*Remark* (Contrapositive quantified statement).

$$\big(\forall c \in (a, b) :\ (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c)\big)$$
$$\implies \neg\,\mathrm{ContinuousOn}(f, [a, b]) \vee \neg\,\mathrm{ContinuousOn}(g, [a, b]) \vee \neg\,\mathrm{DifferentiableOn}(f, (a, b)) \vee \neg\,\mathrm{Differentia}$$

*Remark* (Contrapositive predicate reading)*.* If no interior point makes the derivative pair $(g'(c), f'(c))$ parallel to the chord pair $(g(b) - g(a), f(b) - f(a))$, then at least one of the four regularity hypotheses fails: continuity of $f$ or $g$ on $[a, b]$, or differentiability of $f$ or $g$ on $(a, b)$.

*Remark* (Interpretation). This is the parametric Mean Value Theorem. For the planar curve $t \mapsto (g(t), f(t))$, some interior tangent is parallel to the chord joining its endpoints. It compares two rates of change rather than comparing one function to the horizontal axis. This is the form that later underlies L'Hopital-type ratio arguments and the Cauchy form of Taylor's remainder.

*Remark* (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point
- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

**Corollary (Derivative Bound Implies Lipschitz)**

**Corollary 3.6.4** (Derivative Bound Implies Lipschitz)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on $I$. If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \qquad \text{for every } x \in I,$$

*then $f$ is Lipschitz on $I$ with constant $M$:*

$$|f(x) - f(y)| \leq M|x - y| \qquad \text{for every } x, y \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\big(\exists M \geq 0\ \forall x \in I :\ |f'(x)| \leq M\big) \implies \forall x, y \in I :\ |f(x) - f(y)| \leq M|x - y|.$$

*Remark* (Predicate reading).

$$\mathrm{BoundedDerivativeOn}(f, I, M) \implies \mathrm{Lipschitz}(f, I, M).$$

*Remark* (Failure modes). The derivative bound must hold throughout the interval. A function such as $f(x) = \sqrt{x}$ on $[0, 1]$ is continuous and uniformly continuous, but its derivative is unbounded near 0, so no global Lipschitz constant follows from this test. The interval hypothesis is also essential: across a disconnected gap, the MVT cannot compare the two components.

*Remark* (Interpretation). This corollary converts a pointwise slope bound into a global modulus of continuity. It is the precise bridge from derivative estimates to the Lipschitz condition: controlling the size of $f'$ controls every secant slope on the interval.

*Remark* (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Lipschitz Condition
- Definition: Derivative at a Point

## 3.7 Reading the Graph from $f'$ and $f''$

### 3.7.1 First-Order Shape

**First-Order Shape — Quick Reference**

| Statement | Role |
|---|---|
| Zero derivative | Vanishing derivative forces constant behavior |
| Equal derivatives | Functions differ by a constant |
| Nonnegative derivative | Detects nondecreasing behavior |
| Nonpositive derivative | Detects nonincreasing behavior |
| Positive derivative | Local increasing behavior |
| Negative derivative | Local decreasing behavior |
| First derivative test: max | Sign change detects a local maximum |
| First derivative test: min | Sign change detects a local minimum |

*Remark* (Section guide). The Mean Value Theorem is now available, so the harvest can begin. Vanishing derivatives force constancy, derivative signs force monotonicity, and second derivatives detect bending when enough regularity is present. Darboux and the smoothness hierarchy show that derivatives carry hidden rigidity even when they are not continuous.

**Definition (Increasing at a Point)**

**Definition 3.7.1** (Increasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that $f$ is *increasing at* $c$ if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

*Remark* (Standard quantified statement).

$$\exists \delta > 0\, \forall x \in A \cap (c - \delta, c)\, \forall y \in A \cap (c, c + \delta)\, \big(f(x) \leq f(c) \leq f(y)\big).$$

*Remark* (Interpretation). $f$ is increasing at $c$ if $c$ lies between the values of $f$ on either side in some punctured neighbourhood: $f$ is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about $f$ outside the $\delta$-neighbourhood of $c$. A function can be increasing at $c$ without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of $c$, distinguishing this from the notion of a strict local increase.

*Remark* (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

## Definition (Decreasing at a Point)

**Definition 3.7.2** (Decreasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that $f$ is *decreasing at* $c$ if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \cap (c - \delta, c) \; \forall y \in A \cap (c, c + \delta) \; \big(f(x) \geq f(c) \geq f(y)\big).$$

*Remark* (Interpretation). The function is decreasing at $c$ when nearby values to the left sit above $f(c)$ and nearby values to the right sit below it.

*Remark* (Dependencies).

- Definition: Increasing at a Point

## Theorem (Zero Derivative Implies Constant)

**Theorem 3.7.3** (Zero Derivative Implies Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on $I$ and differentiable on $(a, b)$. If*

$$f'(x) = 0 \qquad \text{for all } x \in (a, b),$$

*then $f$ is constant on $I$. Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in (a, b) \; \mathrm{DerivativeAt}(f, x, 0) \; \Rightarrow \; \exists C \in \mathbb{R} \; \forall x \in I \; (f(x) = C).$$

*Remark* (Contrapositive quantified statement).

$$\neg\big(\exists C \in \mathbb{R} \; \forall x \in I \; (f(x) = C)\big) \; \Rightarrow \; \exists x \in (a, b) \; \neg\,\mathrm{DerivativeAt}(f, x, 0).$$

If $f$ is not constant on $I$, then $f'$ is not identically zero on $(a, b)$.

*Remark* (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f, I) \; \Rightarrow \; \exists x \in (a, b) \; (f'(x) \neq 0).$$

*Remark* (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on $(a, b)$, then for any two points $x_1 < x_2$ in $I$, the MVT supplies a $c$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, $f$ is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

*Remark* (Dependencies).

- Theorem: Mean Value Theorem

**Corollary**

**Corollary 3.7.4** (Equal Derivatives Imply Difference Is Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on $I$ and differentiable on $(a, b)$. If*

$$f'(x) = g'(x) \qquad \text{for all } x \in (a, b),$$

*then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in (a, b)\ (f'(x) = g'(x)) \ \Rightarrow\ \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = g(x) + C).$$

*Remark* (Contrapositive quantified statement).

$$\forall C \in \mathbb{R}\ \exists x \in I\ (f(x) \neq g(x) + C) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then $f'$ and $g'$ disagree somewhere.

*Remark* (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f - g,\ I) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

*Remark* (Interpretation)*.* This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to $h = f - g$, whose derivative is $(f - g)' = f' - g' = 0$.

*Remark* (Dependencies).

- Theorem: Zero Derivative Implies Constant

**Theorem (Nondecreasing Iff Nonnegative Derivative)**

**Theorem 3.7.5** (Nondecreasing Iff Nonnegative Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on $I$. Then $f$ is nondecreasing on $I$ if and only if*

$$f'(x) \geq 0 \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{NondecreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \geq 0).$$

*Remark* (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) < 0)\ \Rightarrow\ \neg\,\mathrm{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

*Remark* (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) < 0)\ \Rightarrow\ \neg\,\mathrm{NondecreasingOn}(f, I).$$

*Remark* (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction ($f$ nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when $f$ is nondecreasing, so the limit is nonnegative. The reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.

Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

*Remark* (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Derivative at a Point

**Theorem**

**Theorem 3.7.6** (Nonincreasing Iff Nonpositive Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on $I$. Then $f$ is nonincreasing on $I$ if and only if*

$$f'(x) \leq 0 \qquad \textit{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{NonincreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \leq 0).$$

*Remark* (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) > 0)\ \Rightarrow\ \neg\,\mathrm{NonincreasingOn}(f, I).$$

*Remark* (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) > 0)\ \Rightarrow\ \neg\,\mathrm{NonincreasingOn}(f, I).$$

*Remark* (Interpretation). Nonpositive derivative everywhere is exactly the infinitesimal signature of a function that never increases.

*Remark* (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

**Lemma**

**Lemma 3.7.7** (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that $f$ is differentiable at $c$. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

*and*

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \wedge L > 0 \;\Rightarrow\; \exists \delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) < f(c)\big) \wedge \big(c < x < c{+}\delta \Rightarrow f(x) >$$

*Remark* (Contrapositive quantified statement).

$$\forall \delta > 0\, \exists x \in I\, \big((c{-}\delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c{+}\delta \wedge f(x) \leq f(c))\big) \;\Rightarrow\; f'(c) \leq 0.$$

*Remark* (Contrapositive predicate reading).

$$\neg\,\mathrm{LocallyIncreasingAt}(f, c) \;\Rightarrow\; f'(c) \leq 0.$$

*Remark* (Interpretation)*.* If the instantaneous rate of change is strictly positive at $c$, the function is genuinely climbing through $c$: it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct $\varepsilon$-$\delta$ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies $\delta$ such that the difference quotient is within $\varepsilon$ of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

*Remark* (Dependencies).

- Definition: Derivative at a Point

**Lemma**

**Lemma 3.7.8** (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that $f$ is differentiable at $c$. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

*and*

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DerivativeAt}(f,c,L) \land L<0 \;\Rightarrow\; \exists \delta>0\, \forall x \in I\; \big(c-\delta<x<c \Rightarrow f(x)>f(c)\big) \land \big(c<x<c+\delta \Rightarrow f(x) <$$

*Remark* (Contrapositive quantified statement).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

*Remark* (Contrapositive predicate reading).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

*Remark* (Interpretation). If the derivative at $c$ is strictly negative, nearby values move downward as the input passes through $c$ from left to right.

*Remark* (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

**Theorem (First Derivative Test: Relative Maximum)**

**Theorem 3.7.9** (First Derivative Test: Relative Maximum). *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on $I$, and let $c \in (a,b)$. Suppose $f$ is differentiable on $(a,c)$ and $(c,b)$. If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \leq 0 \quad \text{for all } x \in (c, c+\delta),$$

*then $f$ has a relative maximum at $c$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \left[(c-\delta, c+\delta) \subseteq I \;\land\; \forall x \in (c-\delta,c)\,(f'(x) \geq 0) \;\land\; \forall x \in (c,c+\delta)\,(f'(x) \leq 0)\right] \\ \Rightarrow\; \text{LocalMaximum}(f,c).$$

*Remark* (Interpretation). If the function is nondecreasing immediately to the left of $c$ and nonincreasing immediately to the right, then $c$ is a local maximum — the function climbs to $c$ and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c-\delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c+\delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on $f'$ replaces the Interior Extremum Theorem's converse, which does not hold in general; this is a sufficient condition, not a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

*Remark* (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Theorem

**Theorem 3.7.10** (First Derivative Test: Relative Minimum). *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on $I$, and let $c \in (a,b)$. Suppose $f$ is differentiable on $(a,c)$ and $(c,b)$. If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \le 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \ge 0 \quad \text{for all } x \in (c, c+\delta),$$

*then $f$ has a relative minimum at $c$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \left[(c-\delta, c+\delta) \subseteq I \ \wedge\ \forall x \in (c-\delta, c)\ (f'(x) \le 0) \ \wedge\ \forall x \in (c, c+\delta)\ (f'(x) \ge 0)\right] \\ \Rightarrow\ \mathrm{LocalMinimum}(f, c).$$

*Remark* (Interpretation). If the derivative changes from nonpositive to nonnegative at $c$, the graph falls into $c$ and then rises away from it, so $c$ is a local minimum.

*Remark* (Dependencies).

- Theorem: First Derivative Test: Relative Maximum
- Definition: Relative Minimum

## 3.7.2 Second-Order Shape

Second-Order Shape — Quick Reference

| **Concept** | **Role** |
| --- | --- |
| Second derivative | Derivative of the derivative |
| Higher derivatives | Iterated derivatives $f^{(n)}$ |
| Convexity test | $f'' \ge 0$ detects convexity |
| Concavity test | $f'' \le 0$ detects concavity |
| Second derivative test | Classifies critical points |
| Inflection condition | Continuous $f''$ changes sign through zero |

## 3.7.3 Higher-Order Derivatives

Definition (Second Derivative)

**Definition 3.7.11** (Second Derivative). Let $f$ be differentiable on a neighbourhood of $c$. If $f'$ is differentiable at $c$, the *second derivative* of $f$ at $c$ is

$$f''(c) := (f')'(c).$$

*Remark* (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f', c, f''(c)) \quad \Longleftrightarrow \quad f''(c) = (f')'(c).$$

*Remark* (Interpretation). The first derivative measures instantaneous slope. The second derivative measures how that slope itself changes.

*Remark* (Dependencies).

- Derivative at a Point

**Definition (Higher Derivatives)**

**Definition 3.7.12** (Higher Derivatives). Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at $c$, define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

*Remark* (Standard quantified statement).

$$\forall n \geq 2 : \quad f^{(n)}(c) = (f^{(n-1)})'(c)$$

whenever the derivative on the right exists.

*Remark* (Interpretation). Higher derivatives are obtained by iterating the derivative operator. Their existence is not automatic: each stage requires differentiability of the previous derivative.

*Remark* (Dependencies).

- Second Derivative

### 3.7.4 Second-Order Shape

**Theorem (Second-Derivative Convexity Test)**

**Theorem 3.7.13** (Second-Derivative Convexity Test). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on $I$. If*

$$f''(x) \geq 0 \qquad \text{for all } x \in I,$$

*then $f$ is convex on $I$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\big(\forall x \in I,\ f''(x) \geq 0\big) \Longrightarrow \mathrm{ConvexOn}(f, I).$$

*Remark* (Interpretation). Nonnegative second derivative means the slopes do not decrease. This is the derivative-based signature of convex bending.

*Remark* (Dependencies).

- Second Derivative
- Convex Function

**Theorem (Second-Derivative Concavity Test)**

**Theorem 3.7.14** (Second-Derivative Concavity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on $I$. If*

$$f''(x) \le 0 \qquad \textit{for all } x \in I,$$

*then $f$ is concave on $I$. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\big(\forall x \in I,\ f''(x) \le 0\big) \implies \mathrm{ConcaveOn}(f, I).$$

*Remark* (Interpretation)*.* Nonpositive second derivative means the slopes do not increase. This is the derivative-based signature of concave bending.

*Remark* (Interpretation)*.* The second-derivative tests are narrower than the chord definitions of convexity and concavity. The function $|x|$ is convex, but it is not twice differentiable at 0.

*Remark* (Dependencies)*.*

- Second Derivative
- Concave Function

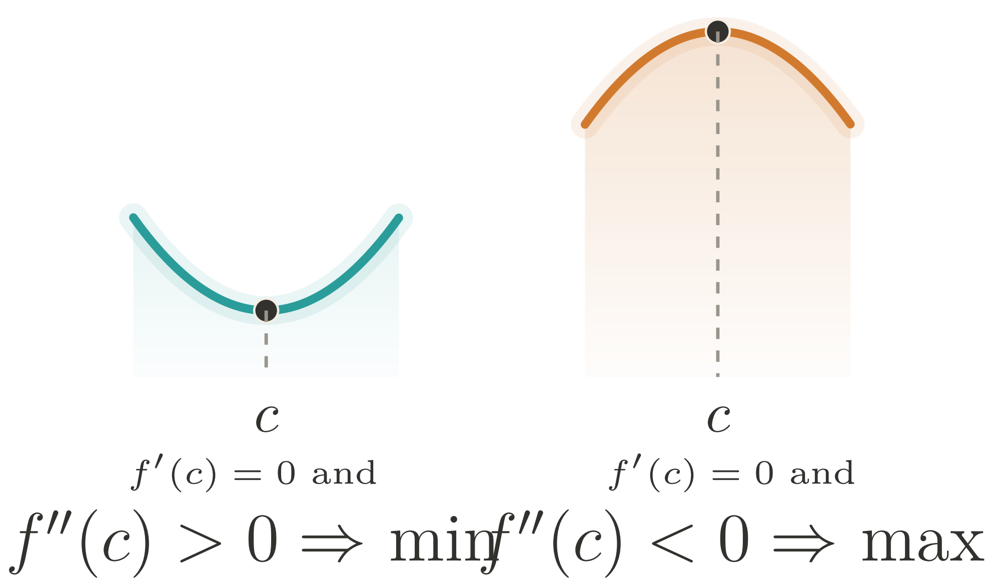


Figure 3.5: Second derivative test: the sign of $f''(c)$ decides the local shape at a critical point.

**Theorem (Second Derivative Test)**

**Theorem 3.7.15** (Second Derivative Test)**.** *Let $f$ be twice differentiable in a neighbourhood of $c$, and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then $f$ has a strict relative minimum at $c$.*
- *If $f''(c) < 0$, then $f$ has a strict relative maximum at $c$.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow \text{StrictRelativeMinimum}(f, c),$$

and

$$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow \text{StrictRelativeMaximum}(f, c).$$

*Remark* (Interpretation). At a critical point, positive second derivative means the graph bends upward; negative second derivative means it bends downward.

*Remark* (Dependencies).

- Second Derivative
- Relative Minimum
- Relative Maximum

**Proposition (Inflection Point Necessary Condition)**

**Proposition 3.7.16** (Inflection Point Necessary Condition)**.** *Let $f$ have an inflection point at $c$. If $f''$ exists on a neighbourhood of $c$ and is continuous at $c$, then*

$$f''(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \wedge \text{ContinuousAt}(f'', c) \implies f''(c) = 0.$$

*Remark* (Interpretation). An inflection point is a change in bending. When the second derivative is continuous, such a change can pass through $c$ only by passing through zero. The converse can fail: $f''(c) = 0$ does not by itself guarantee an inflection point.

*Remark* (Dependencies).

- Inflection Point
- Second Derivative
- Continuity at a Point

### 3.7.5 Darboux's Theorem

**Darboux — Quick Reference**

| Statement | Role |
|---|---|
| Darboux's Theorem | Derivatives have the intermediate value property |

**Theorem (Darboux's Theorem)**

**Theorem 3.7.17** (Darboux's Theorem)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on $I$. If $k$ is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

*Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\forall k \ \big(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \ \Rightarrow \ \exists c \in (a, b) \ (f'(c) = k)\big).$$

*Remark* (Interpretation)*.* The Intermediate Value Theorem says continuous functions have the intermediate value property. Darboux's theorem says derivatives also have that property even when the derivative is not continuous. Thus a derivative cannot have a jump discontinuity. The proof uses differentiability of $f$ to construct an auxiliary function $g(x) = f(x) - kx$ and then applies the Extreme Value Theorem and the Interior Extremum Theorem.

*Remark* (Dependencies)*.*

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

### 3.7.6 Smoothness Classes

**Smoothness Classes — Quick Reference**

| Concept | Meaning |
|---|---|
| Class $C^1$ | Differentiable with continuous first derivative |
| Class $C^k$ | $k$ derivatives with continuous top derivative |
| Class $C^\infty$ | Continuous derivatives of every order |
| Class $C^\omega$ | Equal to local Taylor series |
| Smoothness tower | Strict hierarchy $C^\omega \subsetneq C^\infty \subsetneq \cdots$ |
| Class $C^{1,1}$ | $C^1$ with Lipschitz first derivative |
| Placement of $C^{1,1}$ | Refines the gap between $C^1$ and $C^2$ |
| Bounded $f''$ | Bounded second derivative gives $C^{1,1}$ |
| Example: differentiable not $C^1$ | Oscillatory derivative discontinuity at 0 |

**Definition (Continuously Differentiable; Class $C^1$)**

**Definition 3.7.18** (Continuously Differentiable; Class $C^1$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is *continuously differentiable* on $I$, written $f \in C^1(I)$, if $f$ is differentiable at every point of $I$ and the derivative $f' : I \to \mathbb{R}$ is continuous on $I$.

*Remark* (Standard quantified statement).

$$f \in C^1(I) \iff \big(\forall x \in I : \text{DifferentiableAt}(f, x)\big) \wedge \text{ContinuousOn}(f', I).$$

*Remark* (Interpretation). The condition $f \in C^1(I)$ is strictly stronger than differentiability. A function may be differentiable everywhere while its derivative fails to be continuous; Darboux's Theorem still constrains that failure, because derivatives cannot jump. Thus the gap between differentiability and $C^1$ is oscillatory rather than jump-like.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuity at a Point

*Example* (Differentiable Everywhere, but Not $C^1$). Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At the origin,

$$f'(0) = \lim_{x \to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \to 0} x \sin(1/x) = 0$$

by squeezing. For $x \neq 0$,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to 0, but $\cos(1/x)$ oscillates, so $f'$ is not continuous at 0. Thus $f$ is differentiable on all of $\mathbb{R}$, but $f \notin C^1(\mathbb{R})$. The failure is oscillatory, not a jump, as Darboux's theorem requires.

**Definition (Class $C^k$)**

**Definition 3.7.19** (Class $C^k$)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \to \mathbb{R}$. The function $f$ is of *class* $C^k$ on $I$, written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \ldots, f^{(k)}$ exist on $I$ and $f^{(k)}$ is continuous on $I$. By convention, $C^0(I)$ is the class of continuous functions on $I$.

*Remark* (Standard quantified statement).

$$f \in C^k(I) \iff \big(\forall j \in \{0, 1, \ldots, k\} : f^{(j)} \text{ exists on } I\big) \wedge \text{ContinuousOn}\big(f^{(k)}, I\big).$$

*Remark* (Predicate reading).

$$\mathrm{ClassC}(f, I, k).$$

For a fixed interval $I$, the classes are nested:

$$f \in C^{k+1}(I) \Longrightarrow f \in C^k(I).$$

*Remark* (Negated quantified statement).

$$f \notin C^k(I) \iff \exists x \in I : \ \big[f^{(k)} \text{ fails to exist at } x\big] \vee \big[f^{(k)} \text{ exists on } I \wedge \neg\, \mathrm{ContinuousAt}(f^{(k)}, x)\big].$$

*Remark* (Negation predicate reading). $\neg\,\mathrm{ClassC}(f, I, k)$: somewhere the $k$-th derivative either does not exist or exists but is not continuous. A single point where the top derivative is discontinuous certifies $f \notin C^k(I)$.

*Remark* (Interpretation). The notation $C^k(I)$ records $k$ layers of differentiability together with continuity at the top layer. This is the regularity scale used to state how many times a theorem may differentiate a function without leaving the controlled setting.

*Remark* (Dependencies).

- Definition: Higher Derivatives
- Definition: Continuously Differentiable; Class $C^1$

**Definition (Smooth Function; Class $C^\infty$)**

**Definition 3.7.20** (Smooth Function; Class $C^\infty$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is *smooth*, or of *class* $C^\infty$, written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

*Remark* (Standard quantified statement).

$$f \in C^\infty(I) \iff \forall k \in \mathbb{N} : \ f \in C^k(I).$$

*Remark* (Negated quantified statement).

$$f \notin C^\infty(I) \iff \exists k \in \mathbb{N} : \ f \notin C^k(I).$$

A single order at which differentiability or continuity breaks demotes the function out of $C^\infty$.

*Remark* (Interpretation). Smoothness is the regularity language imported by differential geometry. Smooth manifolds are assembled from $C^\infty$ transition maps, and the tangent-map construction seeded by the differential uses this vocabulary.

*Remark* (Dependencies).

- Definition: Class $C^k$

**Definition (Real-Analytic Function; Class $C^\omega$)**

**Definition 3.7.21** (Real-Analytic Function; Class $C^\omega$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is *real-analytic*, or of *class* $C^\omega$, written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \qquad \text{for every } x \in (a-r, a+r) \cap I.$$

*Remark* (Standard quantified statement)*.*

$$f \in C^\omega(I) \iff \forall a \in I\ \exists r > 0\ \forall x \in (a-r, a+r) \cap I : \ f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

*Remark* (Interpretation)*.* Analytic regularity means that the Taylor series does more than approximate the function: near each point, the series equals the function. The infinite Taylor series and its convergence are developed in Taylor's construction and error analysis, which supplies the series machinery behind this definition.

*Remark* (Dependencies)*.*

- Definition: Smooth Function; Class $C^\infty$
- Definition: Taylor Polynomial at a Point

**Theorem (The Smoothness Tower)**

**Theorem 3.7.22** (The Smoothness Tower)**.** *For every nondegenerate interval $I$, the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

*hold, and each displayed inclusion is strict. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\forall k \in \mathbb{N} : \ C^{k+1}(I) \subsetneq C^k(I), \qquad C^\omega(I) \subsetneq C^\infty(I).$$

*Remark* (Failure modes)*.* The inclusions do not fail; only their strictness needs witnesses.

- $C^k \setminus C^{k+1}$: $\varphi_k(x) = x^k|x|$, whose $k$-th derivative is $(k+1)!\,|x|$, continuous but not differentiable at 0.
- $C^\infty \setminus C^\omega$: the flat function $e^{-1/x^2}$, smooth with all derivatives zero at 0, so its Maclaurin series is zero while the function is not.

These separators prove every displayed containment is a genuine drop.

*Remark* (Interpretation). The inclusions are definitional; strictness is witnessed by examples. On any nondegenerate interval, translated and rescaled copies of $x \mapsto x^k|x|$ separate $C^k$ from $C^{k+1}$. Translated copies of the flat function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

separate $C^\infty$ from $C^\omega$: all derivatives at the flat point vanish, so the Taylor series is identically zero, while the function is positive on one side of that point.

*Remark* (Dependencies).

- Definition: Class $C^k$
- Definition: Smooth Function; Class $C^\infty$
- Definition: Real-Analytic Function; Class $C^\omega$
- Theorem: Darboux's Theorem

*Remark* (Section guide). Between $C^1$ and $C^2$ there is a regularity floor worth naming. A $C^1$ function has a continuous derivative; a $C^2$ function has a derivative that is itself differentiable. Many functions used in analysis live in the gap: their first derivative is continuous and controlled, but a classical second derivative may fail to exist at a few points. The control is a Lipschitz bound on $f'$, and the class is written $C^{1,1}$.

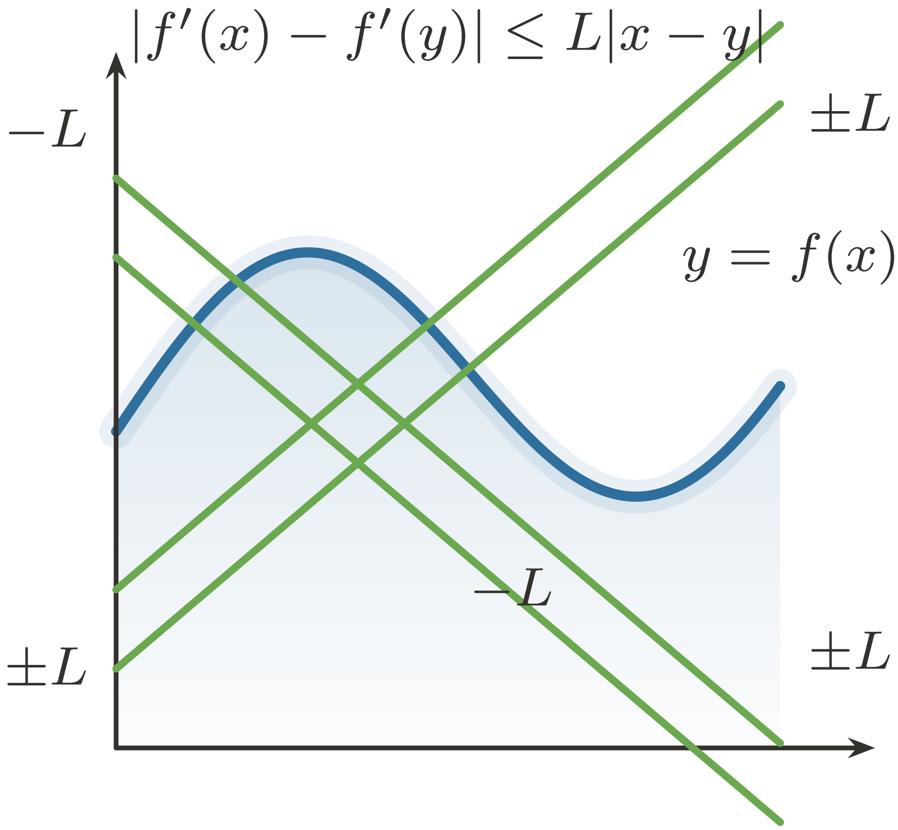


Figure 3.6: A Lipschitz derivative changes at a uniformly bounded rate.

**Definition (Lipschitz-Derivative Class $C^{1,1}$)**

**Definition 3.7.23** (Lipschitz-Derivative Class $C^{1,1}$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function $f$ is of *class* $C^{1,1}$ on $I$, written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and $f'$ is Lipschitz on $I$: there exists $L \ge 0$ such that

$$|f'(x) - f'(y)| \le L|x-y| \qquad \text{for all } x, y \in I.$$

*Remark* (Standard quantified statement)*.*

$$f \in C^{1,1}(I) \iff f \in C^1(I) \wedge \exists L \ge 0\ \forall x, y \in I :\ |f'(x) - f'(y)| \le L|x-y|.$$

*Remark* (Definition predicate reading)*.*

$$\mathrm{ClassC11}(f, I) \iff \mathrm{ClassC}(f, I, 1) \wedge \exists L \ge 0 : \mathrm{Lipschitz}(f', I, L).$$

The smallest admissible $L$ is the Lipschitz constant of $f'$:

$$\sup_{x \ne y} \frac{|f'(x) - f'(y)|}{|x-y|}.$$

*Remark* (Negated quantified statement)*.*

$$f \notin C^{1,1}(I) \iff f \notin C^1(I) \vee \big(\forall L \ge 0\ \exists x, y \in I :\ |f'(x) - f'(y)| > L|x-y|\big).$$

*Remark* (Interpretation). $C^{1,1}$ says that $f'$ behaves as if $|f''| \leq L$, without requiring $f''$ to exist everywhere. It is the exact regularity tracked by many first-order estimates: the derivative is not merely continuous, but changes at a bounded rate.

*Remark* (Dependencies).

- Definition: Continuously Differentiable; Class $C^1$
- Corollary: Derivative Bound Implies Lipschitz

**Theorem (Placement of $C^{1,1}$)**

**Theorem 3.7.24** (Placement of $C^{1,1}$). *For every nondegenerate interval $I$,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

*and $C^{1,1}(I)$ is not contained in $C^2(I)$. If $K$ is a compact interval, then*

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

*and the first inclusion is strict.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\big(f \in C^{1,1}(I) \Rightarrow f \in C^1(I)\big), \qquad \big(K \text{ compact} \wedge f \in C^2(K) \Rightarrow f \in C^{1,1}(K)\big),$$

with the displayed containments strict.

*Remark* (Failure modes). The global inclusion $C^2(I) \subseteq C^{1,1}(I)$ is false on arbitrary intervals: $f(x) = x^3$ is $C^2(\mathbb{R})$, but $f'(x) = 3x^2$ is not Lipschitz on $\mathbb{R}$. Strictness is witnessed by two elementary functions.

- $C^{1,1} \setminus C^2$: $f(x) = x|x|$ has $f'(x) = 2|x|$, which is Lipschitz, but $f''(0)$ does not exist.
- $C^1 \setminus C^{1,1}$: $f(x) = |x|^{4/3}$ has continuous first derivative, but $f'$ has unbounded difference quotients at 0.

*Remark* (Interpretation). $C^{1,1}$ is a genuine intermediate regularity condition, but it is not the same as $C^2$. A bounded second derivative forces the class; a Lipschitz first derivative may have corners; and a continuous first derivative may still vary too sharply to be Lipschitz.

*Remark* (Dependencies).

- Definition: Lipschitz-Derivative Class $C^{1,1}$
- Theorem: The Smoothness Tower
- Corollary: Derivative Bound Implies Lipschitz

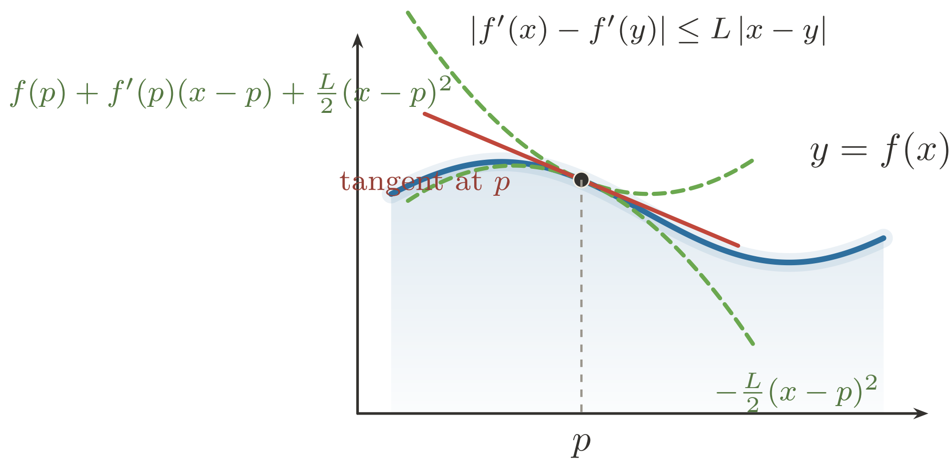


Figure 3.7: $C^{1,1}$ control gives quadratic upper and lower bounds around a tangent line.

**Corollary (Bounded Second Derivative Implies $C^{1,1}$)**

**Corollary 3.7.25** (Bounded Second Derivative Implies $C^{1,1}$)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \le M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and $f'$ is Lipschitz with constant $M$.* *Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\big(f \in C^2(I) \land \forall x \in I :\ |f''(x)| \le M\big) \Rightarrow f \in C^{1,1}(I) \land \mathrm{Lipschitz}(f', I, M).$$

*Remark* (Interpretation)*.* This is Derivative Bound Implies Lipschitz applied one level up, to $f'$ in place of $f$. The converse requires almost-everywhere language and is therefore deferred.

*Remark* (Dependencies)*.*

- Corollary: Derivative Bound Implies Lipschitz
- Definition: Class $C^k$
- Definition: Lipschitz-Derivative Class $C^{1,1}$

*Remark* (Forward flag)*.* A Lipschitz derivative is differentiable at most points, but making that sentence precise requires measure zero. In one variable this passes through bounded variation; in several variables it becomes Rademacher's theorem. Those results belong to the integration and measure-theoretic development, not to this chapter.

## 3.8 The Algebra of Derivatives

**Algebra of Derivatives — Quick Reference**

| Rule | Role |
|---|---|
| Constant multiple | Scalars pass through differentiation |
| Sum | Derivative distributes over addition |
| Product | Differentiates pointwise products |
| Quotient | Differentiates quotients where denominator is nonzero |
| Finite sum | Iterates the sum rule over finitely many terms |
| Extended product | Iterates the product rule over finitely many factors |
| Integer power | Differentiates powers built from products |
| Finite linear combination | Packages linearity in finite form |
| Inverse Function Theorem | Produces a $C^1$ local inverse |
| Inverse derivative | Gives the reciprocal derivative formula |
| L'Hôpital 0/0 | Evaluates indeterminate zero ratios |
| L'Hôpital $\infty/\infty$ | Evaluates indeterminate divergent ratios |

### 3.8.1 From Characterization to Computation

*Remark* (Section guide). With Carathéodory in hand, computing derivatives becomes bookkeeping with continuous slope factors. The basic rules record one structural fact: differentiation is a linear operator that also obeys a product law.

*Remark* (Interpretation). Carathéodory's characterization turns differentiability into continuity of a slope factor. If

$$f(x) - f(c) = (x - c)\phi_f(x), \qquad g(x) - g(c) = (x - c)\phi_g(x),$$

with $\phi_f$ and $\phi_g$ continuous at $c$, then the standard algebraic rules for derivatives follow from the algebra of continuous functions. The recurring move is no longer to manipulate a punctured quotient directly, but to construct a continuous factor whose value at $c$ is the derivative.

*Remark* (Dependencies).

- Carathéodory Characterization of Differentiability
- Algebra of Limits

**Theorem (Constant Multiple Rule)**

**Theorem 3.8.1** (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of $A$, and let $\alpha \in \mathbb{R}$. If $f$ is differentiable at $c$, then $\alpha f$ is differentiable at $c$, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \text{DerivativeAt}(\alpha f,\, c,\, \alpha f'(c)).$$

*Remark* (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar $\alpha$ factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Constant Multiple Rule for Limits

**Theorem (Sum Rule)**

**Theorem 3.8.2** (Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of $A$. If $f$ and $g$ are differentiable at $c$, then $f + g$ is differentiable at $c$, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(f + g,\, c,\, f'(c) + g'(c)).$$

*Remark* (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Sum Rule for Limits

**Theorem (Product Rule)**

**Theorem 3.8.3** (Product Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of $A$. If $f$ and $g$ are differentiable at $c$, then $fg$ is differentiable at $c$, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(fg,\ c,\ f'(c)g(c) + f(c)g'(c)).$$

*Remark* (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of $f$ weighted by the current value of $g$, plus the rate of change of $g$ weighted by the current value of $f$. The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of $f$ and $g$, isolating the individual difference quotients. Continuity of $f$ at $c$ (which follows from differentiability) is used to evaluate $\lim_{x \to c} f(x) = f(c)$ in one of the factors.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits

**Theorem (Quotient Rule)**

**Theorem 3.8.4** (Quotient Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of $A$. Suppose that $f$ and $g$ are differentiable at $c$ and that $g(c) \neq 0$. Then $f/g$ is differentiable at $c$, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \wedge g(c) \neq 0 \Rightarrow \text{DerivativeAt}\left(\tfrac{f}{g},\ c,\ \tfrac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}\right).$$

*Remark* (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near $c$ — continuity of $g$ (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of $c$, so the difference quotient of $f/g$ is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of $f$ and $g$ in the numerator, exactly as in the product rule proof.

*Remark* (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits
- Quotient Rule for Limits

### 3.8.2 Closure

Corollary

**Corollary 3.8.5** (Finite Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of $A$. If each $f_i$ is differentiable at $c$, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at $c$, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \in \{1, \dots, n\}\ \mathrm{DifferentiableAt}(f_i, c) \ \Rightarrow\ \mathrm{DerivativeAt}\big(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\big).$$

*Remark* (Interpretation). Finite sums and scalar weights do not introduce a new kind of differentiation rule; they package repeated use of the Sum Rule and Constant Multiple Rule.

*Remark* (Dependencies).

- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary

**Corollary 3.8.6** (Extended Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of $A$. If each $f_i$ is differentiable at $c$, then $\prod_{i=1}^{n} f_i$ is differentiable at $c$, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i\ \mathrm{DifferentiableAt}(f_i, c) \ \Rightarrow\ \mathrm{DerivativeAt}\Big(\textstyle\prod_i f_i,\ c,\ \sum_k f_k'(c) \prod_{i \neq k} f_i(c)\Big).$$

*Remark* (Interpretation). In a finite product, each term takes one turn being differentiated while all other factors are held at their values.

*Remark* (Dependencies).

- Theorem: Product Rule

Corollary

**Corollary 3.8.7** (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of $A$, and let $n \in \mathbb{N}$. If $f$ is differentiable at $c$, then $f^n$ is differentiable at $c$, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \ \wedge \ n \in \mathbb{N} \ \Rightarrow \ \text{DerivativeAt}(f^n,\, c,\, n(f(c))^{n-1} f'(c)).$$

*Remark* (Interpretation). The integer power rule is the extended product rule applied to $n$ identical factors.

*Remark* (Dependencies).

- Corollary: Extended Product Rule

Theorem

**Theorem 3.8.8** (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of $A$. If each $f_i$ is differentiable at $c$, then*

$$\left( \sum_{i=1}^{n} \alpha_i f_i \right)' (c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \ \text{DifferentiableAt}(f_i, c) \ \Rightarrow \ \text{DerivativeAt}\left(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\right).$$

*Remark* (Interpretation). The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at $c$, then $D_c : \mathcal{D}(I) \to \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

*Remark* (Dependencies).

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Constant Multiple Rule
- Theorem: Sum Rule

### 3.8.3 Global Forms on an Interval

**Theorem**

**Theorem 3.8.9** (Constant Multiple Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $f$ is differentiable on $I$, then $\alpha f$ is differentiable on $I$, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \ \text{DifferentiableAt}(f, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\alpha f,\ x,\ \alpha f'(x)).$$

*Remark* (Interpretation)*.* The pointwise constant-multiple rule becomes an interval statement by applying it at every point of the interval.

*Remark* (Dependencies).

- Theorem: Constant Multiple Rule

**Theorem**

**Theorem 3.8.10** (Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If $f$ and $g$ are differentiable on $I$, then $f + g$ is differentiable on $I$, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \ \big(\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x)\big) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(f+g,\ x,\ f'(x)+g'(x)).$$

*Remark* (Interpretation)*.* The derivative of a sum is computed point by point across the whole interval.

*Remark* (Dependencies).

- Theorem: Sum Rule

**Theorem**

**Theorem 3.8.11** (Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If $f$ and $g$ are differentiable on $I$, then $fg$ is differentiable on $I$, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \ \big(\text{DifferentiableAt}(f,x) \wedge \text{DifferentiableAt}(g,x)\big) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(fg,\ x,\ f'(x)g(x) + f(x)g'($$

*Remark* (Interpretation). The interval product rule records the product rule as an identity of derivative functions.

*Remark* (Dependencies).

- Theorem: Product Rule

**Theorem**

**Theorem 3.8.12** (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that $f$ and $g$ are differentiable on $I$ and that $g(x) \neq 0$ for all $x \in I$. Then $f/g$ is differentiable on $I$, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \textit{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \ \big(\text{DifferentiableAt}(f,x) \wedge \text{DifferentiableAt}(g,x) \wedge g(x) \neq 0\big) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}\Big(\tfrac{f}{g},\ x,\ \tfrac{f'(x)g(x)-f}{(g(x))}$$

*Remark* (Interpretation). The quotient rule on an interval is the pointwise quotient rule with the nonvanishing denominator hypothesis required at every point.

*Remark* (Dependencies).

- Theorem: Quotient Rule

**Corollary**

**Corollary 3.8.13** (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each $f_i$ is differentiable on $I$, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on $I$, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \qquad \textit{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \ \forall x \in I \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)).$$

*Remark* (Interpretation). Finite linear combinations commute with differentiation as functions on the whole interval.

*Remark* (Dependencies).

- Corollary: Finite Sum Rule

**Corollary**

**Corollary 3.8.14** (Extended Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \ldots, f_n : I \to \mathbb{R}$. If each $f_i$ is differentiable on $I$, then $\prod_{i=1}^{n} f_i$ is differentiable on $I$, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \mathrm{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \mathrm{DerivativeAt}\Big(\textstyle\prod_i f_i,\ x,\ \sum_k f_k'(x) \prod_{i \neq k} f_i(x)\Big).$$

*Remark* (Interpretation). The product rule for many factors holds uniformly point by point on the interval.

*Remark* (Dependencies).

- Corollary: Extended Product Rule

**Corollary**

**Corollary 3.8.15** (Power Rule for Integer Powers of a Function on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If $f$ is differentiable on $I$, then $f^n$ is differentiable on $I$, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I\ \ \mathrm{DifferentiableAt}(f, x)\ \wedge\ n \in \mathbb{N}\ \Rightarrow\ \forall x \in I\ \ \mathrm{DerivativeAt}(f^n,\ x,\ n(f(x))^{n-1} f'(x)).$$

*Remark* (Interpretation). The interval power rule says that the pointwise integer-power formula assembles into an equality of derivative functions.

*Remark* (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

**Theorem**

**Theorem 3.8.16** (Finite Linear Combination Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each $f_i$ is differentiable on $I$, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i\ \forall x \in I\ \text{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x))\,.$$

*Remark* (Interpretation). The pointwise statements above — each proved at a single point $c$ — extend to global statements when the hypotheses hold at every point of $I$. If $f$ is differentiable on $I$, then $f' : I \to \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on $I$:

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on $I$. The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \to \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on $I$. The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

*Remark* (Dependencies).

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Finite Linear Combination Rule

When $f$ is strictly monotone and differentiable on $I$ with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \to I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on $J$. This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

### 3.8.4 Inverting the Operation

**Theorem (Inverse Function Theorem, One Variable)**

**Theorem 3.8.17** (Inverse Function Theorem, One Variable)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be of class $C^1$, and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. *$f|_U : U \to V$ is a bijection;*
2. *the inverse $g = (f|_U)^{-1} : V \to U$ is of class $C^1$;*
3. *for every $y \in V$,*
$$g'(y) = \frac{1}{f'(g(y))}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$f \in C^1(I) \wedge c \in I \wedge f'(c) \neq 0$$
$$\implies \exists U \ni c\ \exists V \ni f(c)\ \big[\text{Bijective}(f|_U, U, V) \wedge (f|_U)^{-1} \in C^1(V) \wedge \forall y \in V :\ ((f|_U)^{-1})'(y) = 1/f'((f|_U)^{-1}$$

*Remark* (Predicate reading). LocallyInvertible$(f, c)$ holds with a $C^1$ local inverse. The condition $f'(c) \neq 0$ upgrades local set-theoretic invertibility to local analytic invertibility: the inverse exists as a differentiable function and has controlled derivative.

*Remark* (Failure modes). The hypotheses fail in different ways.

- If $f'(c) = 0$, the inverse may exist without being differentiable. The map $f(x) = x^3$ is globally bijective, but its inverse $x \mapsto x^{1/3}$ is not differentiable at 0.
- If $f'$ is not continuous at $c$, the sign of $f'$ need not persist near $c$. Mere differentiability with $f'(c) \neq 0$ is therefore not enough to force local monotonicity or local invertibility.

*Remark* (Contrapositive quantified statement).

$$\neg\,\text{LocallyInvertible}_{C^1}(f, c) \implies f \notin C^1(\text{near } c)\ \vee\ f'(c) = 0.$$

*Remark* (Contrapositive predicate reading). If $f$ has no $C^1$ local inverse at $c$, then either $f$ fails to be continuously differentiable near $c$, or its derivative vanishes there. The witnesses $x \mapsto x^3$ and $x \mapsto x + 2x^2 \sin(1/x)$ isolate the two failure mechanisms.

*Remark* (Interpretation). This theorem is the existence result behind the inverse-derivative formula. A continuous nonzero derivative keeps its sign on a small interval; constant sign forces strict monotonicity; strict monotonicity gives a local inverse; and the Carathéodory characterization upgrades that inverse to a differentiable map with reciprocal slope. In several variables, the condition $f'(c) \neq 0$ becomes invertibility of the differential.

*Remark* (Dependencies).

- Definition: Continuously Differentiable; Class $C^1$
- Theorem: Monotonicity from the Sign of the Derivative
- Theorem: Continuous Inverse Theorem
- Theorem: Carathéodory Characterization of Differentiability

**Corollary (Derivative of the Inverse Function)**

**Corollary 3.8.18** (Derivative of the Inverse Function)**.** *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse $g$ satisfies*

$$g' = \frac{1}{f' \circ g}$$

*on $V$. More globally, if $f : I \to J$ is a $C^1$ bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \to I$ is $C^1$ and*

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \qquad \text{for every } y \in J.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$g = (f|_U)^{-1} \Longrightarrow \forall y \in V : \ g'(y) = \frac{1}{f'(g(y))}.$$

*Remark* (Interpretation). The formula says that undoing a differentiable change of variable reverses the local stretch factor. If $f$ multiplies small increments near $x$ by $f'(x)$, then its inverse multiplies corresponding increments near $f(x)$ by $1/f'(x)$.

*Remark* (Dependencies).

- Theorem: Inverse Function Theorem, One Variable
- Theorem: Chain Rule

### 3.8.5 Indeterminate Forms: L'Hôpital's Rule

**Theorem (L'Hôpital's Rule, $0/0$ Form)**

**Theorem 3.8.19** (L'Hôpital's Rule, 0/0 Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0 \qquad \text{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

*for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then*

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

*The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.*

*Remark* (Standard quantified statement)*.*

$$\left(\lim_{x\to a^+} f(x) = 0 \wedge \lim_{x\to a^+} g(x) = 0 \wedge \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L\right) \implies \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

*Remark* (Failure modes)*.* The rule is a one-way implication and is routinely misapplied.

- *Form not indeterminate.* If $f/g$ is not genuinely $0/0$, differentiating numerator and denominator can answer a different problem.
- $\lim f'/g'$ *fails to exist.* With $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient $f/g \to 1$, but $f'/g' = 1 + \cos x$ oscillates. Absence of $\lim f'/g'$ says nothing by itself about $\lim f/g$.
- $g'$ *vanishes near the target.* The denominator derivative must stay nonzero so the Cauchy Mean Value Theorem applies and the derivative ratio is defined.

*Remark* (Interpretation)*.* The $0/0$ form is the Cauchy Mean Value Theorem read as a limit law. On a shrinking interval, a ratio of changes in $f$ and $g$ is represented by a ratio of derivatives at an interior point. If the derivative ratio settles, the original quotient is forced to settle to the same value.

*Remark* (Dependencies)*.*

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

**Theorem (L'Hôpital's Rule, $\infty/\infty$ Form)**

**Theorem 3.8.20** (L'Hôpital's Rule, $\infty/\infty$ Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} |g(x)| = +\infty \qquad \text{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

*for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then*

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

*No separate hypothesis on $\lim_{x\to a^+} f(x)$ is required. Go to proof.*

*Remark* (Standard quantified statement).

$$\left(\lim_{x\to a^+} |g(x)| = +\infty \wedge \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L\right) \Longrightarrow \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

*Remark* (Failure modes). L'Hôpital's Rule is a one-way implication and requires an indeterminate form. If the quotient is not of type $0/0$ or $\infty/\infty$, differentiating numerator and denominator can change the problem. The derivative ratio must also have a limit: for $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient $f/g$ tends to 1, but $f'/g' = 1 + \cos x$ has no limit.

*Remark* (Interpretation). The $\infty/\infty$ form compares two functions whose magnitudes escape. The Cauchy Mean Value Theorem controls the quotient after subtracting fixed endpoint values, and the divergent denominator makes those fixed values negligible.

*Remark* (Dependencies).

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

## 3.9 Taylor: Construction and Its Error

**Taylor Expansion — Quick Reference**

| Concept | Meaning |
|---|---|
| Taylor polynomial | Polynomial forced by derivative data at a point |
| Taylor remainder | Exact error after subtracting the Taylor polynomial |
| Maclaurin polynomial | Taylor polynomial centered at 0 |
| Lagrange remainder | Error controlled by an intermediate derivative value |
| Peano remainder | Local asymptotic error smaller than the retained order |
| Example: $(\sin x - x)/x^3$ | Taylor and L'Hôpital compared on one limit |
| Flat function | Smooth function whose Taylor series misses it |

Taylor's theorem is the derivative's most ambitious claim: local behaviour to finite order is encoded by derivatives at one point, with an error term that can be estimated. The first-order case is the seed of the closing reframe, where the derivative stops being only a number $f'(c)$ and becomes the linear map $h \mapsto f'(c)h$.

**Definition (Taylor Polynomial at a Point)**

**Definition 3.9.1** (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \le k \le n$. The **Taylor polynomial of order** $n$ for $f$ at $a$ is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } I \subseteq \mathbb{R},\ a \in I,\ n \in \mathbb{N},\ f : I \to \mathbb{R}. \\ &\text{If } f^{(k)}(a) \text{ exists for every } 0 \le k \le n, \text{ then} \\ &\qquad \forall x \in I \ \left( T_{n,a}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right). \end{aligned}$$

*Remark* (Interpretation). The Taylor polynomial is the unique polynomial of degree at most $n$ whose first $n$ derivatives at $a$ agree with those of $f$. Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

*Remark* (Dependencies).

- Derivative at a Point
- Power Rule for Integer Powers of a Function
- Finite Sum Rule

**Definition (Taylor Remainder)**

**Definition 3.9.2** (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** $n$ at $a$ is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

*Remark* (Standard quantified statement).

$$\forall x \in I \ \big(R_{n,a}f(x) = f(x) - T_{n,a}f(x)\big).$$

*Remark* (Interpretation). The remainder is the exact error made by replacing $f$ with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

*Remark* (Dependencies).

- Taylor Polynomial at a Point

**Definition (Maclaurin Polynomial)**

**Definition 3.9.3** (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let $f$ be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \le k \le n$. The **Maclaurin polynomial of order** $n$ for $f$ is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

*Remark* (Standard quantified statement).

$$\forall x \in I \ \left( T_{n,0}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \right).$$

*Remark* (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

*Remark* (Dependencies).

- Taylor Polynomial at a Point

**Theorem (Taylor's Theorem with Lagrange Remainder)**

**Theorem 3.9.4** (Taylor's Theorem with Lagrange Remainder)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on $(a, b)$. Then, for every $x \in (a, b)$, there exists $c$ strictly between $a$ and $x$ such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in (a, b)\ \exists c \in (a, x) \left( f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \right).$$

*Remark* (Interpretation). Taylor's theorem says that the error after truncating at degree $n$ has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point $c$ is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

*Remark* (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

**Corollary (Taylor Expansion with Peano Remainder)**

**Corollary 3.9.5** (Taylor Expansion with Peano Remainder)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order $n$ in a neighbourhood of $a$. If $f^{(n)}$ is continuous at $a$, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \to a} \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}{(x-a)^n} = 0.$$

*Remark* (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order $n$, and the remaining error is smaller than $(x-a)^n$ near the center.

*Remark* (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

**Corollary (First-Order Peano Remainder)**

**Corollary 3.9.6** (First-Order Peano Remainder)**.** *Let $f$ be differentiable at $c$. Then, as $h \to 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \lim_{h \to 0} \frac{f(c+h) - f(c) - f'(c)h}{h} = 0.$$

*Remark* (Interpretation). The first-order Peano formula is the derivative rewritten as an approximation theorem. It says that the affine approximation $f(c) + f'(c)h$ captures all first-order change, and the remaining error is smaller than $h$ itself.

*Remark* (Dependencies).

- Derivative at a Point

*Example* ($(\sin x - x)/x^3$ Two Ways). Taylor expansion gives

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

so

$$\frac{\sin x - x}{x^3} = \frac{-x^3/6 + o(x^3)}{x^3} \longrightarrow -\frac{1}{6}.$$

L'Hôpital's rule reaches the same value by three differentiations:

$$\lim_{x \to 0} \frac{\sin x - x}{x^3} = \lim_{x \to 0} \frac{\cos x - 1}{3x^2} = \lim_{x \to 0} \frac{-\sin x}{6x} = \lim_{x \to 0} \frac{-\cos x}{6} = -\frac{1}{6}.$$

Taylor exposes the reason for the answer: the limit is the cubic coefficient.

**Proposition (The Flat Function: Smooth but Not Analytic)**

**Proposition 3.9.7** (The Flat Function: Smooth but Not Analytic)**.** *Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

*Then:*

1. $f \in C^\infty(\mathbb{R})$*;*
2. $f^{(n)}(0) = 0$ *for every* $n \geq 0$*;*
3. *the Maclaurin series of* $f$ *is identically* $0$*, so* $f \notin C^\omega$ *on any neighbourhood of* $0$*.*

*In particular,* $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \varnothing$*.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\big(\forall n \in \mathbb{N} : f^{(n)}(0) = 0\big) \wedge (\forall r > 0\ \exists x :\ 0 < |x| < r \wedge f(x) \neq 0) \Longrightarrow f \in C^\infty \setminus C^\omega.$$

*Remark* (Interpretation). The Taylor series sees only derivative data at the center, and all of that data is zero for the flat function at 0. The function is nevertheless positive away from 0. This separates smoothness from analyticity and supplies the strictness witness referenced by the smoothness tower.

*Remark* (Dependencies).

- Definition: Higher Derivatives
- Definition: Smooth Function; Class $C^\infty$
- Definition: Real-Analytic Function; Class $C^\omega$
- Definition: Maclaurin Polynomial
- Theorem: The Smoothness Tower

### 3.9.1 The Differential

*Remark* (Interpretation). The derivative value $f'(c) \in \mathbb{R}$ is a number. The differential $df_c$ is a linear map from increments to first-order changes. In one dimension these two objects are represented by the same scalar, but they play different roles.

**Definition (Differentiability by a Differential)**

**Definition 3.9.8** (Differentiability by a Differential)**.** Let $f$ be defined on a neighbourhood of $c \in \mathbb{R}$. The function $f$ is *differentiable at $c$ in the differential sense* if there exists a linear map $L : \mathbb{R} \to \mathbb{R}$ such that

$$f(c+h) - f(c) = L(h) + o(h) \qquad \text{as } h \to 0.$$

In that case the *differential* of $f$ at $c$ is $df_c := L$.

*Remark* (Standard quantified statement).

$$\exists L : \mathbb{R} \to \mathbb{R} \text{ linear such that } f(c+h) - f(c) = L(h) + o(h).$$

*Remark* (Interpretation). The differential is the linear part of the increment. The function itself is approximated affinely by $f(c) + df_c(x - c)$, while $df_c$ alone is the linear map acting on the displacement.

*Remark* (Dependencies).

- First-Order Peano Remainder

**Theorem (Differential and Derivative Agree)**

**Theorem 3.9.9** (Differential and Derivative Agree)**.** *Let $f$ be defined on a neighbourhood of $c \in \mathbb{R}$. Then $f$ is differentiable at $c$ in the ordinary derivative sense if and only if it is differentiable at $c$ in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists df_c \text{ with } df_c(h) = f'(c)h.$$

*Remark* (Interpretation). This theorem is the bridge from the slope number to the linear map. The one-dimensional derivative is multiplication by the scalar $f'(c)$.

*Remark* (Dependencies).

- Derivative at a Point
- Differentiability by a Differential

**Theorem (Uniqueness of the Differential)**

**Theorem 3.9.10** (Uniqueness of the Differential). *If $L_1, L_2 : \mathbb{R} \to \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \qquad \text{as } h \to 0,$$

*then $L_1 = L_2$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\big(L_1(h) - L_2(h) = o(h)\big) \implies L_1 = L_2.$$

*Remark* (Interpretation). The differential is not a choice of linear approximation among many. If a first-order linear part exists, it is forced.

*Remark* (Dependencies).

- Differentiability by a Differential

**Theorem (Differential Continuity Criterion)**

**Theorem 3.9.11** (Differential Continuity Criterion). *If $f$ is differentiable at $c$ in the differential sense, then $f$ is continuous at $c$.* *Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableByDifferentialAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

*Remark* (Interpretation). From $f(c+h) - f(c) = df_c(h) + o(h)$, both terms on the right tend to zero as $h \to 0$. Continuity is therefore immediate in the differential language.

*Remark* (Dependencies).

- Differentiability by a Differential

**Theorem (Chain Rule for Differentials)**

**Theorem 3.9.12** (Chain Rule for Differentials). *If $f$ is differentiable at $c$ and $g$ is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

*Remark* (Interpretation). In one dimension this says that composing two scalar multiplications multiplies their scalars. In several variables the same formula becomes matrix multiplication.

*Remark* (Dependencies).

- Chain Rule
- Differential and Derivative Agree

**Theorem (Linearity of the Differential)**

**Theorem 3.9.13** (Linearity of the Differential)**.** *If $f$ and $g$ are differentiable at $c$ and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

*Remark* (Interpretation). The differential packages the linearity of differentiation as equality of linear maps.

*Remark* (Interpretation). On a manifold, the same construction becomes a tangent map

$$T_c f : T_c M \to T_{f(c)} N.$$

The one-dimensional differential is the prototype: replace real increments by tangent vectors and replace scalar multiplication by a linear map between tangent spaces.

*Remark* (Dependencies).

- Finite Linear Combination Rule
- Differential and Derivative Agree

# Bibliography